PROTECTED PLANAR CARRIERS AND ACCESS-PRESERVING CONNECTOR CRITERIA

MATTHEW ROBERT WESNEY

Abstract. We develop a finite planar theory for connecting pairwise disjoint full continua while preserving specified prime-end accesses and avoiding a frozen obstacle set. A protected ribbon-routing certificate is realized by a compact connected full carrier whose protected cross-cut chains survive as prime-end chains, remain distinct, and retain their cyclic order; no local connectivity of the connected atoms is required. The conceptual center is an exact free-component criterion: a protected carrier exists precisely when one component of the completely frozen free domain admits simultaneous singleton ports to every required atom. The slit lemma underlying the criterion is stated and proved here in a corrected form, with the boundary-anchored slits mapped into the closed domain and the free endpoints retained for routing. A sufficient augmented-packet criterion, based on the compacta form of Alexander duality, converts the criterion into a source-checkable test; for a fixed typed packet assignment, fullification is order independent, and merger, localization, and port-loss defects are persistent under monotone enlargement. A cellwise criterion gives a directly checkable sufficient condition. Examples separate the individual-access and common-component conditions and exhibit persistent post-fullification packet mergers, and we compare the certificate formalism with the standard continuum-hull constructions it replaces. A final section develops the finite direct-decoration application: inside the pullback compactification of a terminal primitive root, finitely many selected decoration packets admit a protected regulated connector that avoids every unselected finite pullback and every end fiber. The results are purely finite and planar: they do not construct the virtual carrier, Wave horizon, analytic correspondence, or uniform incidence bounds required for Mandelbrot local connectivity.

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1. Introduction

1.1. **The problem.** A recurring problem in complex dynamics is to connect a finite family of dynamical continua while preserving selected accesses to their complements. In renormalization theory the continua are small filled Julia sets or their combinatorial stand-ins: the Douady–Hubbard theory of polynomial-like maps produces them [1]; the a priori bounds of Kahn–Lyubich organize them into decorated configurations whose limbs and accesses carry the analytic estimates [5]; Dudko's roadmap describes virtual Julia sets as connected hulls of Cantor families of small Julia sets carrying a finite part of the critical orbit [2]; and Kahn–Kapiamba–Lyubich work with a Hubbard tree constructed inside the full dynamical carrier of a primitive renormalization [4]. In every one of these settings a proof needs, at some stage, a single compact connected full continuum — a *carrier* — joining finitely many pairwise disjoint full continua (*atoms*, in the terminology of the source manuscript, fixed in Section 2) inside a Jordan localization, avoiding a compact forbidden set of obstacles and critical values, and, crucially, leaving intact certain designated accesses: prime ends of the atoms through which later curve-family, width, or landing arguments will read the picture.

Ordinary connectedness is far too weak a specification for this task, because a connector chosen merely to connect can destroy the data that made the connection worth having. Three failure modes recur. The connector, or an excess set absorbed together with it, can *absorb a protected side*: the enlarged carrier swallows the collar through which a marked access was to be read, and no curve in the new complement lands there from the marked side any more. It can *identify two prime ends*: two designated crosscut chains, distinct in the complement of the atom, can become equivalent in the complement of the enlarged carrier, so that marked access data collapses. And it can *create a separator*: a connector placed at one stage can disconnect the free region needed at a later stage, and monotone enlargement never undoes a separation. All three failures are invisible to the coarse bookkeeping — homeomorphism types, degrees, critical words — that dynamical constructions usually carry, which is what makes them dangerous.

1.2. **Main results.** The first group of results makes the failures precise and shows that they already occur in the simplest possible situations. Theorem 3.1 (manuscript Theorem 6.12) shows that there is no least compact

connected full continuum containing even the two-point package $\{-1, 1\}$: carrier selection by minimization is not merely inconvenient but impossible, since minimal carriers abound while no least one exists. Proposition 3.2 (manuscript Proposition 6.15) exhibits a conformal, critical-value-free, degree-one branch of $z \mapsto z^2$ in which the very first monotone absorption of pulled excess erases a protected prime-end access, together with its conformally transported copy, irreversibly. Between them, the two obstructions close both naive routes: no least hull exists to select, and a proposed hull cannot be blindly completed.

The positive theory replaces hull selection by certification. A *protected ribbon-routing certificate* (Definition 5.1; manuscript Definition 10.1) records, as explicit data: the atoms; the frozen forbidden set; the protected accesses, each presented as a prime-end crosscut chain with its selected sides, truncated so that the terminal collars are pairwise disjoint; free-port arcs attaching to each atom at a single free point; and an embedded ribbon tree joining the ports, with its planar cyclic data declared. Theorem 5.4 (manuscript Theorem 10.4) realizes every certificate by an actual carrier: a compact connected full continuum, compactly contained in the localization, containing the atoms, missing the forbidden set, with singleton attachment traces at the atoms, in whose complement every protected chain survives as a prime-end chain, the survival assignment being injective and cyclic-order preserving. No local connectivity of any atom is assumed at any point: the hypothesis that such constructions can never afford is never used.

The conceptual center of the paper is the exact criterion, Theorem 6.3 (manuscript Theorem 10.8). Freeze, before any connector is chosen, the complete finite future inventory — atoms, closed terminal protected collars, separated obstacles, critical-value exclusions, and the localization-forbidden set — call the union Q, and let $\Omega_{\mathrm{free}} = U \setminus Q$ be the frozen free domain. Then a protected ribbon-routing certificate exists if and only if a single component of Ω_{free} admits simultaneous singleton ports to every required atom. Moreover, the failure mode is monotone: if a first protected collar or obstacle separates two required port classes, no further monotone connector enlargement inside the localization can repair the failure. The criterion converts an existential search over embedded trees into a property of one planar set fixed in advance, and it explains, in hindsight, the design of the certificate: everything the future proof will read must be frozen into Q before routing begins, precisely so that the criterion quantifies over nothing but Q.

The criterion is exact but not yet checkable from partial data, and the next group of results addresses checkability. Theorem 7.4 (manuscript Theorem 10.11) gives a sufficient test in packet form: if the frozen inventory can be assigned to finitely many pairwise disjoint full continua — augmented atoms and separated obstacles, each the filled hull of a connected seed — with one unprotected free port per original atom, then the compacta form

of Alexander duality forces the common complement to be connected, producing the single free component that the exact criterion demands. Theorem 8.3 (manuscript Theorem 10.14) turns the packet clauses into an audit for a fixed typed assignment: fullification is independent of processing order; the clauses reduce to three verifiable conditions (coverage of the frozen inventory by compact connected seeds; pairwise disjoint completions compactly contained in the localization; nonempty unprotected exposure locus for every original atom); and each failure — packet merger, localization escape, port loss — is witnessed by a certificate that persists under every monotone same-label enlargement. Proposition 8.5 (manuscript Proposition 10.15) then gives the most directly checkable sufficient condition: if the seeds sit in pairwise disjoint closed Jordan cells, with one boundary-reaching unprotected arc per atom, every packet clause follows.

The final substantive section applies the framework at full strength. In the direct-decoration setting — a terminal primitive root continuum with two disjoint univalent pullbacks generating a pullback compactification with finite fibers and end fibers — Theorem 9.10 (manuscript Theorem 10.45) produces, for any finite selection of decoration addresses and finitely many protected root prime-end chains with proper total impression, a protected regulated connector whose packet fullifications stay inside pairwise disjoint gap cells, avoid every unselected finite fiber and every end fiber, and preserve the transported prime ends together with their cyclic order. The application exercises every part of the machinery: gap-cell isolation, packet replacement, impression ports, the audit, and the realization theorem.

1.3. **What is new, and what is not.** The ingredients of the proofs are classical: the Janiszewski separation theorem, Carathéodory's theory of prime ends, filled hulls, the Jordan and Schoenflies extension theorems, and Čech cohomology with the compacta form of Alexander duality. Readers should not expect — and will not find — a new theorem of classical planar topology. The contribution is the assembled certification framework: accesses protected explicitly, as crosscut chains with selected sides, rather than implicitly, as intentions; cyclic order carried as data and proved to survive; the complete future inventory frozen before routing, so that existence becomes a property of one planar set; typed packets whose completion is order-independent; and failure certificates that are monotone-persistent, so that a negative answer is as stable and as citable as a positive one. In short, the paper contributes proof technology for access-preserving planar connection, not a new fact about the plane. Section 4 compares this technology in detail with the standard continuum-hull constructions it is designed to replace: least hulls, monotone absorption, regulated and Hubbard trees, Moore-type monotone quotients, and the lamination-lift candidate.

1.4. **Relation to the source manuscript.** All results are reproduced, with notational streamlining only, from the author's research manuscript [7], and every reproduced statement carries its manuscript number in its

bracketed title. There is exactly one intentional mathematical divergence. Manuscript Lemma 10.7, the slit lemma underlying the exact criterion, is formally inconsistent as printed: its slits are declared as maps into the open domain yet are required to take a boundary value at one end, and the removal prescribed there also deletes the free endpoints that the proof of manuscript Theorem 10.8 later joins inside the complement. Lemma 6.1 states and proves the repaired form — slits mapped into the closed domain, boundary-anchored, with only the open slit interiors removed and the free tips retained for routing — and the proof of Theorem 6.3 is rewritten accordingly. Remark 6.2 records the divergence precisely. No other statement has been altered. The comparison Section 4 and the worked examples of Sections 6 and 8 are new expository material for this paper; they promote no manuscript claim.

1.5. **Organization.** Section 2 fixes notation and collects the classical background: full continua and filled hulls, Janiszewski's theorem, prime ends and crosscut chains with their selected sides, the compacta form of Alexander duality, and the Jordan–Schoenflies extension package, together with the manuscript's terms *atom* and *fullification*. Section 3 proves the two motivating obstructions, Theorem 3.1 and Proposition 3.2. Section 4 carries out the comparison with standard continuum-hull constructions. Section 5 defines protected ribbon-routing certificates and proves the realization theorem and its conformal naturality. Section 6 proves the repaired slit lemma and the exact free-component criterion, together with an example separating individual access from the existence of a common free component. Section 7 develops augmented-atom packets and the common-domain theorem, and Section 8 proves order-independent fullification, the port audit with its persistent failure certificates, and the cellwise criterion, together with an example of a persistent post-fullification packet merger. Section 9 develops the finite direct-decoration application, culminating in Theorem 9.10. Section 10 states the scope boundary.

1.6. **Scope.** A firm boundary is maintained throughout, following the source manuscript's own discipline (Warning 10.59 of [7]). This paper works entirely with finite, explicitly given planar data. It does not construct a virtual carrier or virtual component package; it does not construct or use a Wave horizon; it establishes no analytic correspondence and no covering or immersion action; it proves no uniform incidence bound and no compression; and it makes no claim about local connectivity of the Mandelbrot set. The open problems of the source manuscript that would supply such objects — in particular manuscript Open Problems 10.6 and 10.58 — are cited only as open, and nothing here advances them. What is proved is exactly this: once a finite protected certificate or packet system is supplied, the planar realization problem is completely solved, with an exact criterion, checkable sufficient conditions, and persistent failure certificates. Section 10 restates the boundary in detail.

2. Notation and classical background

Everything in this section is classical. It is collected once, in the normalized forms used by the rest of the paper, for three reasons: the statements are invoked repeatedly and in exact form; two of them — the compacta form of Alexander duality and the tameness of plane arcs — are easy to quote in weaker versions that would not support the proofs; and the source manuscript's terminology — *atom*, *package*, *carrier*, *fullification*, *protected access*, *collar* — is fixed here at first use.

2.1. **Plane, sphere, and general notation.** Throughout, $\mathbb{C}$ is the complex plane, $\mathbb{Z}$ the integers, and $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ the Riemann sphere, identified with the round sphere S^2 by stereographic projection and carrying its standard orientation. Compact subsets of $\mathbb{C}$ are exactly the compact subsets of $\widehat{\mathbb{C}}$ missing ∞, and we pass between the two ambient spaces without comment. The symbol dist denotes Euclidean distance in $\mathbb{C}$; diameters of crosscuts (Section 2.4) are measured in the spherical metric, and all other metric quantities are Euclidean. For a set X, $\operatorname{int} X$ is its interior, ∂X its boundary, and $\overline{X}$ its closure, taken in the ambient space clear from context; the relative closure of X in a subspace Y is written $\overline{X}^{\,Y}$, as in the attachment trace (7).

For sets A, B we write $A \Subset B$, and say A is *compactly contained* in B, if $\overline{A}$ is compact and $\overline{A} \subset \operatorname{int} B$; this is the manuscript's relation $\Subset$. A *domain* is a nonempty connected open subset of $\mathbb{C}$ or $\widehat{\mathbb{C}}$; a *plane domain* is a domain in $\mathbb{C}$. Every plane domain is polygonally connected: any two of its points are joined by a polygonal path inside it [8]. A *Jordan curve* is a homeomorphic image of the circle, a *Jordan arc* a homeomorphic image of $[0,1]$, and an *open Jordan arc* a Jordan arc with its two endpoints removed. A *Jordan disk* (synonymously, a *Jordan domain*) is a bounded plane domain whose boundary is a Jordan curve; by the Jordan and Schoenflies theorems recalled in Proposition 2.8 below, a Jordan disk is simply connected, and its closure — a *closed Jordan disk* — is homeomorphic to the closed unit disk. A *continuum* is a nonempty compact connected subset of $\mathbb{C}$ or $\widehat{\mathbb{C}}$; it is *nondegenerate* if it contains more than one point.

2.2. **Full continua, filled hulls, and fullification.** A compact set $K \subset \mathbb{C}$ is *full* if $\mathbb{C} \setminus K$ is connected. Following the source manuscript (equation (9.5) of [7]), the *filled hull* of a compact set $K \subset \mathbb{C}$ is

$$\operatorname{Fill}(K) \;=\; K \cup \bigcup \big\{\, B : B \text{ a bounded component of } \mathbb{C} \setminus K \,\big\}, \qquad (1)$$

that is, K together with the bounded components of its complement. The passage from K to $\operatorname{Fill}(K)$ is called *fullification*; the term is the manuscript's, and it is used as a noun for the operation and, occasionally, for its result. The following properties are used constantly and are recorded with proof.

Lemma 2.1 (Basic properties of filled hulls)**.** *Let $K \subset \mathbb{C}$ be compact, and let $\Omega_\infty(K)$ denote the unbounded component of $\mathbb{C} \setminus K$.*

1. $\mathrm{Fill}(K) = \mathbb{C} \setminus \Omega_\infty(K)$*; in particular* $\mathrm{Fill}(K)$ *is compact, and it is contained in every closed disk containing K.*
2. *The following are equivalent: K is full;* $\widehat{\mathbb{C}} \setminus K$ *is connected;* $\mathrm{Fill}(K) = K$*.*
3. *If K is a continuum, then* $\mathrm{Fill}(K)$ *is a full continuum, and* $\widehat{\mathbb{C}} \setminus \mathrm{Fill}(K) = \Omega_\infty(K) \cup \{\infty\}$ *is a simply connected domain containing ∞.*
4. *The operator* Fill *is extensive ($K \subset \mathrm{Fill}(K)$), monotone ($K \subset L$ implies $\mathrm{Fill}(K) \subset \mathrm{Fill}(L)$), and idempotent ($\mathrm{Fill}(\mathrm{Fill}(K)) = \mathrm{Fill}(K)$).*

Proof. (1) The set $\mathbb{C} \setminus \Omega_\infty(K)$ consists of K together with all bounded components of $\mathbb{C} \setminus K$, which is (1); it is closed because $\Omega_\infty(K)$ is open. If E is a closed disk with $K \subset E$, then $\mathbb{C} \setminus E$ is a connected unbounded subset of $\mathbb{C} \setminus K$, hence contained in $\Omega_\infty(K)$; taking complements gives $\mathrm{Fill}(K) \subset E$. Thus $\mathrm{Fill}(K)$ is closed and bounded.

(2) The components of $\widehat{\mathbb{C}} \setminus K$ are the bounded components of $\mathbb{C} \setminus K$ together with $\Omega_\infty(K) \cup \{\infty\}$: the latter set is open in $\widehat{\mathbb{C}}$ (a spherical neighborhood of ∞ misses the compact set K, and its intersection with $\mathbb{C}$, being connected and unbounded, lies in $\Omega_\infty(K)$), it is connected, and its complement within $\widehat{\mathbb{C}} \setminus K$ is the union of the bounded components, which is open. Hence $\widehat{\mathbb{C}} \setminus K$ is connected exactly when there are no bounded components, which is fullness of K, which by (1) is the identity $\mathrm{Fill}(K) = K$.

(3) Let K be a continuum and let B be a bounded component of $\mathbb{C} \setminus K$. The boundary of B is contained in K: if a boundary point x of B lay in $\mathbb{C} \setminus K$, its component B' in $\mathbb{C} \setminus K$ would be an open set containing x, hence meeting B, hence equal to B; but then $x \in B$, contradicting openness of B at its own boundary point. So $\partial B \subset K$. Thus $\overline{B} = B \cup \partial B$ is a connected set meeting K, and $\mathrm{Fill}(K) = K \cup \bigcup_B \overline{B}$ is a union of connected sets each meeting the connected set K, hence connected. With (1) this makes $\mathrm{Fill}(K)$ a continuum, and its complement in $\widehat{\mathbb{C}}$ is the domain $\Omega_\infty(K) \cup \{\infty\}$, so $\mathrm{Fill}(K)$ is full. Finally, a domain in $\widehat{\mathbb{C}}$ whose complement is connected is simply connected [8, 10], and the complement of the domain $\Omega_\infty(K) \cup \{\infty\}$ is the continuum $\mathrm{Fill}(K)$.

(4) Extensivity is (1). For monotonicity, let $K \subset L$; then $\Omega_\infty(L)$ is a connected unbounded subset of $\mathbb{C} \setminus K$, hence $\Omega_\infty(L) \subset \Omega_\infty(K)$, and taking complements in (1) gives $\mathrm{Fill}(K) \subset \mathrm{Fill}(L)$. For idempotence, the complement of $\mathrm{Fill}(K)$ is the single unbounded component $\Omega_\infty(K)$, so $\mathrm{Fill}(\mathrm{Fill}(K)) = \mathbb{C} \setminus \Omega_\infty(K) = \mathrm{Fill}(K)$. □

The deeper absorption law $\mathrm{Fill}(\mathrm{Fill}(K) \cup L) = \mathrm{Fill}(K \cup L)$, which makes fullification of a typed packet independent of processing order, is Lemma 8.1 below (manuscript Lemma 10.12); it is a formal consequence of the three properties in clause (4).

Definition 2.2 (Package, atoms, carrier, fullification)**.** A *package* is a finite union $S = A_1 \cup \cdots \cup A_s$ of pairwise disjoint full continua, called its *atoms*. Atoms may be degenerate: in Theorem 3.1 the package consists of the two points ± 1. A *carrier* of the package S is a compact connected full continuum $H \subset \mathbb{C}$ with $S \subset H$. Both *atom* — a required full continuum of the package — and *fullification* — the passage (1) — are the source manuscript's terms [7] and are used in its sense throughout.

2.3. **Janiszewski's separation theorem.** A closed set $F \subset \widehat{\mathbb{C}}$ *separates* points $p, q \in \widehat{\mathbb{C}} \setminus F$ if p and q lie in different components of $\widehat{\mathbb{C}} \setminus F$; the set F is *nonseparating* if $\widehat{\mathbb{C}} \setminus F$ is connected. By Lemma 2.1(2), a compact subset of $\mathbb{C}$ is nonseparating in $\widehat{\mathbb{C}}$ exactly when it is full; the two words are used interchangeably for such sets.

Theorem 2.3 (Janiszewski's separation theorem; see [8, 10])**.** *Let $F_1, F_2 \subset \widehat{\mathbb{C}}$ be closed sets whose intersection $F_1 \cap F_2$ is connected and nonempty, and let $p, q \in \widehat{\mathbb{C}} \setminus (F_1 \cup F_2)$. If neither F_1 nor F_2 separates p from q, then $F_1 \cup F_2$ does not separate p from q.*

Corollary 2.4. *If $F_1, F_2 \subset \widehat{\mathbb{C}}$ are closed and nonseparating and $F_1 \cap F_2$ is connected and nonempty, then $F_1 \cup F_2$ is nonseparating. In particular, the union of two full compact subsets of $\mathbb{C}$ whose intersection is connected and nonempty is full.*

Proof. Let $p, q \in \widehat{\mathbb{C}} \setminus (F_1 \cup F_2)$. Since $\widehat{\mathbb{C}} \setminus F_1$ is connected, F_1 does not separate p from q, and likewise for F_2. By Theorem 2.3, $F_1 \cup F_2$ does not separate p from q. As the pair was arbitrary, $\widehat{\mathbb{C}} \setminus (F_1 \cup F_2)$ is connected. The final sentence follows from Lemma 2.1(2). □

This is the form used in the Janiszewski induction of Lemma 5.2, where the intersections are singletons. The complementary situation of finitely many pairwise *disjoint* full compacta, where the stated form of the theorem does not apply, is handled cohomologically, by the duality of Section 2.5 and the nerve argument of Lemma 7.3 (manuscript Lemma 10.10).

2.4. **Prime ends, crosscut chains, and protected accesses.** Informally, an *access* to a compact set K is a way of reaching ∂K from the complement. It can be presented by a landing arc, as the access $p_0^+(m)$ of Proposition 3.2 is presented by the vertical segment $m + it$; but landing presentations presuppose that something lands, and for the arbitrary continua handled in this paper no landing is available. The presentation that requires nothing is Carathéodory's theory of prime ends [9], which we now recall in the normalization used throughout.

Let $\Omega \subset \widehat{\mathbb{C}}$ be a simply connected domain whose complement contains at least two points. The standing example is $\Omega = \widehat{\mathbb{C}} \setminus K$ for a nondegenerate full continuum K, in which case Ω is a simply connected domain containing ∞ by Lemma 2.1(3); Jordan disks are a second example. A *crosscut* of Ω is

an open Jordan arc $\gamma \subset \Omega$ whose closure $\overline{\gamma}$ is a Jordan arc with two distinct endpoints on $\partial\Omega$. A crosscut divides Ω into exactly two components, each a simply connected domain having γ on its boundary [9].

A *crosscut chain* (or *prime-end chain*; Pommerenke's null chains [9], in the following normalization) is a sequence

$$\Gamma = (\gamma_n, D_n)_{n\geq 1}$$

in which the γ_n are pairwise disjoint crosscuts of Ω whose spherical diameters tend to zero, and D_n, the *selected side* of γ_n, is one of the two components of $\Omega\backslash\gamma_n$, subject to the coherence condition $D_{n+1} \subset D_n$ for every n. Coherence and disjointness together force $\gamma_{n+1} \subset D_n$: the crosscut γ_{n+1} lies on the boundary of $D_{n+1} \subset D_n$, so it lies in $\overline{D_n} \cap \Omega = D_n \cup \gamma_n$, and it misses γ_n. The record of a chain thus contains strictly more than the crosscuts: it contains, for every n, the declaration of the selected side, and it is exactly this declaration that the certificates of Section 5 protect. Two chains are *equivalent* if each eventually divides the other, that is, if every selected domain of either chain contains all sufficiently deep selected domains of the other. An equivalence class is a *prime end* of Ω.

Two classical facts about prime ends are used. First, the Carathéodory correspondence: an orientation-preserving Riemann map $\psi\colon \{|z| < 1\} \to \Omega$ induces a bijection between the points of the unit circle and the prime ends of Ω, under which radial limits correspond to chains; see [9, Chapter 2]. Second, the *impression*

$$I(\Gamma) \;=\; \bigcap_{n\geq 1} \overline{D_n}$$

(closures in $\widehat{\mathbb{C}}$) is a nonempty compact connected subset of $\partial\Omega$ — the intersection of a decreasing sequence of nonempty compacta — and depends only on the prime end of Γ, not on the representing chain [9]. The nested closures $\overline{D_n}$ decrease *to* the impression; this is the form in which impressions enter Lemma 9.8.

The prime ends of Ω carry a natural *cyclic order*: the one transported from the unit circle by the Carathéodory correspondence. It is independent of the choice of Riemann map, since two orientation-preserving Riemann maps differ by a disk automorphism, which preserves the cyclic orientation of the circle. For a finite family of prime ends represented by chains with pairwise disjoint selected sides, the cyclic order can be computed inside the domain, with no reference to a uniformization: choose one sufficiently deep oriented crosscut from each chain; the cyclic order of the prime ends is the separation order of these pairwise disjoint crosscuts in Ω. The computation is carried out carefully, by means of embedded triods, in the proof of Lemma 5.3, and that formulation is the one shown to be invariant under carrier enlargement.

Definition 2.5 (Protected access; collar)**.** Let A be a nondegenerate atom. A *protected access* to A is a distinguished prime-end crosscut chain $\Gamma = (\gamma_n, D_n)_{n\geq 1}$ of $\widehat{\mathbb{C}} \setminus A$, recorded together with its selected sides. The terminal

selected domain D_1 is the *collar* of the access, and $D_1 \setminus A$ is its *collar tail*. When finitely many protected accesses appear together inside a Jordan localization, they are *truncated* — an initial segment of each chain is discarded and the indices shifted — so that the terminal collar tails are pairwise disjoint, lie in the localization, and miss the forbidden set and the other atoms; Section 5 records this normalization as part of the certificate data.

The adjective *protected* names a discipline, not a topological property: a protected access is one that every later construction is forbidden to disturb, in the precise sense that no connector may enter its collar tail (Definition 5.1). The reward for the discipline is the survival theorem, Lemma 5.3: the chain then persists verbatim, with the same crosscuts and the same selected sides, as a prime-end chain of the enlarged complement.

2.5. **Čech cohomology and Alexander duality.** For a compact set $K \subset S^2$, $\check{H}^*(K;\mathbb{Z})$ denotes Čech cohomology with integer coefficients. Two of its properties are needed [3]: continuity — the Čech cohomology of K is the direct limit of the singular cohomology of the open neighborhoods of K, so that every class and every relation is realized in some neighborhood — and the Mayer–Vietoris sequence for pairs of compact sets in S^2. Both are used in the nerve computation of Lemma 7.3 and in the second proof of Lemma 5.2. The duality statement is the following.

Theorem 2.6 (Alexander duality, compacta form; [3, pp. 256–257])**.** *For every nonempty proper compact subset $K \subset S^2$,*

$$\widetilde{H}_0\big(S^2 \setminus K;\mathbb{Z}\big) \;\cong\; \check{H}^1(K;\mathbb{Z}). \tag{2}$$

This is equation (9.6) of the manuscript [7]. Reading (2) at the level of connectivity gives the dictionary used throughout the paper: $S^2 \setminus K$ is connected exactly when $\check{H}^1(K;\mathbb{Z}) = 0$, and hence, by Lemma 2.1(2), a compact set $K \subset \mathbb{C}$ is full exactly when $\check{H}^1(K;\mathbb{Z}) = 0$. "Nonseparating" and "vanishing first Čech cohomology" are synonyms, and the two proofs of Lemma 5.2 translate one into the other.

Warning 2.7. The Čech form of (2), valid for an arbitrary nonempty proper compact $K \subset S^2$, is required; the locally contractible special case, in which Čech and singular cohomology agree [3], is not sufficient. The atoms of this paper are arbitrary full continua — in general not locally connected, hence not locally contractible, and admitting no CW approximation compatible with their embedding — and for such compacta duality genuinely fails with singular cohomology in place of Čech. The standard example is the Warsaw circle $W \subset \mathbb{C}$, the closed topologist's sine curve completed by an arc: W separates the plane, so $\widetilde{H}_0(S^2 \setminus W;\mathbb{Z}) \cong \mathbb{Z}$, and accordingly $\check{H}^1(W;\mathbb{Z}) \cong \mathbb{Z}$; but W is path connected with trivial fundamental group — no loop can traverse the accumulation segment — so its singular homology H_1 vanishes and its *singular* cohomology gives $H^1_{\mathrm{sing}}(W;\mathbb{Z}) = 0$. Every duality argument in this paper — Lemma 5.2 (second proof), Lemma 7.3, Lemma 9.2 — is

therefore carried out in Čech cohomology, and the nerve and continuity properties invoked there are the Čech ones.

2.6. **Jordan, Schoenflies, and Carathéodory extensions.** The naturality statements of this paper (Proposition 5.6, the naturality clauses of Theorem 8.3) rest on a package of classical extension theorems, collected here with the exact attributes used later.

Proposition 2.8 (Extension package)**.**

(1) (Jordan; Schoenflies) *Every Jordan curve $J \subset \widehat{\mathbb{C}}$ divides $\widehat{\mathbb{C}}$ into exactly two components, each a simply connected domain with boundary J, and the closure of each is homeomorphic to the closed unit disk. Moreover, every homeomorphism between two Jordan curves in $\widehat{\mathbb{C}}$ extends to a homeomorphism of $\widehat{\mathbb{C}}$.*

(2) (Tameness of plane arcs) *For every Jordan arc $\alpha \subset \mathbb{C}$ there is a homeomorphism of $\mathbb{C}$ carrying α onto the straight segment $[0,1]\times\{0\}$. In particular every Jordan arc extends to a Jordan curve, and no arc separates the plane.*

(3) (Carathéodory extension) *Every conformal homeomorphism $\phi\colon U \to V$ of Jordan disks extends to a homeomorphism $\overline{U} \to \overline{V}$ of their closures.*

(4) (Glued sphere extension) *Every orientation-preserving conformal homeomorphism $\phi\colon U \to V$ of Jordan disks $U, V \subset \mathbb{C}$ extends to an orientation-preserving homeomorphism $\Phi\colon \widehat{\mathbb{C}} \to \widehat{\mathbb{C}}$, and Φ may be chosen with $\Phi(\infty) = \infty$.*

(5) (Fill-equivariance) *Every homeomorphism Φ of $\widehat{\mathbb{C}}$ with $\Phi(\infty) = \infty$ satisfies $\Phi(\mathrm{Fill}(K)) = \mathrm{Fill}(\Phi(K))$ for every compact $K \subset \mathbb{C}$.*

Proof. Clauses (1) and (2) are classical plane topology; see [8, 10]. Clause (3) is Carathéodory's theorem on boundary extension of Riemann maps of Jordan domains; see [9].

For (4), extend ϕ to a homeomorphism $\overline{\phi}\colon \overline{U} \to \overline{V}$ by (3). Its boundary restriction is a homeomorphism $\partial U \to \partial V$ of Jordan curves; the complementary sets $\widehat{\mathbb{C}} \setminus U$ and $\widehat{\mathbb{C}} \setminus V$ are closed Jordan disks by (1), and a homeomorphism between the boundary circles of two closed disks extends over the disks (cone the homeomorphism off an interior point, after identifying each closed disk with the standard one by (1)). Gluing the two extensions along the boundary curves yields a homeomorphism $\Phi\colon \widehat{\mathbb{C}} \to \widehat{\mathbb{C}}$ with $\Phi|_U = \phi$. A sphere homeomorphism that preserves orientation on a nonempty open set preserves it globally, so Φ is orientation preserving. Finally, ∞ lies in the interior of the complementary disk $\widehat{\mathbb{C}} \setminus U$, and $\Phi(\infty)$ in the interior of $\widehat{\mathbb{C}} \setminus V$; composing Φ with a homeomorphism of the closed disk $\widehat{\mathbb{C}} \setminus V$ that fixes ∂V pointwise and carries $\Phi(\infty)$ to ∞ (such a homeomorphism exists, interiors of disks being homogeneous relative to the boundary), extended by the identity on $\overline{V}$, arranges $\Phi(\infty) = \infty$ without disturbing $\Phi|_U = \phi$.

For (5), the restriction $\Phi|_{\mathbb{C}}$ is a homeomorphism of $\mathbb{C}$ onto itself. It maps $\mathbb{C} \setminus K$ homeomorphically onto $\mathbb{C} \setminus \Phi(K)$ and permutes complementary components. A homeomorphism of $\mathbb{C}$ is proper, so the image of the unbounded component of $\mathbb{C} \setminus K$ is unbounded and the images of bounded components are bounded; hence $\Phi(\Omega_\infty(K)) = \Omega_\infty(\Phi(K))$, and taking complements as in Lemma 2.1(1) gives $\Phi(\mathrm{Fill}(K)) = \mathrm{Fill}(\Phi(K))$. □

Clause (4) is the mechanism behind every naturality statement in the paper: a univalent change of Jordan localization extends to a sphere homeomorphism, and a sphere homeomorphism preserves all the notions the certificates are built from — compactness, connectedness, fullness (by clause (5) and Lemma 2.1), crosscuts, selected sides, null chains, intersections and disjointness, and, when orientation is preserved, the cyclic order of prime ends and the ribbon structure of an embedded tree. Clause (2) is used in the slit-straightening steps of the repaired slit lemma (Lemma 6.1).

This completes the classical inventory. The dynamical vocabulary of the source manuscript — frozen inventory, free domain, ports, packets, audits — is introduced where it is first needed: the frozen free domain Ω_{free} in Section 6, packets and their audit in Sections 7 and 8.

3. Obstructions to naive carriers

Before any positive theory is developed, it is worth recording exactly why the two most tempting shortcuts fail. The first shortcut is a *selection* rule: given the finite package of marked continua that a carrier must contain, simply take the least compact connected full set containing it. The second is a *completion* rule: given a proposed carrier and the excess produced by pulling it back through the dynamics, simply absorb that excess monotonically until the process stabilizes. Both rules are natural, both are implicit in informal uses of connected hulls, and both fail — the first because the minimizer does not exist, the second because absorption destroys precisely the access data that a protected carrier is supposed to preserve. The two obstructions below are taken from the source manuscript [7, Theorem 6.12, Proposition 6.15]; they are elementary, but they dictate the architecture of everything that follows, and in particular the certificate formalism of Section 5. The comparison with standard continuum-hull constructions in Section 4 should be read with both examples in mind.

3.1. **Failure of least carriers.** Recall from Section 2 that a compact set $K \subset \mathbb{C}$ is *full* if $\mathbb{C} \setminus K$ is connected, and that a *carrier* for a finite package $S \subset \mathbb{C}$ is a compact connected full continuum containing S. Order the carriers of a fixed package by inclusion. A *least* carrier would be a carrier contained in every other carrier of the same package. The following theorem shows that a least carrier fails to exist already for the simplest nondegenerate package: two points, with the identity map as dynamics and with no obstacle set whatsoever.

Theorem 3.1 (No least connected full carrier; manuscript Theorem 6.12)**.** *There is no least compact connected full plane continuum containing the two-point package $S = \{-1, 1\}$, even with identity dynamics and no obstacles.*

Proof. Let

$$A_+ = \{e^{i\theta} : 0 \le \theta \le \pi\}, \qquad A_- = \{e^{i\theta} : -\pi \le \theta \le 0\}$$

be the upper and lower semicircles of the unit circle joining -1 to 1. Each is an arc, that is, a homeomorphic image of $[0, 1]$; hence each is compact and connected. An arc does not separate the plane [8], so $\mathbb{C} \setminus A_\pm$ is connected and each $A_\pm$ is full. Each contains S, so each is a carrier for S; in particular the family of carriers is nonempty. Finally, the semicircles meet exactly in their common endpoints:

$$A_+ \cap A_- = S.$$

Suppose a least carrier H existed. By definition of leastness, H is contained in every carrier of S; applying this to the two carriers A_+ and A_- gives

$$H \subset A_+ \cap A_- = S.$$

On the other hand $S \subset H$, since H is a carrier of the package S. Hence $H = S = \{-1, 1\}$. But the two-point set S is disconnected, while a carrier is by definition connected. This contradiction shows that no least carrier exists. □

The failure is one of *leastness*, not of minimality: minimal carriers exist in abundance. Indeed, a subcontinuum of an arc that contains both endpoints is the whole arc, so every arc from -1 to 1 — in particular each of A_+ and A_-, and every circular arc through -1 and 1 — is a minimal element of the family of carriers. The order-theoretic structure is thus the worst possible for a canonical selection: a large family of pairwise distinct minimal elements and no least one.

The moral drawn in the manuscript is the one that governs this paper: "take the minimal connected full hull" is not a carrier rule. Even for two points and trivial dynamics, the carrier is underdetermined, and the ambiguity is not a set-theoretic accident but a genuine topological degree of freedom — the two semicircles pass on opposite sides of the origin, and in any configuration with obstacles, additional atoms, or protected accesses, that choice of side is visible to the rest of the construction. A regulated connector system, together with its access or isotopy class and a forward-coherence proof, is therefore *additional data*, beyond anything determined by the package itself. Recording that data explicitly, and saying precisely when it can be realized by an actual carrier, is the task of the certificate formalism of Section 5.

3.2. **An analytic protected-access collision.** Theorem 3.1 rules out selecting the carrier by minimization. The remaining naive rule is completion:

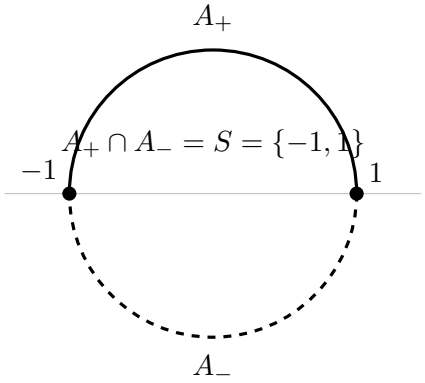


FIGURE 1. Two minimal carriers for the package $S = \{-1, 1\}$: the upper semicircle A_+ (solid) and the lower semicircle A_- (dashed). Each is an arc, hence a compact connected full continuum containing S, and they meet only in the package points. A least carrier would be contained in their intersection S, which is disconnected; hence no least carrier exists (Theorem 3.1).

accept an arbitrary proposed carrier and then absorb, monotonically, whatever excess the dynamics pulls back onto it. The manuscript's Section 6.4 shows that this rule fails as well, and fails in the most favorable analytic circumstances imaginable: a single conformal, critical-value-free, degree-one branch of the quadratic map, acting on an interval. The absorption step erases a protected prime-end access, and monotonicity then propagates the erasure forever.

We reproduce the analytic setup exactly. Let

$$f(z) = z^2, \qquad I = [1, 2], \qquad m = \tfrac{3}{2},$$

and let

$$Q = \Big\{ z : \big|z - \tfrac{3}{2}\big| \le \tfrac{1}{2},\ \operatorname{Im} z \ge 0 \Big\} \tag{3}$$

be the closed upper half-disk of radius $\frac{1}{2}$ centered at m. Set

$$U^- = \big\{z : \big|z - \tfrac{3}{2}\big| < 1\big\}, \qquad U^+ = f(U^-).$$

The disk U^- lies in the half-plane $\operatorname{Re} z > \frac{1}{2}$. Two consequences follow. First, $0 \notin U^-$, so $f'(z) = 2z \neq 0$ on U^- and the critical point of f is outside the domain; since $f(z) = f(w)$ forces $w = \pm z$, and U^- is disjoint from its antipode $-U^- \subset \{\operatorname{Re} z < -\frac{1}{2}\}$, the restriction

$$F = f|_{U^-} : U^- \longrightarrow U^+ \tag{4}$$

is injective, hence a conformal homeomorphism onto U^+. Second, the critical value $0 = f(0)$ is not in U^+, so the branch is critical-value-free. Take

$$T_0 = I, \qquad T_0' = Q, \qquad T_1 = f(I) = [1, 4], \qquad T_1' = f(Q). \tag{5}$$

Within this branch the stage-0 data are the pullbacks of the stage-1 data:

$$T_0' = F^{-1}(T_1'), \qquad T_0 = F^{-1}(T_1). \tag{6}$$

Thus $T_0 = I$ is the proposed carrier at stage 0, and $T_0' = Q$ is the actual pulled carrier: the completion rule under scrutiny would replace T_0 by an enlargement absorbing the difference. Two pieces of terminology from the manuscript's completion formalism are needed, and we define them at first use. The *pulled excess* at stage 0 is the difference $X_0 = T_0' \setminus T_0$ between the pulled carrier and the proposed carrier. The *attachment trace* of the excess on the proposed carrier is

$$\tau_{T_0}(X_0) := \overline{X_0}^{T_0'} \cap T_0, \tag{7}$$

where $\overline{X_0}^{T_0'}$ denotes the closure of X_0 taken in the actual pulled carrier T_0': the trace records where the excess attaches to the proposed carrier after closure, not where it meets it as a set.

Proposition 3.2 (Quadratic protected-access collision; manuscript Proposition 6.15)**.** *In the setup* (3)–(6)*, the pulled excess has the single component*

$$X_0 = Q \setminus I$$

with attachment trace

$$\tau_{T_0}(X_0) = \overline{X_0}^{T_0'} \cap T_0 = I.$$

The closure is taken in the actual pulled carrier T_0'; in particular, the trace is not $X_0 \cap T_0$, which is empty by the definition of the excess. The complete one-stage itinerary of the excess is

$$X_0 \longmapsto f(X_0),$$

on the right sheet with critical word 0*. If the upper access $p_0^+(m)$, represented by the vertical segment $m+it$, is protected, then every monotone attachment-complete enlargement absorbing X_0 erases that access and its transported access $F_* p_0^+(m)$.*

Proof. The excess and its trace. A point of Q lies in $I = [1, 2]$ exactly when it is real, so

$$X_0 = Q \setminus I = \left\{ z : \left|z - \tfrac{3}{2}\right| \le \tfrac{1}{2},\ \operatorname{Im} z > 0 \right\}$$

is the relative upper half-disk with its real diameter removed. It is the intersection of the closed disk of radius $\frac{1}{2}$ about m with the open upper half-plane, hence convex, hence connected: the excess has a single component. Its closure in $\mathbb{C}$ is the entire closed half-disk Q, since every point $x \in I$ is the limit of the points $x + it \in X_0$ as $t \downarrow 0$. As $Q = T_0'$, the closure of X_0 in T_0' is all of Q, and therefore

$$\tau_{T_0}(X_0) = \overline{X_0}^{T_0'} \cap T_0 = Q \cap I = I :$$

the excess attaches to the proposed carrier along the entire interval, even though $X_0 \cap T_0 = \varnothing$ by construction. This is the reason the completion

formalism must work with attachment traces rather than intersections: the naive intersection sees nothing, while the closure sees an attachment along all of I.

The itinerary. Since $0 \notin U^-$, we have $F'(z) = 2z \neq 0$ throughout U^-, and the selected branch avoids the critical value $0 \notin U^+$. The itinerary of the excess therefore consists of the single right-sheet, degree-one step $X_0 \mapsto f(X_0)$ with no critical encounter; its critical word is 0. There is no hidden multivalence, folding, or critical intrusion anywhere in the example.

Erasure of the protected access. The access $p_0^+(m)$ is the access to I at the interior point m from the upper side, represented by the curve $t \mapsto m+it$, $t \in (0, \epsilon]$, which lies in $U^- \setminus I$ and lands at m. Suppose a monotone attachment-complete enlargement absorbs X_0 while retaining I. The enlarged carrier is compact and contains $X_0 \cup I$; since it is closed and $\overline{X_0} = Q$, it contains the entire half-disk Q. But Q contains an upper relative half-neighborhood of m: the set

$$\left\{ z : \left|z - \tfrac{3}{2}\right| < \tfrac{1}{2},\ \operatorname{Im} z \geq 0 \right\} \subset Q$$

is a neighborhood of m in the closed upper half-plane. Consequently, any curve in U^- landing at m from the upper side eventually lies in this half-neighborhood, hence in Q, hence in the enlarged carrier. No curve in the complement $U^- \setminus Q$ can therefore land at m from the upper side: the access $p_0^+(m)$ no longer exists as an access from the complement of the enlarged carrier. Monotonicity makes the erasure permanent: further monotone enlargement only adds points to the carrier and so cannot restore a complement access through a set already absorbed.

Transport to stage 1. The branch F of (4) is a conformal homeomorphism of U^- onto U^+ carrying I to $T_1 = [1, 4]$, Q to T_1', X_0 to the stage-1 excess $T_1' \setminus T_1$, and the representing curve $m + it$ to a curve in $U^+ \setminus T_1$ landing at $f(m) = \frac{9}{4}$ from the F-selected side. A homeomorphism of the localization carries complement curves to complement curves, so the same argument applied to the image data erases the transported access $F_* p_0^+(m)$ at stage 1. Conformality thus transports the identical conclusion to stage 1, as claimed. □

3.3. **What the obstructions force.** The two obstructions are complementary and together close off every naive route. Theorem 3.1 shows that the carrier cannot be *selected* canonically: there is no least connected full hull, even for two points with identity dynamics, so any construction that speaks of "the" minimal carrier is speaking of an object that does not exist. Proposition 3.2 shows that the carrier cannot be *completed* blindly either: the very first monotone absorption of a pulled excess can erase a protected prime-end access, and monotonicity then makes the loss irreversible at every later stage of the enlargement, as well as at the conformally transported stage.

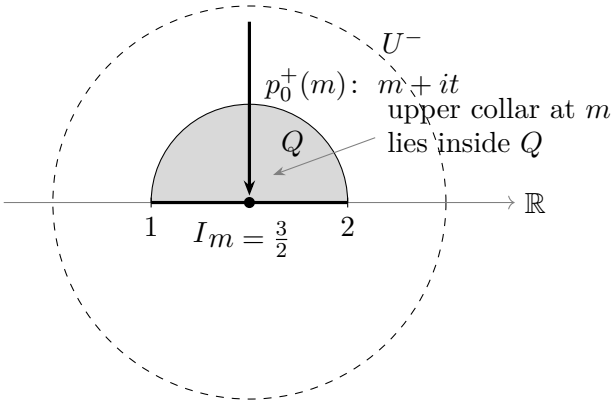


FIGURE 2. The quadratic protected-access collision (Proposition 3.2). The proposed carrier is the interval $I = [1, 2]$ (thick); the pulled carrier is the closed upper half-disk Q (shaded) of radius $\frac{1}{2}$ about $m = \frac{3}{2}$, inside the localization disk U^- (dashed) on which $f(z) = z^2$ restricts to a conformal homeomorphism. The protected upper access $p_0^+(m)$, represented by the vertical segment $m + it$, must traverse Q: absorbing the excess $X_0 = Q \setminus I$ forces the carrier to contain the entire upper collar at m and erases the access.

The second obstruction is the sharper of the two, because of what it does *not* involve. In the manuscript's collision bookkeeping [7], which records sheet mergers, critical intrusions, obstacle collisions, access identifications, and localization escapes in the order $(\chi_{\text{sheet}}, \chi_{\text{crit}}, \chi_{\text{obs}}, \chi_{\text{access}}, \chi_{\text{loc}})$, the collision tuple of the completed enlargement in this example is

$$\big(\varnothing,\ \varnothing,\ \varnothing,\ \{p_0^+(m),\, F_* p_0^+(m)\},\ \varnothing\big)\,:$$

every coordinate vanishes except the access coordinate. The branch is conformal, degree one, and critical-value-free; there is no sheet merger, no obstacle, and no localization escape. Consequently no amount of critical-point bookkeeping — however carefully one tracks critical words, critical values, or degrees — can detect, let alone excuse, the collision. Access protection is invisible to critical data and must be carried as explicit data of its own, with its own survival theorem.

This is exactly what the remainder of the paper provides. The certificate formalism of Section 5 records the protected accesses as prime-end crosscut chains [9] with pairwise disjoint terminal collars, together with free-port arcs carrying singleton attachment traces and a ribbon tree with declared sides; Theorem 5.4 then realizes any such certificate by an actual compact connected full carrier in which every protected chain survives with unchanged cyclic order. The selection ambiguity of Theorem 3.1 is resolved not by minimization but by the certificate itself: the connector system, its access

class, and its routing are supplied as data, and the exact free-component criterion of Section 6 determines precisely when such data can be supplied at all.

4. Comparison with standard continuum-hull constructions

The certificate formalism developed from Section 5 onward is offered as a replacement for a family of standard constructions, and it is fair to ask what, exactly, is being replaced and why. This section compares the framework with the five constructions a practitioner would reach for first: least or minimal connected full hulls; monotone absorption of pulled excess; regulated and Hubbard trees in uniformized or locally connected coordinates; Moore decompositions and monotone quotients; and the lamination-lift candidate of the source manuscript. The two obstructions of Section 3 already dispose of the first two; the remaining three fail for structural reasons that are worth stating precisely, because they explain several design decisions of the certificate formalism that would otherwise look pedantic. Throughout, the comparison is with respect to the task fixed in the introduction: join finitely many pairwise disjoint full atoms inside a Jordan localization, avoid a frozen forbidden set, and *preserve designated prime-end accesses together with their cyclic order*. Constructions that solve a different problem well are not thereby criticized; the claim is only that none of them solves this one, and that the last of them — the lamination lift — is an open candidate, not a construction.

4.1. **Least and minimal connected full hulls.** The most tempting rule is extremal: "take the least connected full hull of the required package." Theorem 3.1 (manuscript Theorem 6.12) shows the rule names no object, already for the two-point package $S = \{-1, 1\}$ with identity dynamics and no obstacles: the upper and lower semicircles are carriers meeting exactly in S, so a least carrier would be contained in their disconnected intersection. The failure is specific to *leastness*: minimal carriers exist in abundance — every arc from -1 to 1 is one — and form a large antichain. An extremal selection principle therefore cannot even break the tie between the two semicircles, and the tie is not innocent: the two choices pass on opposite sides of the origin, and in any configuration with obstacles or protected collars that choice of side is visible to every later routing step. The source manuscript sharpens the point dynamically [7, Proposition 6.11]: a package, full connected carriers, critical-value-free degree-one selected-carrier dynamics, and identical coarse orbit labels are realized simultaneously by an interval and by a triod, and attachment-freeness holds in only one of the two models, so no data of the listed kinds can select between them.

The published connected-hull descriptions in the renormalization literature are not at fault here, and the comparison should not be read as a criticism of them. Dudko's roadmap describes virtual Julia sets as connected hulls of Cantor families of small Julia sets carrying a finite part of the critical

orbit [2]; that description specifies *which* continua a carrier must contain, and it does not purport to specify *how* they are joined. Theorem 3.1 shows that the routing is genuinely additional data — in the manuscript's words, a regulated connector system, together with its access or isotopy class and a forward-coherence proof, is additional data [7] — and the certificate of Definition 5.1 is exactly a format for that data, with Theorem 6.3 deciding when the format can be filled in at all.

4.2. **Monotone absorption of pulled excess.** The second rule is completion: accept any proposed carrier and absorb, monotonically, whatever excess the dynamics pulls back onto it, until the process stabilizes. Unlike the least hull, this rule is well defined: the manuscript proves that the finite inflationary completion operator stabilizes at a least fixed point, unique and conjugation-equivariant [7, Theorem 6.14]. The defect is not existence but what completion destroys. Proposition 3.2 (manuscript Proposition 6.15) exhibits a conformal, critical-value-free, degree-one branch of $z \mapsto z^2$ in which the very first absorption of pulled excess — the relative upper half-disk attaching to the interval along its entire length — erases a protected prime-end access and its conformally transported copy. Monotonicity, the rule's defining virtue, is precisely what makes the loss permanent: enlargement can only add points, and no addition restores a complement access through a set already absorbed. Moreover, every coordinate of the manuscript's collision bookkeeping except the access coordinate vanishes in this example, so no amount of critical-value, degree, or sheet bookkeeping detects the erasure (Section 3.3).

The certificate formalism keeps monotone completion but reverses the order of commitments. The protected collars are frozen into the inventory Q *before* any absorption is contemplated; a proposed absorption that would enter a collar tail is not a step of a completion but a named violation, and the persistence clause of Theorem 6.3, together with the audit of Theorem 8.3, converts each such violation into a certificate — a merger, a port loss, a separation — that survives every further monotone enlargement. Absorption remains available for everything unprotected; what changes is that "protected" is a constraint checked by the plane, not an intention held by the author.

4.3. **Regulated trees in uniformized coordinates and Hubbard trees.** When the continuum to be traversed is locally connected, canonical connectors exist and the present framework is not needed. In a locally connected filled Julia set every pair of points is joined by a unique regulated arc, and the union of the regulated arcs through a finite invariant set is the classical Hubbard tree of the Douady–Hubbard theory [1]; equivalently, in uniformized coordinates, one takes hyperbolic geodesics in the complement or in the interior components and closes up, and Carathéodory's theorem — continuity of the Riemann map up to the boundary, which *is* local connectivity of the boundary — guarantees that the geodesics land and that

the resulting tree is compact, canonical, and forward invariant. The decoration constructions of Kahn–Lyubich [5] organize configurations of small Julia sets and their limbs under a priori bounds by explicit combinatorial data; there too the combinatorial layer is carried by objects — puzzle pieces, rays, equipotentials — whose availability is part of the hypotheses.

Every step of this construction is unavailable for the atoms of the present paper. The atoms are arbitrary full continua; no Riemann map is asserted to extend, no external or internal ray is asserted to land, and regulated arcs need not exist. This is not a hypothetical worry but the operating regime of the intended applications, in which local connectivity of the relevant continua is exactly what is not known. The closest published object is instructive as a boundary marker. Kahn–Kapiamba–Lyubich construct a native Hubbard tree inside the full dynamical carrier of a primitive renormalization, and in the non-locally-connected case they lift it from the locally connected rational-lamination model [4, §§2.5–2.6]. As the source manuscript's own source boundary records [7], that construction is a positive control in its published primitive category — where the lamination model and the lifting semiconjugacy are theorems — and not a theorem for virtual-Molecule packages. The framework of this paper is calibrated to the complementary regime: Theorem 5.4(6) uses no local connectivity of any atom, the tree of the certificate is produced by the exact criterion of Theorem 6.3 from planar data alone, and no canonicity of the spanning tree is claimed anywhere — the tree is data, chosen in a free component whose existence is the actual content. What is lost relative to the Hubbard-tree regime is canonicity; what is gained is that the construction exists at all without the locally connected model.

4.4. **Moore decompositions and monotone quotients.** A third classical route collapses instead of routing. By Moore's theorem, the quotient of the sphere by an upper semicontinuous decomposition into nonseparating continua is again a sphere [6]. One is therefore tempted to collapse every atom (or every completed packet) to a point, join the finitely many image points by arcs in the quotient sphere — where the joining problem is trivial — and pull the picture back. For the connection problem *without* protected data this is a perfectly serviceable device, and it is exactly the mechanism by which locally connected models are related to their originals: a monotone quotient map realizes the model, and the fibers record what was collapsed.

For the access-preserving problem the device fails at the first step, because the quotient map destroys the very data being protected. All prime ends of an atom have a single image point: a protected crosscut chain, its selected sides, and the cyclic order of several protected accesses at one atom have no image in the quotient at all, so no construction performed downstairs can certify their survival, and distinct protected accesses cannot even be distinguished there. The pullback of an arc through the quotient map is a union of fibers, not a connector: it carries no singleton attachment traces,

and nothing prevents a fiber from entering a protected collar — the pullback must take whatever the fiber contains. In the vocabulary of Section 5, a monotone quotient forgets precisely the collar and side data of Definition 5.1, and what it cannot express it cannot preserve. The certificate formalism therefore works in the original plane throughout: atoms are never collapsed, accesses are carried as crosscut chains with selected sides, and the realization theorem attaches the connector at declared free ports with singleton traces (Theorem 5.4(2)) — the attachment discipline that a quotient construction cannot see, let alone enforce.

4.5. **The lamination-lift candidate.** The most serious competitor is the manuscript's own candidate, recorded in its Section 10.3 [7], and the comparison must be exact: the candidate is presented there, and here, as a candidate only. Suppose, following the non-locally-connected construction of Kahn–Kapiamba–Lyubich [4], that an ambient carrier admits a monotone semiconjugacy

$$\pi\colon K \longrightarrow K_{\mathrm{com}}$$

onto a locally connected combinatorial model, together with a forward-coherent regulated tree $T_{\mathrm{com}} \subset K_{\mathrm{com}}$ containing every root shadow. The canonical candidate connector is

$$H_{\mathrm{lam}} \;=\; \mathrm{Fill}\big(\pi^{-1}(T_{\mathrm{com}})\big), \tag{8}$$

equation (9.3) of [7]. Part of what is wanted is genuinely true, by soft arguments: a monotone preimage of a compact connected set is compact and connected, so (8) is a full continuum containing the required shadows; the semiconjugacy provides forward coherence before fullification; and fullification preserves genuine polynomial forward invariance.

The missing assertion is not connectedness. Measured against the certificate clauses, two failure channels are named in the manuscript, and both remain unresolved. First, a projection fiber $\pi^{-1}(x)$ over a point of the tree, or a component added by fullification, may enter a protected collar otherwise than at a declared free singleton port; this is the erasure mechanism of Proposition 3.2 wearing different clothes, and it is instructive that the manuscript rejects the upper-half-disk collision at exactly this clause, the excess occupying the protected upper collar with an interval attachment trace [7]. Second, a disconnected stage package may still split across iterated pullback components, so that containing every root shadow does not yet realize the complete package routing. The manuscript collects the clauses a proof would have to establish — authorized localization and complete proof-use inventory, collar avoidance except at declared free singleton ports, realization of every package component and route, obstacle handling inside the localization, complete access marking, and an incidence bound or compression — as its Open Problem 10.6 (Virtual Protected Routing Certificate). That problem is open, and nothing in this paper resolves any part of it; consistently

with the manuscript's Warning 10.59, the candidate (8) is discussed here only to locate it on the map.

What this paper contributes to the candidate is the deciding instrument, not a verdict. For any frozen inventory that a future argument supplies, Theorem 6.3 decides certificate existence exactly; Theorems 7.4 and 8.3 and Proposition 8.5 make the decision checkable packet by packet; and the two failure channels above fall under the typed, monotone-persistent defect certificates of Section 8 — a fiber in a collar is a port-loss or merger defect, a split package is a separation in the sense of the persistence clause of Theorem 6.3. Whether the candidate passes the test is precisely what is open.

4.6. **Summary.** Table 1 compresses the comparison. The pattern is uniform: each standard construction either fails to exist (the least hull), destroys the protected data (absorption, Moore quotients), requires the hypothesis whose absence is the whole difficulty (regulated and Hubbard trees), or is an explicitly open candidate (the lamination lift). The certificate framework renounces canonicity — the connector is data, not an invariant — and in exchange obtains, for arbitrary full atoms and with no local connectivity anywhere: realization with access survival and cyclic-order preservation (Theorem 5.4); an exact existence criterion on frozen data (Theorem 6.3); checkable sufficient tests (Theorem 7.4, Proposition 8.5); order-independent completion with a three-clause audit (Theorem 8.3); and failure certificates that persist under monotone enlargement. The effect is to convert "choose a hull," an instruction that Theorem 3.1 shows to be ill posed and Proposition 3.2 shows to be destructive, into a checkable planar test with a stable answer in both directions. This is the precise sense in which the contribution of the paper is proof technology: no new theorem of classical planar topology is claimed, and every statement about the open constructions touched on above — virtual carriers, lamination lifts, and their dynamical certificates — remains exactly as open here as in the source manuscript.

5. Protected ribbon-routing certificates and realization

The two obstructions of the preceding section calibrate what a realization theorem can promise. Theorem 3.1 shows that even the two-point package with identity dynamics and no obstacles admits no least compact connected full carrier, so no canonical minimal choice is available and a connector must be *supplied*, not selected by an extremal principle. Proposition 3.2 shows that a monotone, attachment-complete enlargement can permanently erase a protected prime-end access even along a conformal, critical-value-free branch: once a set occupying a protected side has been absorbed, no further monotone enlargement restores an access through it. Together these force the shape of the present section. The connector data must be recorded *in advance*, as an explicit finite certificate that names every protected access, reserves a collar around each one, and routes the connecting tree through the

Construction	Object exists	Protected accesses	Failure record
least connected full hull	no (Thm. 3.1)	—	—
monotone absorption	least fixed point [7]	erasable (Prop. 3.2)	post hoc only
regulated/Hubbard tree	needs l.c. model [1, 4]	native category only	—
Moore monotone quotient	always [6]	collapsed	—
lamination lift (8)	candidate [7]	open	open
protected certificate	iff-criterion (Thm. 6.3)	preserved (Thm. 5.4)	persistent (Thm. 8.3)

TABLE 1. The five standard constructions against the certificate framework, measured on the access-preserving connection task. "Object exists" asks whether the construction produces a well-defined carrier for arbitrary pairwise disjoint full atoms in a Jordan localization; "protected accesses" asks whether designated prime-end data survive with their cyclic order; "failure record" asks whether a negative outcome is witnessed by a certificate that persists under monotone enlargement. The lamination-lift row records candidacy, not failure: its clauses are the manuscript's open problem, and no verdict is implied in either direction.

reserved-free region. The theorem then owed to the reader is purely topological: every such certificate is realized by a compact connected full carrier, the protected accesses survive as prime ends of the enlarged complement, distinct accesses stay distinct, their cyclic order is unchanged, and the sides created by the connector itself appear in the boundary order dictated by the ribbon structure of the tree. This is the content of Theorem 5.4 below. The section closes with the naturality of the construction under univalent changes of the Jordan localization (Proposition 5.6). The question of when a certificate *exists* is deferred to Section 6, where it receives an exact answer.

Setup. Throughout this section $U \subset \mathbb{C}$ is a Jordan disk, that is, a bounded simply connected domain whose boundary is a Jordan curve. A *continuum* is a nonempty compact connected subset of $\mathbb{C}$; a continuum K is *full* if $\mathbb{C} \setminus K$ is connected, equivalently if $\widehat{\mathbb{C}} \setminus K$ is a simply connected domain containing ∞. Following the source manuscript [7], the full continua to be joined are called *atoms*. Let

$$A_1, \dots, A_s \Subset U$$

be pairwise disjoint atoms, and let

$$Z \Subset U \setminus \bigcup_i A_i$$

be a compact set, the frozen *forbidden obstacle and critical-value set*. Nothing in this section uses any dynamical structure on the A_i or on Z; the data are purely planar.

Since each A_i is full, $\Omega_i = \widehat{\mathbb{C}} \setminus A_i$ is a simply connected domain, and the prime-end formalism recalled in Section 2 applies to it [9]. We use it in the following normalized form. A *crosscut* of Ω_i is an open Jordan arc $\gamma \subset \Omega_i$ whose closure is a closed arc with two distinct endpoints on $\partial\Omega_i = \partial A_i$; the crosscut divides Ω_i into exactly two subdomains. A *prime-end chain* (null chain) is a sequence $(\gamma_n)_{n\geq 1}$ of pairwise disjoint crosscuts whose spherical diameters tend to zero, together with a coherent choice of complementary domains D_n, the *selected sides*: D_n is one of the two components of $\Omega_i \setminus \gamma_n$, and $D_{n+1} \subset D_n$ for every n. Two chains are *equivalent* if each eventually divides the other, that is, every selected domain of either chain contains all sufficiently deep selected domains of the other; an equivalence class is a *prime end* of Ω_i. The record of a chain therefore contains strictly more than the crosscuts themselves: it contains, for every n, the declaration of which side of γ_n is selected. It is exactly this declaration that the certificates below protect.

For every protected access to A_i, indexed by a finite set of labels a, choose a prime-end crosscut chain

$$\Gamma_{i,a} = (\gamma_{i,a,n},\, D_{i,a,n})_{n\geq 1}$$

of $\widehat{\mathbb{C}} \setminus A_i$. The chains are truncated, that is, an initial segment of each is discarded and the indices shifted, so that the terminal *collar tails*

$$D_{i,a,1} \setminus A_i$$

are pairwise disjoint, lie in U, and miss Z and the other atoms. Such a truncation is possible because the finitely many chains define distinct accesses with shrinking selected domains, and because Z and the atoms are compact and disjoint from the impressions being approached. The set $D_{i,a,1}$ is called the (terminal, or first) *protected collar* of the access (i, a); every deeper selected domain $D_{i,a,n}$ lies inside it. Figure 3 shows a single atom, one protected chain, its nested selected domains, and its collar.

Definition 5.1 (Protected ribbon-routing certificate; manuscript Definition 10.1)**.** A *protected ribbon-routing certificate* consists of the preceding data—the Jordan disk U, the atoms $A_1, \dots, A_s$, the forbidden set Z, and the truncated protected chains $\Gamma_{i,a}$—together with pairwise disjoint *free-port arcs* $\alpha_i \colon [0,1] \to U$ satisfying

$$\alpha_i \cap A_i = \{x_i\}, \qquad \alpha_i((0,1]) \cap \Big(Z \cup \bigcup_j A_j \cup \bigcup_{j,a} D_{j,a,1}\Big) = \varnothing,$$

where $x_i = \alpha_i(0)$, and an embedded finite tree $G \subset U$ joining the outer port endpoints $\alpha_1(1), \dots, \alpha_s(1)$ while missing the same protected and forbidden sets $Z \cup \bigcup_j A_j \cup \bigcup_{j,a} D_{j,a,1}$. The planar cyclic order of half-edges at every vertex of G is retained as part of the certificate (its *ribbon structure*). Every

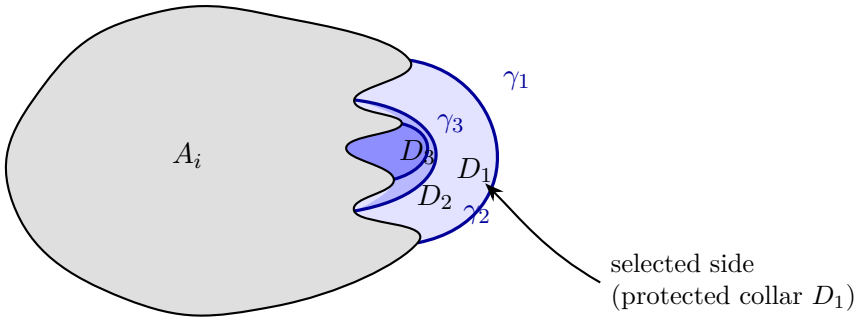


FIGURE 3. A protected prime-end crosscut chain at a single atom A_i. The crosscuts $\gamma_1, \gamma_2, \gamma_3$ are pairwise disjoint arcs of $\widehat{\mathbb{C}} \setminus A_i$ with endpoints on ∂A_i and shrinking diameters; the shaded regions are the nested selected sides $D_1 \supset D_2 \supset D_3$, drawn with increasing opacity toward the prime end. The outermost selected domain D_1 is the protected collar of the access: after truncation the collar tail $D_1 \setminus A_i$ is disjoint from every other collar tail, lies in U, and misses the obstacle set Z and the other atoms. The boundary of the atom is drawn deliberately oscillating: no local connectivity of ∂A_i is assumed anywhere in this section, and the access is carried entirely by the chain, not by any landing curve.

access read by a proof must be a protected chain $\Gamma_{i,a}$ or a declared side of this ribbon tree.

Given a certificate R, define its *carrier*

$$H_R \;=\; G \cup \bigcup_i \alpha_i \cup \bigcup_i A_i. \tag{9}$$

Four reading notes. First, the port arc α_i touches its atom at the single *free port* x_i, a point deliberately not lying in any protected collar: the second displayed condition includes a-indexed collars of the *same* atom i, so the port enters the atom away from every protected access. Second, "joining the outer port endpoints" is read as part of the embeddedness of the certificate: the tree G meets each port arc α_i in exactly the point $\alpha_i(1)$, so that $G \cup \bigcup_i \alpha_i$ is again an embedded finite tree, with the port arcs attached as pendant edges. This is the configuration produced by the construction of Section 6, where the routing tree is built in the complement of the slit port arcs, and it is used in the proof of the realization theorem below. Third, the tree G is a combinatorial object with a planar enhancement: recording the cyclic order of half-edges at each vertex fixes, up to isotopy, the two sides of every edge, and it is these *declared ribbon sides* that the realization theorem converts into new accesses with a prescribed boundary order. Fourth, the final sentence of the definition is a discipline on all later arguments, not a property of

the data: any subsequent proof that consults an access of the carrier must consult either a protected chain or a declared ribbon side, never an incidental access created by the geometry of a particular embedding. Figure 4 shows a certificate with three atoms.

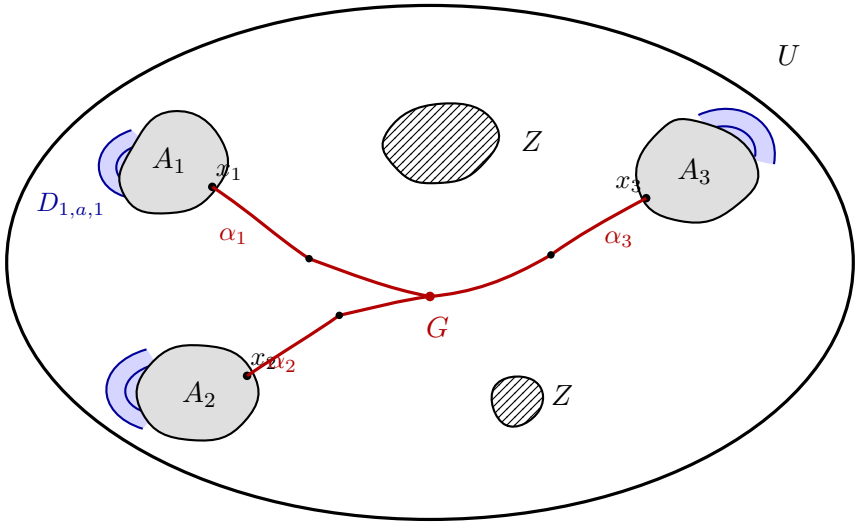


FIGURE 4. A protected ribbon-routing certificate with three atoms inside the Jordan disk U. Each atom A_i carries a protected collar (shaded, with two nested crosscuts indicated) and a free-port arc α_i (heavy) landing at the single free point x_i, chosen away from every protected collar. The embedded finite tree G (heavy) joins the outer port endpoints; the interiors of the α_i and all of G avoid the hatched obstacle set Z, all atoms, and all first collars $D_{j,a,1}$. The planar cyclic order of the half-edges at the vertex of G is part of the certificate. The carrier is $H_R = G \cup \bigcup_i \alpha_i \cup \bigcup_i A_i$.

Fullness of one-point tree attachments. The first step of the realization is to verify that the carrier (9) is a full continuum. The union $T = G \cup \bigcup_i \alpha_i$ is itself a finite plane tree: G is a finite tree and, by the attachment convention recorded after Definition 5.1, each α_i meets G in exactly the point $\alpha_i(1)$, so adjoining the arcs one at a time adds pendant edges and creates no cycle. By the port conditions of Definition 5.1, T meets each atom A_i in exactly the point x_i and otherwise misses every atom. The relevant general fact is the following lemma, for which we record two proofs: a direct induction on Janiszewski's separation theorem, and a cohomological argument by Alexander duality and Mayer–Vietoris. The first is elementary and self-contained; the second is the pattern that recurs, for pairwise disjoint packets, in the packet lemmas of Section 7.

Lemma 5.2 (One-point tree attachments are full; manuscript Lemma 10.2)**.** *If a finite plane tree meets each of finitely many pairwise disjoint full continua in exactly one point and otherwise misses them, then their union is a full continuum.*

First proof (Janiszewski induction). Let T be the finite plane tree and $A_1, \dots, A_s$ the pairwise disjoint full continua, with $T \cap A_i = \{p_i\}$. All sets are regarded as compact subsets of the sphere $\widehat{\mathbb{C}}$, and *nonseparating* means that the complement in $\widehat{\mathbb{C}}$ is connected; for a compact subset of $\mathbb{C}$ this is equivalent to fullness, since the complement in $\widehat{\mathbb{C}}$ is open and connected exactly when the bounded complementary components in $\mathbb{C}$ are absent.

A finite plane tree is nonseparating. Indeed, T is compact, connected, and simply connected, and one may argue by induction on the number of edges: a single closed arc does not separate the sphere, and if $T = T' \cup e$ with e a closed pendant edge meeting T' in one endpoint, then T' and e are nonseparating compacta with connected (one-point) intersection, so Janiszewski's theorem, quoted below, makes T nonseparating.

We now attach the full continua one at a time. Put

$$B_0 = T, \qquad B_k = T \cup A_1 \cup \cdots \cup A_k \quad (1 \le k \le s),$$

so that B_s is the union in question. Each B_k is compact, and it is connected because T is connected and each attached A_j meets T. We claim every B_k is nonseparating. The case $k = 0$ was just proved. Assume B_{k-1} is nonseparating and consider $B_k = B_{k-1} \cup A_k$. The intersection is

$$B_{k-1} \cap A_k = (T \cap A_k) \cup \bigcup_{j<k} (A_j \cap A_k) = \{p_k\},$$

a singleton, hence connected: the atoms are pairwise disjoint and T meets A_k only at p_k. Both B_{k-1} and A_k are compact and nonseparating. Janiszewski's separation theorem [8, 10] states that if two closed subsets of the sphere have connected intersection and neither separates a given pair of points, then their union does not separate that pair. Applying this to every pair of points of $\widehat{\mathbb{C}} \setminus B_k$ shows that B_k separates no pair of points of its complement, that is, B_k is nonseparating. By induction, B_s is a compact connected set with connected complement in $\widehat{\mathbb{C}}$, hence a full continuum. □

Second proof (Alexander duality and Mayer–Vietoris). Work again in $\widehat{\mathbb{C}} = S^2$ and retain the notation B_k of the first proof. For a compact set $X \subset S^2$, Alexander duality in its Čech form [3] gives an isomorphism

$$\widetilde{H}_0\big(S^2 \setminus X; \mathbb{Z}\big) \;\cong\; \check{H}^1(X; \mathbb{Z}),$$

so X is nonseparating precisely when its first reduced Čech cohomology vanishes. (Čech cohomology is essential here: the atoms are arbitrary continua, in general not locally connected, and no CW-approximation of them is available; Čech theory has the required continuity under inverse limits of

neighborhoods and satisfies the Mayer–Vietoris sequence for compact pairs in a normal ambient space.)

The tree T is contractible, so $\check{H}^1(T;\mathbb{Z}) = 0$; and $\check{H}^1(A_i;\mathbb{Z}) = 0$ for every i because each A_i is full, by the same duality read in the other direction. Now induct on k. Suppose $\check{H}^1(B_{k-1};\mathbb{Z}) = 0$ and write $B_k = B_{k-1} \cup A_k$ with $B_{k-1} \cap A_k = \{p_k\}$. The Mayer–Vietoris sequence in reduced Čech cohomology for the compact pair (B_{k-1}, A_k) reads

$$\widetilde{\check{H}}^0(\{p_k\};\mathbb{Z}) \longrightarrow \check{H}^1(B_k;\mathbb{Z}) \longrightarrow \check{H}^1(B_{k-1};\mathbb{Z}) \oplus \check{H}^1(A_k;\mathbb{Z}).$$

The left group vanishes because the one-point intersection is connected, and the right group vanishes by the inductive hypothesis and fullness of A_k. Exactness forces $\check{H}^1(B_k;\mathbb{Z}) = 0$. By induction $\check{H}^1(B_s;\mathbb{Z}) = 0$, and Alexander duality converts this back into connectedness of $S^2 \setminus B_s$. Since B_s is compact and connected, it is a full continuum. □

The two proofs establish literally the same induction step, and the translation between them is the dictionary used throughout this paper: "nonseparating" equals "vanishing first Čech cohomology," "connected intersection" equals "vanishing zeroth reduced cohomology of the overlap." The cohomological form is the one that survives when the finitely many one-point overlaps are replaced by the pairwise *disjoint* packets of Section 7, where the nerve argument of Lemma 7.3 takes over.

Survival of protected chains. The second step is the heart of the realization: enlarging an atom to a full continuum that stays out of a protected collar does not damage the protected access. The chain survives verbatim—the same crosscuts, the same selected sides—as a prime-end chain of the complement of the enlarged continuum, and finitely many protected accesses with disjoint collars stay distinct and keep their cyclic order.

Lemma 5.3 (Protected crosscut survival and order; manuscript Lemma 10.3)**.** *Let $H \supset A_i$ be a full continuum and let $(\gamma_n, D_n)_{n\ge1}$ be one of the protected chains above. If*

$$(H \setminus A_i) \cap (D_1 \setminus A_i) = \varnothing,$$

then a tail of the same crosscuts is a prime-end chain in $\widehat{\mathbb{C}} \setminus H$. Its endpoints remain boundary points and its selected sides contain the old D_n. A finite family of such chains with pairwise disjoint terminal selected domains induces distinct prime ends of $\widehat{\mathbb{C}} \setminus H$ with unchanged cyclic order.

Proof. Since H is a full continuum, $\widehat{\mathbb{C}} \setminus H$ is a simply connected domain and has its own prime-end theory. We verify that the tail $(\gamma_n)_{n\ge2}$, with the same selected sides, is a null chain for it.

The selected domains avoid H entirely. For every n the selected domain D_n lies in $\widehat{\mathbb{C}} \setminus A_i$, so $D_n \setminus A_i = D_n$, and it is contained in D_1 by nestedness.

Hence

$$H \cap D_n = (A_i \cap D_n) \cup \big((H \setminus A_i) \cap D_n\big) \subset \varnothing \cup \big((H \setminus A_i) \cap (D_1 \setminus A_i)\big) = \varnothing,$$

so $D_n \subset \widehat{\mathbb{C}} \setminus H$ for every n.

Each deep crosscut remains a crosscut. Fix $n \geq 2$. The open arc γ_n lies in $D_{n-1} \subset D_1$: indeed γ_n lies in the closure of $D_n \subset D_{n-1}$ and in $\widehat{\mathbb{C}} \setminus A_i$, and the part of ∂D_{n-1} inside $\widehat{\mathbb{C}} \setminus A_i$ is γ_{n-1}, which γ_n misses. So by the preceding paragraph the arc γ_n misses H; its two distinct endpoints lie on ∂A_i, hence in $A_i \subset H$. Every neighborhood of either endpoint meets the crosscut interior, which consists of points of $\widehat{\mathbb{C}} \setminus H$, so both endpoints remain boundary points of the enlarged complement $\widehat{\mathbb{C}} \setminus H$, and they are still distinct. Thus γ_n is again an open Jordan arc in $\widehat{\mathbb{C}} \setminus H$ whose closure is an arc with two distinct endpoints on $\partial(\widehat{\mathbb{C}} \setminus H)$, that is, a crosscut of $\widehat{\mathbb{C}} \setminus H$.

The selected sides are unchanged. For $n \geq 2$ let D'_n denote the component of $(\widehat{\mathbb{C}} \setminus H) \setminus \gamma_n$ containing D_n; this is well defined because D_n is connected and disjoint from $H \cup \gamma_n$. We claim $D'_n = D_n$, so that the containment $D'_n \supset D_n$ asserted in the statement holds with equality. The set D_n is open, and it is relatively closed in $(\widehat{\mathbb{C}} \setminus H) \setminus \gamma_n$: as a component of the open set $(\widehat{\mathbb{C}} \setminus A_i) \setminus \gamma_n$ it satisfies

$$\partial D_n \subset \partial A_i \cup \overline{\gamma_n} \subset A_i \cup \gamma_n \subset H \cup \gamma_n,$$

since the closure of the open arc γ_n adds only its two endpoints, which lie in A_i; hence no boundary point of D_n belongs to $(\widehat{\mathbb{C}} \setminus H) \setminus \gamma_n$, and every point of that set lying in $\overline{D_n}$ already lies in D_n. A nonempty connected subset of an open set that is both open and relatively closed is a component of it, so $D'_n = D_n$. Nestedness of the new sides is now inherited from the original chain: $D'_{n+1} = D_{n+1} \subset D_n = D'_n$ for every $n \geq 2$, and $\gamma_{n+1} \subset D_n = D'_n$ lies in the selected side of γ_n, as a null chain requires.

The diameters still tend to zero. The crosscuts are unchanged as point sets, so their spherical diameters are the same numbers as before and still tend to zero. Consequently $(\gamma_n, D'_n)_{n \geq 2}$ is a prime-end chain of $\widehat{\mathbb{C}} \setminus H$, which proves the first two assertions.

Distinctness. Now consider a finite family of protected chains satisfying the hypothesis, with pairwise disjoint terminal selected domains. Recall that two null chains of the same domain define the same prime end if and only if they eventually divide one another: every selected domain of either chain contains all sufficiently deep selected domains of the other. Suppose two chains of the family, with terminal domains $D_1^{(1)}$ and $D_1^{(2)}$, induced the same prime end of $\widehat{\mathbb{C}} \setminus H$. Then some deep selected domain of the first chain—by the preceding step, the same set $D_m^{(1)}$ as before the enlargement—would be contained in a selected domain of the second chain, hence in $D_1^{(2)}$; but that deep domain is nonempty and contained in $D_1^{(1)}$, contradicting $D_1^{(1)} \cap D_1^{(2)} = \varnothing$. Disjoint terminal domains therefore prevent mutual division, and the

induced prime ends are pairwise distinct. In particular the survival map, from protected chains to prime ends of $\widehat{\mathbb{C}} \setminus H$, is injective on such a family.

Cyclic order. The prime ends of the simply connected domain $\widehat{\mathbb{C}} \setminus H$ carry a natural cyclic order, the one transported from the unit circle by a Riemann map [9]. For a finite family of prime ends represented by pairwise disjoint crosscuts whose selected sides are also pairwise disjoint, this order can be read off inside the domain, as follows. Choose one deep crosscut from each chain, deep enough that the chosen crosscuts lie in the pairwise disjoint terminal collars; then the chosen crosscuts are pairwise disjoint and their selected sides are pairwise disjoint. Fix three chains of the family, with chosen crosscuts c_1, c_2, c_3 and selected sides S_1, S_2, S_3. Call an embedded triod Y—three arcs with a common branch point, disjoint except at that point—*computing* if Y lies in the domain and meets the configuration $\bigcup_{k=1}^{3}(c_k \cup S_k)$ in exactly its three leg tips, one tip on each c_k. Transporting by a Riemann map turns the configuration into three pairwise disjoint crosscuts of the unit disk cutting off pairwise disjoint caps, and a computing triod into an embedded triod tipping on the three crosscuts and otherwise avoiding crosscuts and caps. In the disk, computing triods exist, and the cyclic order of the three legs at the branch point of any computing triod agrees with the cyclic order of the three caps on the circle, that is, with the cyclic order of the three prime ends: the complement of the three open caps in the closed disk is a closed Jordan region whose boundary traverses the three crosscuts in the cyclic order of the caps, and an embedded triod in a Jordan region whose legs end on three disjoint boundary arcs realizes at its branch point exactly the cyclic order of those arcs. Hence, back in the original domain, computing triods exist for every triple of chains, and each one realizes the cyclic orientation of its prime-end triple at its branch point.

Now compare the two domains. By the preceding steps, the chosen crosscuts and their selected sides are *literally the same sets* for $\widehat{\mathbb{C}} \setminus A_i$ and for $\widehat{\mathbb{C}} \setminus H$. Fix a triple of chains and choose a computing triod Y in the smaller domain $\widehat{\mathbb{C}} \setminus H$. Since $\widehat{\mathbb{C}} \setminus H \subset \widehat{\mathbb{C}} \setminus A_i$ and the configuration is unchanged, Y is simultaneously a computing triod in $\widehat{\mathbb{C}} \setminus A_i$. Reading the branch point of this one triod in both domains shows that the prime-end triple has the same cyclic orientation in $\widehat{\mathbb{C}} \setminus H$ as in $\widehat{\mathbb{C}} \setminus A_i$. Every ordered triple therefore retains its orientation, and the cyclic order of the whole family is unchanged. □

The realization theorem. Assembling the two lemmas yields the main result of this section.

Theorem 5.4 (Protected regulated connector realization; manuscript Theorem 10.4)**.** *Every protected ribbon-routing certificate R has an actual carrier H_R, given by* (9)*, with the following properties:*

(1) *$H_R \Subset U$ is compact, connected, and full;*
(2) *it contains every A_i, misses Z, and has singleton attachment trace $\{x_i\}$ at each atom;*

(3) *every protected crosscut chain remains a prime-end chain in* $\widehat{\mathbb{C}} \setminus H_R$*;*
(4) *the survival map is injective and preserves cyclic order;*
(5) *the connector-created sides have the boundary order of the retained ribbon tree; and*
(6) *no local connectivity of any* A_i *is required.*

Proof. *Clause* (1). As observed before Lemma 5.2, the set $T = G \cup \bigcup_i \alpha_i$ is a finite plane tree meeting each atom A_i in exactly the point x_i and otherwise missing every atom. Lemma 5.2 applies to T and $A_1, \dots, A_s$ and shows that $H_R = T \cup \bigcup_i A_i$ is a full continuum. Each constituent is compactly contained in U, and there are finitely many, so $H_R \Subset U$.

Clause (2). Containment of the atoms is part of (9). The certificate conditions of Definition 5.1 state that the arcs $\alpha_i((0,1])$ and the tree G miss Z, and Z misses every atom by hypothesis, so $H_R \cap Z = \varnothing$. The attachment trace at A_i is the intersection of A_i with the connector: $T \cap A_i = \{x_i\}$ because $\alpha_i \cap A_i = \{x_i\}$, the other port arcs and G miss A_i, and the atoms are pairwise disjoint.

Clause (3). Fix a protected chain $\Gamma_{i,a}$. We verify the hypothesis of Lemma 5.3 for $H = H_R$ and the atom A_i. The difference $H_R \setminus A_i$ is contained in

$$G \cup \bigcup_j \alpha_j((0,1]) \cup \bigcup_{j \neq i} A_j.$$

The tree G and every half-open port arc $\alpha_j((0,1])$ miss *all* first collars $D_{j',a',1}$, in particular $D_{i,a,1}$, by Definition 5.1; and each atom A_j with $j \neq i$ misses the collar tail $D_{i,a,1} \setminus A_i$ by the truncation normalization. Hence $(H_R \setminus A_i) \cap (D_{i,a,1} \setminus A_i) = \varnothing$: the connector misses the terminal collar tail, and therefore misses every later closed collar tail $D_{i,a,n} \subset D_{i,a,1}$ as well. Lemma 5.3 now shows that a tail of $\Gamma_{i,a}$ is a prime-end chain in $\widehat{\mathbb{C}} \setminus H_R$.

Clause (4). The truncation normalization makes the terminal selected domains of distinct protected chains pairwise disjoint. For each atom A_i, the final part of Lemma 5.3 with $H = H_R$ gives distinctness and preservation of cyclic order for the subfamily of chains protecting A_i. The distinctness argument of the lemma uses only disjointness of the terminal selected domains and survival of both chains in $\widehat{\mathbb{C}} \setminus H_R$, so it applies verbatim to a pair of chains protecting different atoms as well. The survival map is therefore injective on the full family over all (i,a), and it preserves the cyclic order of every protected family with a common source domain.

Clause (5). Let N be a small regular neighborhood of the finite connector tree T in the plane, chosen so thin that N misses Z and meets each atom only near its port point. A regular neighborhood of a finite plane tree is a closed disk, and its boundary circle traverses every edge of T on two sides: following ∂N, one passes along each edge once in each direction, and at a vertex of valence d the boundary visits the d half-edges in exactly their planar cyclic order. The resulting cyclic *boundary word*, in which every edge appears twice with opposite orientations, is precisely the boundary word of

the ribbon structure retained in the certificate. The accesses to H_R created by the connector—the two sides of every edge of G and of every port arc—are read along this boundary circle, so their boundary order is the boundary order of the retained ribbon tree, as claimed.

Clause (6). The argument uses crosscut chains only: atoms enter through their prime-end chains and through the single point x_i, never through a boundary parametrization. No Euclidean boundary extension of any A_i—in particular no continuity of a Riemann map up to the boundary, equivalently no local connectivity of any atom—is assumed at any step. □

Remark 5.5. Clause (6) is the point of the whole construction and deserves emphasis. Realization arguments for connectors frequently assume that accesses land, that is, that radial limits of Riemann maps exist at the relevant boundary points; by Carathéodory's theorem this amounts to local connectivity of the atom boundaries. The certificate formalism removes that hypothesis: an access is a chain of crosscuts with declared sides, an object available for an arbitrary full continuum, and Lemma 5.3 shows it is preserved verbatim under any enlargement respecting the collar. This is what allows the framework to be applied, in Section 9, to atoms that are not known—and are not claimed—to be locally connected.

Univalent naturality. The certificate data and the carrier construction are natural under univalent change of the Jordan localization.

Proposition 5.6 (Univalent naturality; manuscript Proposition 10.5)**.** *If $\phi\colon U \to V$ is an orientation-preserving conformal homeomorphism of Jordan disks and every certificate entry is transported by ϕ, then*

$$H_{\phi_* R} = \phi(H_R). \tag{10}$$

The protected survival map and the ribbon boundary order commute with the prime-end map induced by ϕ.

Proof. Since U and V are Jordan disks, Carathéodory's extension theorem [9] extends ϕ to a homeomorphism $\overline{U} \to \overline{V}$ of the closed disks. By the Schoenflies theorem the restricted boundary homeomorphism $\partial U \to \partial V$ extends to a homeomorphism of the complementary closed disks $\widehat{\mathbb{C}} \setminus U \to \widehat{\mathbb{C}} \setminus V$, and gluing the two extensions along the boundary curves produces an orientation-preserving homeomorphism $\Phi\colon \widehat{\mathbb{C}} \to \widehat{\mathbb{C}}$ with $\Phi|_U = \phi$.

A sphere homeomorphism preserves every notion entering Definition 5.1 and Theorem 5.4: it carries compact connected sets to compact connected sets and complements to complements, hence preserves fullness; it carries open Jordan arcs with endpoints on a boundary to the same, hence crosscuts to crosscuts; it maps the two complementary sides of a crosscut to the two sides of the image crosscut, hence selected sides to selected sides; it carries null chains to null chains (spherical diameters are not literally preserved, but by uniform continuity of Φ the image crosscuts of a null chain again have diameters tending to zero); it preserves intersections, so the port conditions

$\alpha_i \cap A_i = \{x_i\}$ and all disjointness requirements transport; and, being orientation preserving, it preserves the planar cyclic order of half-edges at the tree vertices and the cyclic order of prime ends. Consequently the transported data—the disk V, the atoms $\phi(A_i)$, the obstacle $\phi(Z)$, the chains $\Phi(\Gamma_{i,a})$, the ports $\phi \circ \alpha_i$, and the tree $\phi(G)$ with its transported ribbon structure—form a protected ribbon-routing certificate in V. The defining union (9) then transports term by term:

$$H_{\phi_* R} = \phi(G) \cup \bigcup_i \phi(\alpha_i) \cup \bigcup_i \phi(A_i) = \phi\Big(G \cup \bigcup_i \alpha_i \cup \bigcup_i A_i\Big) = \phi(H_R),$$

which is (10). Because Φ maps $\widehat{\mathbb{C}} \setminus H_R$ onto $\widehat{\mathbb{C}} \setminus H_{\phi_* R}$ carrying null chains to null chains and preserving their division relation, it induces a bijection of prime ends under which the survival map of Lemma 5.3 for H_R corresponds to the survival map for $H_{\phi_* R}$, and the ribbon boundary words of Theorem 5.4(5) correspond letter by letter. □

Consequently one actual anchor certificate transports through the complete degree-one corridor of the source manuscript [7]. We record the manuscript's caveat: this transport does *not* prove equality of the transported carrier with any carrier copy supplied by the ψ-ql immersion. Naturality moves a certificate and its realization along a univalent map; it does not identify the result with an independently constructed carrier, and no such identification is claimed anywhere in this paper.

With realization settled, the burden shifts to existence: when do the port arcs and the routing tree demanded by Definition 5.1 exist at all? Section 6 answers this question exactly, in terms of a single component of the frozen free domain Ω_{free} admitting simultaneous singleton ports to every required atom.

6. The exact free-component criterion

Theorem 5.4 converts a protected ribbon-routing certificate into an actual carrier, but it leaves open the question of when such a certificate exists: the embedded finite tree of Definition 5.1 looks like an arbitrary choice, and one might fear that its existence depends on delicate features of the atoms or of the forbidden data. The present section removes the arbitrary tree choice. We prove that the existence of a protected ribbon-routing certificate is governed by a single, exact, purely planar condition: one component of a frozen free domain must reach every required atom by a singleton port. The section carries the one intentional mathematical divergence of this paper from the source manuscript [7]: the slit lemma below is stated and proved in a repaired form, and the proof of the criterion is rewritten to match. The divergence is recorded precisely in Remark 6.2.

6.1. **The frozen free domain.** Freeze the complete finite future inventory before choosing a connector. Concretely, let U be the Jordan localization and let $Q \subset U$ be the union of all root-shadow atoms $A_1, \dots, A_s$,

the closed terminal protected collars of every protected prime-end chain in the inventory, the separated obstacles, the critical-value exclusion, and the localization-forbidden set. As a standing convention, following the compact-containment hypotheses of the certificate setup, we assume each constituent of Q is closed in U and that the atoms, collars, and obstacles are compactly contained in U; this guarantees that the free domain defined next is open. Put

$$\Omega_{\mathrm{free}} \;=\; U \setminus Q \tag{11}$$

(equation (9.4) of the manuscript [7]). The set Ω_{free} is open, and its connected components are plane domains, the *free components* of the frozen data.

A component Ω_0 of Ω_{free} *has simultaneous singleton ports* to the required atoms if there are pairwise disjoint Jordan arcs $\alpha_1, \dots, \alpha_s \colon [0,1] \to U$ such that, for every i, the arc α_i meets A_i exactly in the single point $x_i = \alpha_i(0)$ and satisfies $\alpha_i\big((0,1]\big) \subset \Omega_0$. The landing point x_i is called a *free point* of A_i because it is reached from the free domain; no further property of x_i is assumed at this stage. (Compare the unprotected exposure loci of Lemma 8.2, where an additional protected-impression exclusion is imposed; the interaction between the two notions is illustrated in Example 6.4 below.)

The adjective *frozen* is load-bearing, and we emphasize it once. The set Q lists the complete finite future inventory: every atom that will ever have to be joined, the closed terminal collar of every protected chain that any future step of a proof will read, every obstacle that must remain separated, the full critical-value exclusion, and the localization-forbidden set. This list is fixed *before* any connector is chosen. Nothing may be added to Q afterwards, and nothing may be removed from it; all later monotone operations act on proposed connectors, never on the inventory. This discipline is exactly what gives the criterion of Theorem 6.3 its if-and-only-if form, and what makes its failure clause persistent: a separation already visible in the frozen data can never be talked away by enlarging a connector. The horizon-first closure results of the manuscript [7] address how such an inventory stabilizes; here the inventory is simply given, finite, and immutable.

6.2. **The repaired slit lemma.** The sufficiency direction of the criterion rests on the fact that finitely many pairwise disjoint one-ended slits do not disconnect a plane domain. This is Lemma 10.7 of the manuscript. Its printed statement contains a formal inconsistency (a map into an open domain cannot take a boundary value) together with an endpoint mismatch with its intended application; both defects are repaired in the statement below, and Remark 6.2 records the divergence. The mathematical core of the manuscript's argument — the transverse-rectangle detour around the free endpoint — is retained in full.

Lemma 6.1 (One-ended slits do not disconnect a plane domain; manuscript Lemma 10.7, repaired)**.** *Let $\Omega \subset \mathbb{C}$ be a domain and let $\delta_1, \dots, \delta_s \colon [0,1] \to$*

$\overline{\Omega}$ *be Jordan arcs with pairwise disjoint images such that* $\delta_i(0) \in \partial\Omega$ *and* $\delta_i\big((0,1]\big) \subset \Omega$ *for every* i*. Then*

$$\Omega \setminus \bigcup_{i=1}^{s} \delta_i\big((0,1)\big)$$

is path connected; in particular every free endpoint $\delta_i(1)$ *lies in this set and can be joined inside it to every other point of it.*

Proof. Call δ_i the *i-th slit*, write $\sigma_i := \delta_i\big((0,1]\big)$ for its *half-open* part and $\sigma_i^\circ := \delta_i\big((0,1)\big)$ for its *open* part, and call $\delta_i(1)$ its *tip*. The set to be proved path connected is $\Omega \setminus \bigcup_i \sigma_i^\circ$, which contains all s tips. If $s = 0$ there is nothing to prove, so assume $s \geq 1$; then $\partial\Omega \neq \varnothing$.

Step 0: two preliminary observations. First, each half-open part σ_i is relatively closed in Ω. Indeed, the closure of σ_i in $\mathbb{C}$ is the compact arc $\delta_i([0,1])$, so the only point of $\overline{\sigma_i} \setminus \sigma_i$ is $\delta_i(0) \in \partial\Omega$, which does not lie in Ω. Consequently $\Omega \setminus \sigma_i$ is open. By contrast, the open part σ_i° is *not* relatively closed in Ω: its closure inside Ω also contains the tip $\delta_i(1) \in \Omega$. This distinction is precisely the point at which the manuscript's induction required repair; the proof below removes the half-open parts first and reattaches the tips afterwards.

Second, every Jordan arc in the plane is tame: it can be extended to a Jordan curve, and the Schoenflies theorem then supplies a homeomorphism of $\mathbb{C}$ carrying the curve to a round circle and the arc to a circular subarc, hence, after one more homeomorphism, onto the straight segment $[0,1] \times \{0\}$; see, e.g., [8, 10]. Since path connectivity is a topological invariant and a homeomorphism of $\mathbb{C}$ carries domains to domains and boundaries to boundaries, we may straighten any *single* slit whenever convenient. (Straightening one slit will in general bend the others; each step below straightens only the slit it is processing.)

Step 1: removing a single half-open slit leaves a domain. Let Ω be a domain and let δ be a single arc as in the statement, with half-open part $\sigma = \delta\big((0,1]\big)$. We claim that $\Omega' := \Omega \setminus \sigma$ is again a domain. Openness was proved in Step 0, so only path connectivity is at issue. By tameness we may assume, after a homeomorphism of $\mathbb{C}$, that

$$\delta(t) = (t, 0) \quad (0 \leq t \leq 1),$$

so that Ω is a domain containing $S := (0,1] \times \{0\}$, with $(0,0) \in \partial\Omega$, and we must join two arbitrary points $x, y \in \Omega \setminus S$ by a path in $\Omega \setminus S$.

(a) A polygonal start missing an initial tail. Since a plane domain is polygonally connected [8], choose a polygonal path $P \subset \Omega$ from x to y. The path P is compact and does not contain the boundary point $(0,0)$, so

$$r := \operatorname{dist}\big((0,0), P\big) > 0.$$

Fix $\varepsilon := \frac{1}{2}\min(r, 1)$. Every point of the initial tail $[0,\varepsilon] \times \{0\}$ lies within distance $\varepsilon < r$ of the origin, so P misses $[0,\varepsilon] \times \{0\}$. This is the only part

of the closed slit that a compact subset of Ω cannot in general avoid by compactness alone: the closed slit image leaves every compact subset of Ω only along points accumulating at $\delta(0) = (0,0) \in \partial\Omega$, and near that end the continuity of δ at 0 confines the slit to a small disk around the origin. The remaining part $\delta([\varepsilon,1]) = [\varepsilon,1] \times \{0\}$ is a compact subset of Ω and will be handled by explicit detours.

(b) A thin rectangle around the compact subarc. The segment $[\varepsilon,1] \times \{0\}$ lies in Ω, and $(1,0) \in \Omega$ is an interior point of Ω; hence there is $\eta \in (0, \frac{1}{2})$ with $[1, 1+\eta] \times \{0\} \subset \Omega$. The segment $L := [\varepsilon/2,\, 1+\eta] \times \{0\}$ is then a compact subset of Ω, so $\operatorname{dist}(L, \mathbb{C} \setminus \Omega) > 0$, and we may choose $h \in \big(0, \min(\varepsilon/2, \operatorname{dist}(L, \mathbb{C} \setminus \Omega)/2)\big)$ and set

$$R := [\varepsilon/2,\, 1+\eta] \times [-h, h] \subset \Omega.$$

The rectangle R meets the closed slit $[0,1] \times \{0\}$ exactly in the core $c := [\varepsilon/2,\, 1] \times \{0\}$: the slit enters R through the midpoint of its left short side and terminates at the tip $(1,0)$, an interior point of R. In particular $R \cap S = [\varepsilon/2, 1] \times \{0\} = c$.

We record the key connectivity property of the slit rectangle: *$R \setminus c$ is path connected.* Indeed, let $V := \{1 + \eta/2\} \times [-h, h]$, a vertical segment in R missing c. A point of $R \setminus c$ with ordinate $y \neq 0$ travels horizontally to the point of V with the same ordinate; the horizontal path misses the axis $\mathbb{R} \times \{0\}$ entirely, hence misses c. A point of $R \setminus c$ with ordinate 0 has abscissa in $(1, 1+\eta]$ and travels along the axis to $(1+\eta/2, 0)$; this path keeps abscissa > 1, hence misses c. Since V is connected, the claim follows. Geometrically: the two long sides of R are joined to each other inside *$R \setminus c$ around the tip*, by crossing the axis at abscissae beyond 1.

(c) Transverse perturbation and detour repair. Perturb the interior vertices of P, keeping the endpoints x, y fixed, so that no interior vertex lies on the axis $\mathbb{R} \times \{0\}$ and no edge of P is collinear with the axis; if an endpoint happens to lie on the axis (necessarily off S), perturb its adjacent vertex so that the incident edge is not horizontal. Since P is compact with $\operatorname{dist}(P, \mathbb{C} \setminus \Omega) > 0$ and $\operatorname{dist}\big(P, [0,\varepsilon] \times \{0\}\big) \geq r - \varepsilon > 0$, a sufficiently small such perturbation keeps P inside Ω and away from the initial tail. After the perturbation, every edge of P meets the axis in at most one point, and every meeting point interior to an edge is a transverse crossing. Hence P meets S in finitely many points, all of the form $P(t_j)$ with parameters $0 < t_1 < \cdots < t_m < 1$ interior to edges, and all lying in $(\varepsilon, 1] \times \{0\} \subset \operatorname{int} R$. (If $m = 0$, the path P already joins x to y in $\Omega \setminus S$ and Step 1 is complete.)

By transversality, for each j there is $\rho_j > 0$ such that the parameter interval $[t_j - \rho_j,\, t_j + \rho_j]$ has image in $\operatorname{int} R$, meets the axis only at t_j, and the two punctured halves of the image lie strictly on opposite sides of the axis; shrink the ρ_j so that these intervals are pairwise disjoint and avoid 0 and 1. Put $p_j^- := P(t_j - \rho_j)$ and $p_j^+ := P(t_j + \rho_j)$; both lie in $R \setminus c$. Now modify P as follows: on each interval $[t_j - \rho_j,\, t_j + \rho_j]$, replace the sub-path of P by a path from p_j^- to p_j^+ inside $R \setminus c$, which exists by (b); keep P unchanged

elsewhere. Each detour lies in $R \subset \Omega$ and misses S, because $S \cap R = c$ and the detour misses c, while $S \setminus R$ is disjoint from R. The unchanged pieces of P carry parameters outside the intervals $(t_j - \rho_j,\, t_j + \rho_j)$ and therefore avoid S, since the crossings t_j were the only parameters at which P met S. The concatenation is a path from x to y in $\Omega \setminus S$.

Observe that each detour crosses the axis only at abscissae strictly beyond 1: it travels *around* the tip, through points arbitrarily close to $\delta(1)$ but never through any point of the closed slit; the tip itself, which here has been removed together with the rest of σ, is not needed as a way-station. This completes Step 1: $\Omega' = \Omega \setminus \sigma$ is a domain.

Step 2: removing all half-open slits leaves a domain. Set $\Omega_{(0)} := \Omega$ and, inductively, $\Omega_{(j)} := \Omega_{(j-1)} \setminus \sigma_j$ for $1 \le j \le s$. We claim that each $\Omega_{(j)}$ is a domain and that δ_j is a one-ended slit of $\Omega_{(j-1)}$ in the sense of the statement. Indeed, suppose $\Omega_{(j-1)}$ is a domain. Because the slit images are pairwise disjoint, σ_j is disjoint from $\sigma_1, \dots, \sigma_{j-1}$, so $\sigma_j \subset \Omega_{(j-1)}$. The point $\delta_j(0)$ lies outside $\Omega \supset \Omega_{(j-1)}$, and it is the limit of the points $\delta_j(t) \in \Omega_{(j-1)}$ as $t \downarrow 0$; hence $\delta_j(0) \in \partial\Omega_{(j-1)}$ and $\delta_j \colon [0,1] \to \overline{\Omega_{(j-1)}}$. Step 1, applied to the domain $\Omega_{(j-1)}$ and the single slit δ_j, now shows that $\Omega_{(j)}$ is a domain. After s steps we conclude: the set

$$\Omega_{(s)} \;=\; \Omega \setminus \bigcup_{i=1}^{s} \sigma_i \;=\; \Omega \setminus \bigcup_{i=1}^{s} \delta_i\big((0,1]\big)$$

is open and path connected.

Step 3: reattaching the tips. The set whose path connectivity is asserted differs from $\Omega_{(s)}$ exactly by the s tips:

$$\Omega \setminus \bigcup_{i=1}^{s} \sigma_i^{\circ} \;=\; \Omega_{(s)} \cup \{\delta_1(1), \dots, \delta_s(1)\},$$

and the tips are pairwise distinct points not in $\Omega_{(s)}$. Since $\Omega_{(s)}$ is path connected, it suffices to join each tip to some point of $\Omega_{(s)}$ by a path which, apart from the tip itself, lies in $\Omega_{(s)}$.

Fix i and straighten the i-th slit: let Φ be a homeomorphism of $\mathbb{C}$ with $\Phi\big(\delta_i([0,1])\big) = [0,1] \times \{0\}$ and $\Phi(\delta_i(t)) = (t, 0)$. The union $K := \bigcup_{k \neq i} \delta_k([0,1])$ of the other closed slit images is compact and disjoint from $\delta_i([0,1])$, so $\Phi(K)$ is compact and disjoint from $[0,1] \times \{0\}$. The model tip $(1,0)$ lies in the open set $\Phi(\Omega)$ and outside the closed set $\Phi(K)$, so

$$\mu := \operatorname{dist}\Big((1,0),\, \Phi(K) \cup \big(\mathbb{C} \setminus \Phi(\Omega)\big)\Big) > 0.$$

Choose $a \in \big(0, \min(\mu/2, \tfrac{1}{2})\big)$ and consider the square $R_i := [1-a,\, 1+a] \times [-a, a]$. Every point of R_i lies within distance $a\sqrt{2} < \mu$ of the model tip, so $R_i \subset \Phi(\Omega)$ and $R_i \cap \Phi(K) = \varnothing$; moreover R_i meets the model slit $[0,1] \times \{0\}$ exactly in $[1-a, 1] \times \{0\}$. The *escape segment*

$$E_i := [1,\, 1 + a/2] \times \{0\}$$

runs from the model tip into R_i; every point of E_i other than the tip has abscissa > 1, hence lies on none of the straightened slits, and lies in R_i, hence off $\Phi(K)$. Pulling back, $\beta_i := \Phi^{-1}(E_i)$ is a Jordan arc from $\delta_i(1)$ to a point $z_i \in \Omega_{(s)}$ with $\beta_i \setminus \{\delta_i(1)\} \subset \Omega_{(s)}$. Thus the tip $\delta_i(1)$ lies in the same path component as z_i, hence in the path component of $\Omega_{(s)}$.

The same square makes the two-sided accessibility of the tip transparent: inside $R_i \setminus \big([1-a,1] \times \{0\}\big)$ the two long sides of R_i are joined around the tip by the vertical segment at abscissa $1 + a/2$, so the tip is accessible from either side of the slit, and both accesses land in one and the same path component of the complement. Finally, note that the escape arc β_i can be made arbitrarily small: shrinking a shrinks E_i to the model tip, and continuity of Φ^{-1} then confines β_i to any prescribed neighborhood of $\delta_i(1)$.

Step 4: conclusion. By Steps 2 and 3 the set $\Omega \setminus \bigcup_i \sigma_i^\circ = \Omega_{(s)} \cup \{\delta_1(1), \dots, \delta_s(1)\}$ is path connected, and it contains every free endpoint $\delta_i(1)$; each of these can therefore be joined inside it to every other point of it. □

Two by-products of the proof will be used in the proof of Theorem 6.3 and are worth recording explicitly. First, by Step 2, removing the *half-open* slits $\delta_i((0,1])$ leaves an open, path connected set — indeed a plane domain, hence a polygonally connected one. Second, by Step 3, every tip $\delta_i(1)$ admits arbitrarily small escape arcs: for every neighborhood N of $\delta_i(1)$ there is a Jordan arc $\beta \subset N$ from $\delta_i(1)$ into the slit complement with $\beta \setminus \{\delta_i(1)\} \subset \Omega \setminus \bigcup_k \delta_k\big((0,1]\big)$.

Remark 6.2 (The repair). We record the divergence from the printed manuscript statement [7]. Manuscript Lemma 10.7 states the hypothesis in the form $\delta_i \colon [0,1] \to \Omega$ with $\delta_i(0) \in \partial\Omega$, which is formally inconsistent: a map into the open domain Ω cannot take a boundary value. It then removes the half-open slits $\delta_i\big((0,1]\big)$, *including* the free endpoints $\delta_i(1)$, while the manuscript's proof of Theorem 10.8 (reproduced as Theorem 6.3 below) joins precisely those outer endpoints by paths inside the complement — an endpoint mismatch: the points to be joined have just been deleted. The statement of Lemma 6.1 repairs both defects at once: the codomain is $\overline{\Omega}$, so the initial boundary value is legitimate, and only the open arcs $\delta_i\big((0,1)\big)$ are removed, so the free endpoints remain available for routing.

A third, smaller point is repaired inside the proof. The manuscript's induction asserts that the removed slit is relatively closed in Ω, so that the complement is again a domain and the one-slit argument iterates. For the half-open slit $\delta_i\big((0,1]\big)$ this is true (Step 0), but for the open slit $\delta_i\big((0,1)\big)$ it is false: the closure inside Ω adds the tip, so the complement of the open slit is not open at the tip and the inductive step cannot be applied verbatim to the repaired statement. The proof above therefore removes the half-open slits one at a time (Steps 1 and 2), where relative closedness does hold, and reattaches the s retained tips afterwards by the tip-accessibility argument (Step 3).

The repair changes no application. Every invocation of the lemma in the source manuscript — and every invocation in this paper — uses exactly the repaired form: the slits to be removed are open port interiors, and the retained outer endpoints are then joined inside the complement. The transverse-rectangle detour, which is the mathematical core of the manuscript's one-slit argument, is unchanged.

6.3. **The criterion.**

Theorem 6.3 (Protected Free-Component Criterion; manuscript Theorem 10.8)**.** *A protected ribbon-routing certificate exists if and only if one component of Ω_{free} has simultaneous singleton ports to every required atom. If a first protected collar or obstacle separates two required port classes, no further monotone connector enlargement inside the current localization can repair the failure.*

Proof. Necessity. Let a protected ribbon-routing certificate be given, with free-port arcs $\alpha_1, \dots, \alpha_s$, landing points $x_i = \alpha_i(0) \in A_i$, and embedded finite tree G joining the outer port endpoints (Definition 5.1). Delete the atoms from the connector and consider

$$X \;:=\; G \cup \bigcup_{i=1}^{s} \alpha_i\big((0,1]\big).$$

The tree G is connected, and each half-open port arc $\alpha_i\big((0,1]\big)$ is a connected set containing the outer endpoint $\alpha_i(1) \in G$; hence X is connected. Moreover $X \subset \Omega_{\mathrm{free}}$: by Definition 5.1, the sets $\alpha_i\big((0,1]\big)$ and G lie in U and avoid every atom, the obstacle and critical-value set, and every closed terminal protected collar. The localization-forbidden data are avoided by the frozen-inventory discipline: the manuscript treats them as part of the forbidden set that certificate data must miss, and we adopt the same reading here. The avoided sets are exactly the constituents of Q. A connected subset of the open set Ω_{free} lies in a single component Ω_0. The certificate's own port arcs are then simultaneous singleton ports of Ω_0: they are pairwise disjoint Jordan arcs, α_i meets A_i exactly in $\{x_i\}$, and $\alpha_i\big((0,1]\big) \subset X \subset \Omega_0$.

Sufficiency. Conversely, let Ω_0 be a component of Ω_{free} with simultaneous singleton ports $\alpha_1, \dots, \alpha_s$. Fix $\tau \in (0,1]$ and shorten each port away from the atoms, replacing α_i by its initial subarc $\alpha_i|_{[0,\tau]}$, reparametrized over $[0,1]$ and still denoted α_i; the shortened arcs remain pairwise disjoint simultaneous singleton ports, and the shortening emphasizes that only the germ of each port at its atom matters. Write $y_i := \alpha_i(1)$ for the retained outer endpoints.

The shortened ports are one-ended slits of the domain Ω_0 in the sense of Lemma 6.1. Indeed, their images are pairwise disjoint; the interiors satisfy $\alpha_i\big((0,1]\big) \subset \Omega_0$; and the landing point $x_i = \alpha_i(0)$ lies in $A_i \subset Q$, hence outside Ω_0, while being the limit of the points $\alpha_i(t) \in \Omega_0$, so that $x_i \in \partial\Omega_0$ and $\alpha_i \colon [0,1] \to \overline{\Omega_0}$. The retained port interiors are therefore finitely many

pairwise disjoint one-ended slits of Ω_0, and Lemma 6.1 applies to remove their *open* parts: the set

$$W \;:=\; \Omega_0 \setminus \bigcup_{i=1}^{s} \alpha_i\big((0,1)\big)$$

is path connected, and it contains the outer endpoints $y_1, \dots, y_s$ — which is exactly what the repaired statement guarantees.

We now upgrade the joining paths supplied by path connectivity of W to an embedded polygonal tree, using the two by-products recorded after the proof of Lemma 6.1. First, by Step 2 of that proof,

$$\Omega' \;:=\; \Omega_0 \setminus \bigcup_{i=1}^{s} \alpha_i\big((0,1]\big)$$

is a plane domain. Second, by Step 3, each tip y_i admits arbitrarily small escape arcs into Ω'. If $s = 1$, the single shortened port together with the degenerate one-vertex tree $\{z_1\}$, obtained from any escape arc, is already a certificate, so assume $s \geq 2$. Let $\rho := \frac{1}{3}\min_{i\neq j} |y_i - y_j| > 0$, and choose for each i an escape arc $\beta_i \subset B(y_i, \rho)$ with $\beta_i(0) = y_i$ and $\beta_i\big((0,1]\big) \subset \Omega'$; the balls $B(y_i, \rho)$ are pairwise disjoint, so the arcs β_i are pairwise disjoint. Put $z_i := \beta_i(1) \in \Omega'$.

Since the domain Ω' is polygonally connected, choose polygonal paths $P_2, \dots, P_s \subset \Omega'$ joining z_1 to $z_2, \dots, z_s$. Perturb them, keeping their endpoints fixed, into general position inside Ω': after an arbitrarily small perturbation, any two edges taken from distinct paths, or nonadjacent edges of one path, meet in at most one point and transversely, and no vertex of one path lies on another path except at shared endpoints. The union $\Gamma := P_2 \cup \cdots \cup P_s$ is then a connected finite plane graph in Ω' containing $z_1, \dots, z_s$: its vertices are the path endpoints together with the finitely many transverse crossings, at which we subdivide. Let $G_0 \subset \Gamma$ be an inclusion-minimal connected subgraph containing $\{z_1, \dots, z_s\}$. Then G_0 is a finite tree: were it to contain a cycle, deleting one edge of the cycle would leave a connected subgraph still containing every z_i, contradicting minimality. Being a subgraph of an embedded graph, G_0 is embedded, and $G_0 \subset \Omega'$.

The escape arcs may meet G_0 before their endpoints, so we trim them by a first-hit argument. For each i the set $\{t \in [0,1] : \beta_i(t) \in G_0\}$ is closed, and it is nonempty because $\beta_i(1) = z_i \in G_0$; let t_i^* be its minimum. Since $\beta_i(0) = y_i \notin \Omega' \supset G_0$ we have $t_i^* > 0$. Set $\beta_i' := \beta_i|_{[0,t_i^*]}$ and $z_i' := \beta_i(t_i^*) \in G_0$, so that $\beta_i' \cap G_0 = \{z_i'\}$.

Assemble the certificate. Define the rerouted port arcs $\tilde{\alpha}_i := \alpha_i * \beta_i'$ for $1 \leq i \leq s$, the concatenation of α_i and β_i', reparametrized over $[0,1]$. Each $\tilde{\alpha}_i$ is a Jordan arc: both pieces are Jordan arcs meeting exactly at $y_i = \alpha_i(1) = \beta_i'(0)$, because $\beta_i' \setminus \{y_i\} \subset \Omega'$ while the image $\alpha_k([0,1])$ of every port lies in $Q \cup \bigcup_j \alpha_j\big((0,1]\big)$, a set disjoint from Ω'. The same disjointness, together with the pairwise disjoint balls containing the β_i, shows

that $\tilde{\alpha}_1, \dots, \tilde{\alpha}_s$ are pairwise disjoint. Each $\tilde{\alpha}_i$ meets A_i exactly in $\{x_i\}$, and its interior satisfies $\tilde{\alpha}_i\big((0,1]\big) \subset \Omega_0 \subset \Omega_{\mathrm{free}}$: it avoids every atom, every closed terminal protected collar, the obstacle and critical-value set, and the localization-forbidden set — exactly the avoidance clauses that Definition 5.1 imposes on free-port arcs. Next, subdivide G_0 so that each z_i' is a vertex, and call the resulting embedded finite tree G. Then $G \subset \Omega'$ avoids the same forbidden sets, joins the outer port endpoints $z_1', \dots, z_s'$, and meets each port arc $\tilde{\alpha}_i$ exactly in its outer endpoint z_i', by the first-hit property and $G \subset \Omega'$. Consequently $G \cup \bigcup_i \tilde{\alpha}_i$ is again a finite tree: it is connected, and attaching finitely many pairwise disjoint arcs to a tree, each along a single endpoint, adds one vertex and one edge at a time and so preserves the tree characterization $(\text{vertices}) - (\text{edges}) = 1$ for connected finite graphs. The planar cyclic order of half-edges at every vertex is the one induced by the embedding, and it is retained as the certificate's ribbon datum. Finally, the protected prime-end crosscut chains, with their pairwise disjoint closed terminal collars, belong to the frozen inventory and are avoided as recorded above. All clauses of Definition 5.1 are satisfied: the rerouted ports $\tilde{\alpha}_i$ together with the spanning tree G constitute a protected ribbon-routing certificate. Theorem 5.4 now realizes it by a compact connected full carrier with singleton attachment traces, protected crosscut survival, and preserved cyclic order.

Persistence. For the final assertion, suppose that at some stage of the frozen inventory a first protected collar or obstacle $C_0 \subset Q$ separates two required port classes: after the insertion of C_0, every component of Ω_{free} admitting a singleton port to A_i is distinct from every component admitting a singleton port to A_j, for some pair of required atoms A_i, A_j. Then no single component has simultaneous singleton ports to both atoms, so by the criterion just proved no protected ribbon-routing certificate exists. This failure is permanent under monotone connector enlargement inside the current localization, for two cumulative reasons. First, enlargement acts on proposed connectors, never on the inventory: Q is frozen, so Ω_{free} and its partition into components are unchanged, and the necessity argument applies verbatim to every enlarged connector — its routing part is a connected set which, were the certificate discipline respected, would lie in one component of Ω_{free} while porting to atoms in two different components; this is impossible, since a connected subset of $U \setminus Q$ lies in a single component. Second, the only way for a connected set to join the two components is to meet Q across the separating part, and if that part is protected or forbidden, the intersection itself violates the certificate: absorbing a protected collar erases the protected access it certifies, and meeting an obstacle or the critical-value exclusion is excluded outright. (Compare Proposition 3.2 for an analytic incarnation of the first mechanism.) Hence no further monotone connector enlargement inside the current localization can repair the failure; the only conceivable cure — deleting the separator C_0 from the inventory — is a revision of the frozen data, which the formalism forbids. □

The construction is natural under univalent changes of the Jordan localization. An orientation-preserving conformal homeomorphism of Jordan disks extends over the Jordan boundaries, and it carries the frozen set Q, the free domain (11) and its components, the port arcs, the escape arcs, and the spanning tree to their transported counterparts; the transported data form a certificate for the transported inventory, and the realized carriers correspond as in Proposition 5.6. At the same time the criterion is qualitative: the number of ports and of ribbon sides is not bounded uniformly, and no width, modulus, or incidence estimate enters at any point. Nothing in this section constructs the frozen inventory of a dynamical situation; the corresponding virtual free-component questions are stated in the source manuscript only as open problems [7], and no part of this paper bears on them.

6.4. **A separator between individually accessible atoms.** The criterion of Theorem 6.3 is strictly stronger than every atom-by-atom access condition. The following example, promised in the introduction, exhibits two atoms each of which passes every individual-access audit, while no common free component exists; it is the minimal model of the failure mode that the persistence clause certifies. See Figure 5.

Example 6.4 (Individual accesses without a common free component)**.** Let $U := \{z \in \mathbb{C} : |z| < 2\}$ and take the two atoms

$$A_1 := \left[-\tfrac{1}{4}, \tfrac{1}{4}\right], \qquad A_2 := \left[\tfrac{5}{4}, \tfrac{7}{4}\right],$$

closed segments of the real axis, each a full continuum compactly contained in U. Freeze exactly one further inventory item, the round circle

$$C := \{z : |z| = 1\},$$

recorded as a separated obstacle; all that matters below is that $C \subset Q$, so any recording of C in the frozen inventory would do. There are no critical values and no further forbidden data, so the frozen set is $Q = A_1 \cup A_2 \cup C$ and

$$\Omega_{\text{free}} = U \setminus (A_1 \cup A_2 \cup C).$$

The free domain has exactly two components, one per atom. Write

$$\Omega_{\text{in}} := \{|z| < 1\} \setminus A_1, \qquad \Omega_{\text{ann}} := \{1 < |z| < 2\} \setminus A_2,$$

so that $\Omega_{\text{free}} = \Omega_{\text{in}} \sqcup \Omega_{\text{ann}}$, a disjoint union of open sets separated by the circle C. Each is connected, by an explicit radial–angular path argument. In Ω_{in}, every point can be joined to the circle $\{|z| = \frac{1}{2}\}$, which misses A_1: a point with $\arg z \notin \{0, \pi\}$ travels radially to modulus $\frac{1}{2}$, and the radial path keeps its argument, hence stays off the real axis and misses the real segment A_1; a real point of Ω_{in} has modulus in $(\frac{1}{4}, 1)$ and travels radially to modulus $\frac{1}{2}$ through real points of modulus $> \frac{1}{4}$, which also miss A_1. In Ω_{ann}, every point can be joined to the circle $\{|z| = \frac{9}{8}\}$, which misses A_2: a point with $\arg z \neq 0$ travels radially inward keeping its argument; a positive real point

of Ω_{ann} has modulus in $(1, \frac{5}{4}) \cup (\frac{7}{4}, 2)$, and in the first case it travels radially inward to modulus $\frac{9}{8}$ through positive real points of modulus $< \frac{5}{4}$, while in the second case it first travels along its own circle $\{|z| = \text{const}\}$ (which misses A_2, as $\frac{7}{4} < |z| < 2$) to argument π and then proceeds radially inward as before. Both sets are therefore domains, and they are exactly the two components of Ω_{free}.

Every individual-access audit passes. Exhibit the singleton ports

$$\alpha_1(t) := \tfrac{1}{4} + \tfrac{t}{2}, \qquad \alpha_2(t) := \tfrac{7}{4} + \tfrac{t}{8} \qquad (0 \le t \le 1) :$$

the arc α_1 runs from $x_1 = \frac{1}{4} \in A_1$ to $\frac{3}{4}$ and meets A_1 only at x_1, with $\alpha_1\big((0,1]\big) = (\frac{1}{4}, \frac{3}{4}] \subset \Omega_{\text{in}}$; the arc α_2 runs from $x_2 = \frac{7}{4} \in A_2$ to $\frac{15}{8}$ and meets A_2 only at x_2, with $\alpha_2\big((0,1]\big) = (\frac{7}{4}, \frac{15}{8}] \subset \Omega_{\text{ann}}$. Thus each atom has a singleton port into *its own* free component. In the language of the port audit, with empty protected marking and with the atoms as their own completed packets, both unprotected exposure loci are nonempty, so the hypotheses of Lemma 8.2 hold and its conclusion is correct: the two arcs can be (and are) chosen pairwise disjoint with interiors in the common complement of the packets. Atom by atom, nothing is wrong.

No common free component exists. Every port of A_1 runs in Ω_{in}: if α is an arc with $\alpha \cap A_1 = \{\alpha(0)\}$ and $\alpha((0,1])$ contained in one component of Ω_{free}, then $\alpha(t) \to \alpha(0) \in A_1 \subset \{|z| \le \frac{1}{4}\}$ forces $|\alpha(t)| < 1$ for small t, so the component is Ω_{in}. Likewise every port of A_2 runs in Ω_{ann}, since points of Ω_{free} near A_2 have modulus in $(1, 2)$. As $\Omega_{\text{in}} \neq \Omega_{\text{ann}}$, no component of Ω_{free} has simultaneous singleton ports to both atoms, and Theorem 6.3 yields: *no protected ribbon-routing certificate exists* for this frozen inventory. By the persistence clause, the failure is permanent: the separator C is the first (indeed the only) separating insertion, the two required port classes are $\{\Omega_{\text{in}}\}$ and $\{\Omega_{\text{ann}}\}$, and no monotone connector enlargement inside U can repair the separation. The only cure would be to delete C from the frozen inventory — a revision of the frozen data, which the formalism forbids.

The example separates two layers of the theory cleanly. The hypotheses of Lemma 8.2 — individual exposure of every atom — are statements about each atom in isolation, and they hold here; the criterion of Theorem 6.3 is a statement about one common component, and it fails. This is also precisely why Definition 7.1 demands that the completed packets be pairwise disjoint *and cover* Q: the separator C must be assigned to some packet, and every typed assignment then exhibits a persistent defect of the audit of Theorem 8.3. Assigning C to a separated-obstacle packet gives $\text{Fill}(C) = \{|z| \le 1\} \supset A_1$, a packet merger with the completed packet of A_1. Assigning C to the atom seed of A_2 (necessarily together with a connecting arc, to keep the seed connected) again puts the closed unit disk, hence A_1, inside the completed packet of A_2: the same merger. Assigning C to the atom seed of A_1 produces a completed packet containing the closed unit disk with A_1 in its interior, so the unprotected exposure locus of A_1 becomes empty and the

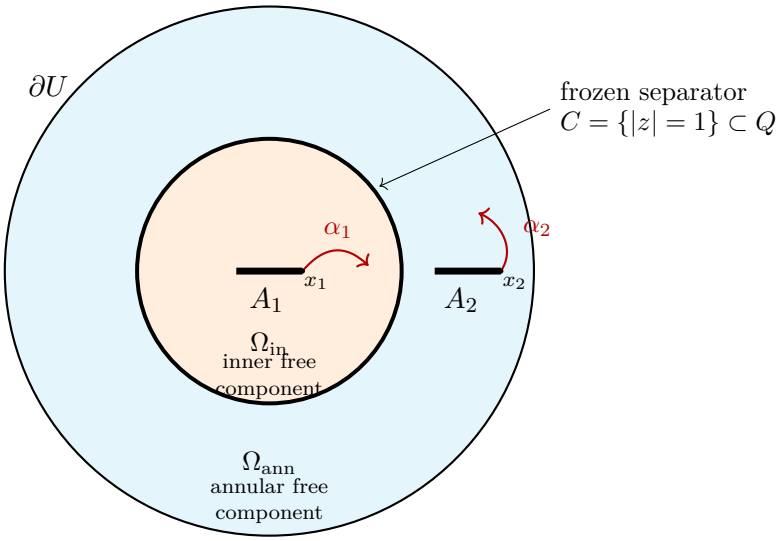


FIGURE 5. The separation example of Example 6.4. The Jordan localization is the disk $U = \{|z| < 2\}$; the frozen inventory consists of the two atoms $A_1 = [-\frac{1}{4}, \frac{1}{4}]$ and $A_2 = [\frac{5}{4}, \frac{7}{4}]$ and the bold circle C, a frozen separator belonging to Q. The free domain $\Omega_{\text{free}} = U \setminus Q$ has exactly two components, shaded in different tints: the inner component Ω_{in} and the annular component Ω_{ann}. Each atom has a singleton port arc (α_1, α_2) into its own component, so every individual-access audit passes; but no single component ports to both atoms, so by Theorem 6.3 no protected ribbon-routing certificate exists, and by the persistence clause no monotone connector enlargement inside U can repair the failure while C stays in the frozen inventory.

port clause fails. Merger and empty exposure are exactly the defects that Theorem 8.3 proves persistent under monotone same-label enlargement, so the packet formalism detects the obstruction at the audit level, consistently with the exact criterion.

7. Augmented atoms and the common domain

The free-component criterion of Theorem 6.3 is exact, and its persistence clause makes failures permanent; but as a *positive* test it is expensive. Deciding whether one component of Ω_{free} reaches every required atom demands a global analysis of how each of the finitely many collars, obstacles, and exclusions of the frozen inventory sits relative to all of the others, and Example 6.4 shows that no atom-by-atom inspection can substitute for that analysis: individual accessibility of every atom is compatible with an unrepairable separation. From a *partial* package — the situation in which the

criterion would actually be invoked, with the inventory delivered piecemeal by an ambient proof program — the exact free-domain test is therefore difficult to verify directly [7].

The present section proves a sufficient criterion that is *source-checkable*: each of its clauses is verified against named, finite source data rather than against the component structure of an open set. The idea is to reverse the direction of the analysis. Instead of asking how the frozen set Q decomposes the localization, we package Q itself into finitely many pairwise disjoint full continua — each atom augmented by the inventory pieces assigned to it, the remainder grouped into separated-obstacle packets — while reserving one free port for each original atom. Once the completed packages are pairwise disjoint, sphere duality forces their common complement to be *connected*; connectedness, not component bookkeeping, then delivers the common free component that Theorem 6.3 demands.

The section has three parts. Definition 7.1 fixes the packet structure and its typed assignment record; Lemma 7.3 is the duality argument; and Theorem 7.4 assembles the sufficient criterion. How the completed packets are actually computed from raw assigned pieces, and how their defining clauses are audited with persistent failure certificates, is the subject of Section 8; Example 8.4 there shows why the audit must be run on the completions and not on the raw pieces.

7.1. **Filled hulls and augmented packets.** Recall from Section 2 the filled hull of a compact plane set K:

$$\mathrm{Fill}(K) \;=\; K \,\cup\, \bigcup\big\{\, B : B \text{ a bounded component of } \mathbb{C}\setminus K \,\big\} \tag{12}$$

(equation (9.5) of [7]). Since K is compact, $\mathbb{C}\setminus K$ has exactly one unbounded component $\Omega_\infty(K)$, and (12) says

$$\mathrm{Fill}(K) \;=\; \mathbb{C}\setminus \Omega_\infty(K).$$

Consequently $\mathrm{Fill}(K)$ is closed, and it is bounded because every closed round disk containing K has its exterior inside $\Omega_\infty(K)$; thus $\mathrm{Fill}(K)$ is compact. Its complement is the single connected set $\Omega_\infty(K)$, so $\mathrm{Fill}(K)$ is *full*, and it is the smallest full compact set containing K: if $F \supset K$ is full and compact, then $\mathbb{C}\setminus F$ is a connected unbounded subset of $\mathbb{C}\setminus K$, hence contained in $\Omega_\infty(K)$, which gives $\mathrm{Fill}(K) \subset F$ after taking complements. Finally, if K is a continuum, then so is $\mathrm{Fill}(K)$: every bounded component B of $\mathbb{C}\setminus K$ has $\partial B \subset K$, so its closure $\overline{B}$ is a connected set meeting K, and $\mathrm{Fill}(K) = K \cup \bigcup \overline{B}$ is a union of connected sets all meeting the connected set K. The passage from K to $\mathrm{Fill}(K)$ is what the manuscript calls *fullification*; its algebra — and in particular the fact that interleaved adjoin-and-fill operations compute a well-defined completion — is developed in Section 8.

Throughout, the *atoms* $A_1, \dots, A_s \Subset U$ are the pairwise disjoint full continua of Section 5 that the connector must join, and Q is the frozen set of Section 6 (equation (9.4) of [7]): the union of the atoms, the closed terminal

protected collars, the separated obstacles, the critical-value exclusion, and the localization-forbidden set, fixed once and for all before any connector is chosen.

Definition 7.1 (Augmented-atom packet; manuscript Definition 10.9)**.** For the frozen set Q, an *augmented-atom packet* consists of full continua

$$\widehat{A}_1, \dots, \widehat{A}_s,\ \widehat{O}_1, \dots, \widehat{O}_t \ \Subset\ U$$

and pairwise disjoint port arcs $\alpha_i \colon [0,1] \to U$ such that:

(1) every protected collar, obstacle, critical exclusion, and other component of Q has one typed assignment, either to a connected atom seed $A_i \cup E_i$ or to a connected separated-obstacle seed O_ℓ;
(2) $\widehat{A}_i = \mathrm{Fill}(A_i \cup E_i)$ for every i, and $\widehat{O}_\ell = \mathrm{Fill}(O_\ell)$ for every ℓ;
(3) the completed packets $\widehat{A}_1, \dots, \widehat{A}_s, \widehat{O}_1, \dots, \widehat{O}_t$ are pairwise disjoint and cover Q; and
(4) $\alpha_i(0) = x_i \in A_i$, $\alpha_i \cap \widehat{A}_i = \{x_i\}$, and

$$\alpha_i\big((0,1]\big)\ \subset\ U \setminus \Big(\bigcup_k \widehat{A}_k \cup \bigcup_\ell \widehat{O}_\ell\Big),$$

and x_i is not the impression of a protected chain.

The assignment records the component name, the proof-use stages, the attachment trace, the return word, the selected sheet, and the absorb-or-separate decision.

Some commentary on the four clauses. In clause (1), E_i denotes the union of the inventory pieces assigned to the i-th atom; it may be empty, and the clause demands that the resulting *atom seed* $A_i \cup E_i$ be connected, so that an assigned piece disjoint from its atom must be accompanied by connecting material assigned to the same label. The assignment is *typed*: it is a function from the finitely many named constituents of the frozen inventory to the packet labels, and each assigned piece carries the six recorded fields listed after clause (4). Those fields — which component of the ambient package the piece is, at which proof-use stages it is read, its attachment trace, its return word, its selected sheet, and whether the decision taken for it was absorption or separation — are bookkeeping data of the ambient proof program; the planar topology of this section reads only the geometry of the seeds. The record is nevertheless load-bearing: it is what makes a later change of assignment an explicit, auditable re-decision rather than a silent repair, a point Example 8.4 makes concrete.

In clause (2), fullness of the completed packets is automatic: the seeds are compact and connected, so their filled hulls are full continua by the discussion following (12). Clause (3) is the substantive geometric hypothesis, and it is stated for the *completed* packets $\widehat{A}_i, \widehat{O}_\ell$, not for the raw seeds: fullification can create intersections that the seeds do not have, and an audit run at seed level certifies nothing (Example 8.4). Coverage of Q means that every item of the frozen inventory lies inside some completed packet, so that

outside the packets there is no frozen material at all — this is the inclusion that will place the common complement inside Ω_{free}. Clause (4) is the port clause. It is deliberately stronger than the port clause of Definition 5.1: the arc's positive-parameter part must avoid every completed hull, whereas the arcs of Definition 5.1 avoid only the obstacle set, the atoms, and the terminal collar tails, and the arc here must meet even its own augmented atom $\widehat{A}_i$ only at the landing point x_i. The final requirement, that x_i not be the impression of a protected chain, keeps the port compatible with the protected marking; it reappears in Section 8 as the requirement that x_i lie in the unprotected exposure locus, outside the forbidden impression set P_i.

7.2. **Nonseparation of finite disjoint full families.** The engine of the common-domain theorem is a nonseparation statement for finitely many disjoint full continua on the sphere. Here a compact set $K \subset S^2$ is called full when $S^2 \setminus K$ is connected; for a compact subset of the plane this agrees with the planar notion, because adjoining the point at infinity to the unbounded complementary component does not change connectivity, so a plane-full compact set is sphere-full and conversely. The proof uses the compacta form of Alexander duality [3, pp. 256–257]:

$$\widetilde{H}_0\big(S^2 \setminus K; \mathbb{Z}\big) \;\cong\; \check{H}^1(K; \mathbb{Z}) \tag{13}$$

for an arbitrary nonempty proper compact subset $K \subset S^2$ (equation (9.6) of [7]). The Čech form, rather than the locally contractible special case, is required here: the atoms and the assigned inventory pieces are arbitrary continua, no local connectivity is assumed anywhere in this paper, and for such compacta singular cohomology need not compute the group on the right of (13).

Remark 7.2. The insistence on the Čech form is not pedantry but a structural feature of the framework. Clause (6) of Theorem 5.4 promises that no local connectivity of any atom is ever required, and the packets of Definition 7.1 inherit whatever boundary pathology their atoms and assigned pieces possess. Čech cohomology is the invariant that remains dual to the complement's component count for every compact subset of the sphere, without any tameness hypothesis; every duality argument below is stated so that it factors through (13) alone.

Lemma 7.3 (Finite disjoint full packets; manuscript Lemma 10.10)**.** *If $K_0, \dots, K_N \subset S^2$ are pairwise disjoint nonempty full continua, then*

$$S^2 \setminus \bigcup_{r=0}^{N} K_r$$

is connected.

Proof. Write $X := \bigcup_{r=0}^{N} K_r$. We may assume that X is a proper subset of S^2, so that (13) applies to X and to each K_r: were $X = S^2$, the sets $K_0, \dots, K_N$ would be finitely many pairwise disjoint nonempty closed sets

covering S^2; each would then be open as well, being the complement of the union of the others, and connectedness of S^2 would force $N = 0$ and $K_0 = S^2$, in which case the complement is empty and there is nothing to prove.

Step 1: each packet has vanishing first Čech cohomology. Fix r. Fullness of K_r makes $S^2 \setminus K_r$ connected, and it is nonempty because K_r is proper. Being open in S^2, the complement is locally path connected, so connectedness makes it path connected and its reduced zeroth singular homology vanishes. Hence (13), applied to the nonempty proper compact set K_r, gives

$$\check{H}^1(K_r;\mathbb{Z}) \;\cong\; \widetilde{H}_0\big(S^2 \setminus K_r;\mathbb{Z}\big) \;=\; 0.$$

Step 2: disjoint neighborhoods split fine covers. The sets $K_0, \dots, K_N$ are finitely many pairwise disjoint compacta, so their mutual distances in the spherical metric are positive; choose $\varepsilon > 0$ smaller than half the minimum of these distances and let N_r be the open ε-neighborhood of K_r. The N_r are pairwise disjoint open sets with $K_r \subset N_r$.

Call a cover of X by relatively open subsets *fine* when every member is contained in one of the sets N_r. The fine covers are cofinal among all relatively open covers of X: an arbitrary cover $\mathcal{U}$ is refined by the fine cover $\{\, V \cap N_r : V \in \mathcal{U},\ 0 \le r \le N \,\}$, since $X \subset \bigcup_r N_r$. Let $\mathcal{V}$ be a fine cover with nonempty members. Each member lies in exactly one N_r, because the N_r are pairwise disjoint; write $\mathcal{V}_r$ for the members contained in N_r. Since $X \cap N_r = K_r$, the collection $\mathcal{V}_r$ is a relatively open cover of K_r. Two members taken from distinct collections $\mathcal{V}_r, \mathcal{V}_{r'}$ are disjoint, lying in the disjoint sets $N_r, N_{r'}$; hence no simplex of the nerve of $\mathcal{V}$ mixes the collections, and the nerve of $\mathcal{V}$ is the disjoint union of the nerves of $\mathcal{V}_0, \dots, \mathcal{V}_N$. Simplicial cohomology of a finite disjoint union is the direct sum, so

$$H^1\big(\operatorname{nerve}\mathcal{V};\mathbb{Z}\big) \;\cong\; \bigoplus_{r=0}^{N} H^1\big(\operatorname{nerve}\mathcal{V}_r;\mathbb{Z}\big).$$

Step 3: pass to the direct limit. Čech cohomology of X is the direct limit of the nerve cohomologies over the directed system of relatively open covers, and the fine covers are cofinal in that system, so the limit may be computed over fine covers only. A refinement of a fine cover is again fine after intersecting with the N_r, and its restriction to N_r refines $\mathcal{V}_r$; the splitting of Step 2 is therefore compatible with the refinement maps, and restriction to N_r is cofinal among the covers of K_r because any relatively open cover of K_r extends to a fine cover of X by adjoining covers of the other packets. Since a direct limit commutes with the finite direct sum, Step 1 gives

$$\check{H}^1(X;\mathbb{Z}) \;\cong\; \bigoplus_{r=0}^{N} \check{H}^1(K_r;\mathbb{Z}) \;=\; 0.$$

Step 4: apply duality a second time. The set X is compact, nonempty, and proper, so (13) applies to X itself:

$$\widetilde{H}_0\big(S^2 \setminus X;\mathbb{Z}\big) \;\cong\; \check{H}^1(X;\mathbb{Z}) \;=\; 0.$$

The complement $S^2 \setminus X$ is nonempty, and a nonempty space with vanishing reduced zeroth homology is connected. This is the assertion. □

Lemma 7.3 is the disjoint counterpart of Lemma 5.2. There, unions of nonseparating continua meeting in connected singletons were handled by the Janiszewski separation lemma, one attachment at a time; here the continua are pairwise disjoint, the count is arbitrary, and the duality argument dispenses with any hypothesis on how the packets face one another. Nothing is assumed about the boundaries of the K_r — they may be nowhere locally connected — and no width, diameter, or separation estimate enters.

7.3. **The common-domain theorem.**

Theorem 7.4 (Augmented-Atom Common-Domain Theorem; manuscript Theorem 10.11)**.** *Every augmented-atom packet determines one component of Ω_{free} with simultaneous singleton ports to every original atom. Hence it supplies a protected regulated connector.*

Proof. Let an augmented-atom packet be given, with completed packets $\widehat{A}_1,\dots,\widehat{A}_s,\widehat{O}_1,\dots,\widehat{O}_t \Subset U$ and port arcs $\alpha_1,\dots,\alpha_s$ as in Definition 7.1.

The exterior packet. Adjoin to the family the set

$$K_0 \;:=\; S^2 \setminus U.$$

The Jordan-disk hypothesis makes K_0 a full continuum: by the Jordan–Schoenflies theorem the Jordan curve ∂U divides the sphere into two open Jordan disks with common boundary ∂U, the closure of the one containing ∞ is the closed Jordan disk $S^2 \setminus U$, hence a continuum, and its complement is the connected set U, so K_0 is full in the sphere sense used in Lemma 7.3. Each completed packet is a plane-full continuum, hence sphere-full as noted in Section 7.2, and compact containment $\widehat{A}_i, \widehat{O}_\ell \Subset U$ makes every completed packet disjoint from K_0; together with clause (3) of Definition 7.1, the family

$$K_0,\ \widehat{A}_1,\dots,\widehat{A}_s,\ \widehat{O}_1,\dots,\widehat{O}_t$$

consists of $1+s+t$ pairwise disjoint nonempty full continua in S^2.

The augmented complement is connected. Lemma 7.3 applies to this family and makes its common complement connected. Since $S^2 \setminus K_0 = U$, that complement is

$$\Omega_{\mathrm{aug}} \;:=\; U \setminus \Big(\bigcup_i \widehat{A}_i \cup \bigcup_\ell \widehat{O}_\ell\Big) \;=\; S^2 \setminus \Big(K_0 \cup \bigcup_i \widehat{A}_i \cup \bigcup_\ell \widehat{O}_\ell\Big), \tag{14}$$

a connected subset of $\mathbb{C}$, open because U is open and the packets are compact. It is nonempty: clause (4) places the positive-parameter part of each port arc inside it.

The augmented complement lies in one free component. Clause (3) of Definition 7.1 makes the completed packets cover Q, so

$$\Omega_{\mathrm{aug}} \;=\; U \setminus \Big(\bigcup_i \widehat{A}_i \cup \bigcup_\ell \widehat{O}_\ell\Big) \;\subset\; U \setminus Q \;=\; \Omega_{\mathrm{free}}.$$

A nonempty connected subset of the open set Ω_{free} lies in exactly one of its components; let Ω_0 be the component containing Ω_{aug}. This is the component that the packet determines.

The ports are simultaneous singleton ports of Ω_0. By clause (4), each port interior satisfies

$$\alpha_i\big((0,1]\big) \;\subset\; U \setminus \Big(\bigcup_k \widehat{A}_k \cup \bigcup_\ell \widehat{O}_\ell\Big) \;=\; \Omega_{\mathrm{aug}} \;\subset\; \Omega_0.$$

Moreover $A_i \subset A_i \cup E_i \subset \widehat{A}_i$, so

$$\alpha_i \cap A_i \;\subset\; \alpha_i \cap \widehat{A}_i \;=\; \{x_i\},$$

and $x_i = \alpha_i(0) \in A_i$ gives $\alpha_i \cap A_i = \{x_i\}$: the arc meets its atom exactly in the one free point x_i. The arcs are pairwise disjoint Jordan arcs by Definition 7.1. Thus Ω_0 has simultaneous singleton ports to every original atom, in the sense of Section 6.

Conclusion. Theorem 6.3 converts the simultaneous singleton ports of Ω_0 into a protected ribbon-routing certificate, and Theorem 5.4 realizes that certificate by a compact connected full carrier with singleton attachment traces, protected crosscut survival, and preserved cyclic order. The packet therefore supplies a protected regulated connector. □

Remark 7.5. Three remarks of the manuscript [7] deserve emphasis. First, no local connectivity or width estimate occurs anywhere in the argument: the packets may have arbitrarily wild boundaries, the duality step of Lemma 7.3 is insensitive to them, and no modulus, diameter, or incidence bound is invoked at any point. Second, the construction is natural under an orientation-preserving conformal homeomorphism of Jordan disks: the boundary extensions can be glued to a sphere homeomorphism, and such a map carries filled packets to filled packets, ports to ports, the common domain (14) to the corresponding common domain, and any selected connector to a connector for the transported data, exactly as in Proposition 5.6; the commutation of such a sphere homeomorphism with the operator Fill is proved in detail in the naturality paragraph of Theorem 8.3. Third, the spanning tree produced inside the common component is not claimed canonical: Theorem 6.3 supplies *some* embedded tree through the free component, and different choices give different carriers with the same protected marking. Only the component Ω_0, the ports, and the survival data are determined by the packet.

Theorem 7.4 consumes completed packets: it presupposes that the filled hulls $\widehat{A}_i, \widehat{O}_\ell$ have already been computed from the typed assignment and checked against clauses (1)–(4). In practice the assignment delivers raw pieces one at a time, and both the computation of the completions and the

verification of the clauses raise questions of their own — whether the result depends on the order of processing, and whether individually verified ports must be untangled globally. The next section resolves both questions and equips the verification with persistent failure certificates.

8. Fixed-packet closure and the port audit

Theorem 7.4 consumes completed packets, and its hypotheses leave two apparent degrees of freedom in producing them. The first is a processing order. The typed assignment of Definition 7.1 delivers each packet as a finite list of compact pieces, and a completion built incrementally — adjoin a piece, fill, adjoin the next piece carrying the same label, fill again — seems to depend on the order in which the pieces arrive. The second is a global untangling. Clause (4) demands pairwise disjoint port arcs avoiding every completed hull simultaneously, whereas what an ambient proof program can realistically certify is one access arc per atom, each verified in isolation; one might fear that the individual arcs must be coordinated globally before they can serve together. This section proves that both dependences are illusory once the typed assignment is fixed. Fullification is order-independent (Lemma 8.1); individual ports localize simultaneously without any coordination (Lemma 8.2); and the resulting audit (Theorem 8.3) characterizes the geometric packet clauses by three checkable conditions whose failures are certified by witnesses that *persist* under every monotone same-label enlargement. Example 8.4, required by the structure of the theory, then shows that the audit must be run on the completions and not on the raw seeds, and Proposition 8.5 packages the whole verification into a cellwise checklist.

8.1. **Order-independent fullification.**

Lemma 8.1 (Filled-hull absorption; manuscript Lemma 10.12)**.** *For compact plane sets K, L,*

$$\mathrm{Fill}\big(\mathrm{Fill}(K) \cup L\big) \;=\; \mathrm{Fill}(K \cup L) \tag{15}$$

(equation (9.7) of [7]*). Consequently, sequentially adjoining and filling finitely many compact pieces gives the filled hull of their union, independently of processing order.*

Proof. Step 1: the operator Fill *is extensive, monotone, and idempotent.* Extensivity, $K \subset \mathrm{Fill}(K)$, is immediate from the definition (12). For monotonicity, recall from Section 7.1 that

$$\mathrm{Fill}(K) \;=\; \mathbb{C} \setminus \Omega_\infty(K),$$

where $\Omega_\infty(K)$ is the unique unbounded component of $\mathbb{C}\setminus K$, and that $\mathrm{Fill}(K)$ is compact. If $K \subset L$, then $\Omega_\infty(L)$ is a connected unbounded subset of $\mathbb{C} \setminus L \subset \mathbb{C} \setminus K$, hence is contained in the unique unbounded component of $\mathbb{C} \setminus K$:

$$\Omega_\infty(L) \;\subset\; \Omega_\infty(K).$$

Taking complements in $\mathbb{C}$ gives $\mathrm{Fill}(K) \subset \mathrm{Fill}(L)$. For idempotence, the complement of $\mathrm{Fill}(K)$ is the single set $\Omega_\infty(K)$, which is connected and unbounded; a compact set whose complement has exactly one component, an unbounded one, has no bounded complementary components, so $\mathrm{Fill}(\mathrm{Fill}(K)) = \mathrm{Fill}(K)$. We call a compact set of the form $\mathrm{Fill}(M)$, equivalently a compact set equal to its own filled hull, *full*; by idempotence, $\mathrm{Fill}(K \cup L)$ is full.

Step 2: the two inclusions. Extensivity applied to the summand K gives $K \cup L \subset \mathrm{Fill}(K) \cup L$, and monotonicity then gives the first inclusion,

$$\mathrm{Fill}(K \cup L) \;\subset\; \mathrm{Fill}\big(\mathrm{Fill}(K) \cup L\big).$$

Conversely, monotonicity gives $\mathrm{Fill}(K) \subset \mathrm{Fill}(K \cup L)$, and extensivity gives $L \subset K \cup L \subset \mathrm{Fill}(K \cup L)$; hence

$$\mathrm{Fill}(K) \cup L \;\subset\; \mathrm{Fill}(K \cup L),$$

and the set on the right is already full by Step 1. Applying Fill to this second inclusion and using monotonicity and idempotence,

$$\mathrm{Fill}\big(\mathrm{Fill}(K) \cup L\big) \;\subset\; \mathrm{Fill}\big(\mathrm{Fill}(K \cup L)\big) \;=\; \mathrm{Fill}(K \cup L).$$

The two displayed inclusions prove (15).

Step 3: finite order independence. Let $P_1, \dots, P_n$ be compact pieces and let π be any permutation of $\{1, \dots, n\}$. Define the incremental completions

$$X_1 := \mathrm{Fill}\big(P_{\pi(1)}\big), \qquad X_j := \mathrm{Fill}\big(X_{j-1} \cup P_{\pi(j)}\big) \quad (2 \le j \le n).$$

We claim $X_j = \mathrm{Fill}\big(P_{\pi(1)} \cup \cdots \cup P_{\pi(j)}\big)$ for every j, by induction. The base case is the definition of X_1. For the step, write $K := P_{\pi(1)} \cup \cdots \cup P_{\pi(j-1)}$ and $L := P_{\pi(j)}$; the inductive hypothesis and (15) give

$$X_j = \mathrm{Fill}\big(\mathrm{Fill}(K) \cup L\big) = \mathrm{Fill}(K \cup L) = \mathrm{Fill}\big(P_{\pi(1)} \cup \cdots \cup P_{\pi(j)}\big).$$

In particular $X_n = \mathrm{Fill}(P_1 \cup \cdots \cup P_n)$, which does not depend on π. The same identity shows that filling may be interleaved arbitrarily, and even that two partially completed runs of the same label may be merged: by (15) applied twice, once on each side of the union,

$$\mathrm{Fill}\big(\mathrm{Fill}(K) \cup \mathrm{Fill}(L)\big) = \mathrm{Fill}\big(K \cup \mathrm{Fill}(L)\big) = \mathrm{Fill}(K \cup L)$$

for all compact K, L. Every schedule of adjoin-and-fill operations on a fixed finite set of pieces therefore computes the same completion. □

8.2. **Seeds, completions, and persistent witnesses.** For a fixed packet label ℓ, let S_ℓ be the union of its original atom, when present, and all compact pieces assigned to that label; S_ℓ is the *raw seed* of the packet. Its unique completion is

$$K_\ell \;=\; \mathrm{Fill}(S_\ell) \tag{16}$$

(equation (9.8) of [7]); by Lemma 8.1 this is what every processing order computes. Note that when the label carries an atom, the convention makes $A_i \subset S_i \subset K_i$.

Failures of the packet clauses are recorded by explicit geometric witnesses. For $x \in K_\ell$, record its *chamber*,

$$\mathrm{ch}_\ell(x) \;:=\; \begin{cases} \mathsf{seed} & \text{if } x \in S_\ell, \\ \text{the bounded component of } \mathbb{C} \setminus S_\ell \text{ containing } x & \text{otherwise.} \end{cases}$$

A completed-packet merger then has the geometric witness

$$\big(\ell,\ \ell',\ x,\ \mathrm{ch}_\ell(x),\ \mathrm{ch}_{\ell'}(x)\big), \qquad x \in K_\ell \cap K_{\ell'} \tag{17}$$

(display (9.9) of [7]): the tuple names the two colliding labels, a common point, and, for each label, whether the point is seed material or lies in a named bounded complementary chamber of the seed. Monotonicity of Fill shows that this witness remains a merger after any same-label enlargement $S_\ell \subset S'_\ell$: enlarging seeds enlarges completions, so $x \in K_\ell \cap K_{\ell'} \subset K'_\ell \cap K'_{\ell'}$, and the same point witnesses the merger of the enlarged packets. (The recorded chambers may upgrade — a bounded chamber of S_ℓ may be absorbed into the seed of S'_ℓ, in which case the record becomes seed — but the intersection never empties.) The same argument makes a *localization escape*

$$K_\ell \cap (\mathbb{C} \setminus U) \;\neq\; \varnothing$$

persistent: a point of K_ℓ outside U lies in K'_ℓ outside U for every same-label enlargement.

The third audited datum concerns ports. Let $P_i \subset A_i$ be the union of the impressions forbidden by the protected marking at the last proof use; the protected marking forbids exactly the impressions of the protected chains, so P_i is the union of those impressions inside A_i, and we use this identification in both directions of the audit below. Define the *unprotected exposure locus*

$$\begin{aligned} \mathrm{Exp}(A_i, K_i; P_i) \;=\; \big\{\, x \in A_i \setminus P_i \;:\; & \text{there is a Jordan arc } \gamma\colon [0,1] \to U \\ & \text{with } \gamma(0) = x \text{ and } \gamma \cap K_i = \{x\} \,\big\} \end{aligned} \tag{18}$$

(display (9.10) of [7]). This is actual access data; boundary membership alone is not substituted for (18). The distinction matters precisely because no local connectivity is assumed: a point of $\partial K_i \setminus P_i$ need not be the landing point of any arc — the boundary of a full continuum can consist almost entirely of non-accessible points [9] — so (18) is an existence statement about arcs, and it is this statement, not membership in a boundary set, that every clause below reads.

8.3. **Simultaneous localization of individual ports.**

Lemma 8.2 (Individual ports localize simultaneously; manuscript Lemma 10.13)**.** *Suppose the completed atom packets $K_1, \dots, K_s \Subset U$ and obstacle packets $L_1, \dots, L_t \Subset U$ form a pairwise disjoint family and* $\mathrm{Exp}(A_i, K_i; P_i) \neq \varnothing$ *for every original atom. Then the access arcs can be chosen pairwise*

disjoint, with positive-parameter parts in

$$U \setminus \Big(\bigcup_i K_i \cup \bigcup_\ell L_\ell\Big).$$

If $K_i \subset K_i'$, then

$$\mathrm{Exp}(A_i, K_i'; P_i) \;\subset\; \mathrm{Exp}(A_i, K_i; P_i) \tag{19}$$

(display (9.11) of [7]*).*

Proof. Disjoint localization neighborhoods. The family $K_1, \dots, K_s, L_1, \dots, L_t$ consists of finitely many pairwise disjoint compacta, each compactly contained in the open set U. Their mutual distances are positive, and so is the distance from each to $\mathbb{C} \setminus U$. Choose $\varepsilon > 0$ smaller than half the minimum of all these distances, and let N_i be the open ε-neighborhood of K_i. Then $N_1, \dots, N_s$ are pairwise disjoint open sets, each compactly contained in U, each containing its K_i, and each missing every obstacle packet L_ℓ and every other atom packet K_j $(j \neq i)$.

Localizing one arc. Fix i and choose, by hypothesis, a point $x_i \in \mathrm{Exp}(A_i, K_i; P_i)$ together with a Jordan arc $\gamma_i \colon [0,1] \to U$ from (18): $\gamma_i(0) = x_i$ and $\gamma_i \cap K_i = \{x_i\}$. The landing point satisfies $x_i \in A_i \subset K_i \subset N_i$, and N_i is open, so continuity of γ_i at 0 supplies $t_i \in (0,1]$ with $\gamma_i([0,t_i]) \subset N_i$. Replace γ_i by its initial subarc $\gamma_i|_{[0,t_i]}$, reparametrized over $[0,1]$ and still denoted γ_i. Restriction can only shrink the intersection with K_i while retaining the landing point, so the restricted arc still satisfies $\gamma_i \cap K_i = \{x_i\}$, and its image now lies in N_i.

Simultaneity for free. The restricted arcs are pairwise disjoint because the N_i are, and no coordination between the choices was needed: each arc was shortened using only its own neighborhood. Each positive-parameter part satisfies

$$\gamma_i\big((0,1]\big) \;\subset\; N_i \setminus K_i,$$

which misses K_i by the singleton-intersection clause and misses every other packet because N_i does; hence it lies in $U \setminus (\bigcup_i K_i \cup \bigcup_\ell L_\ell)$, as asserted.

Monotonicity. Let $K_i \subset K_i'$ and let $x \in \mathrm{Exp}(A_i, K_i'; P_i)$, witnessed by an arc γ with $\gamma(0) = x$ and $\gamma \cap K_i' = \{x\}$. Every arc avoiding the larger packet off its landing point also avoids the smaller one: $\gamma \cap K_i \subset \gamma \cap K_i' = \{x\}$, while $x \in A_i \subset K_i$ keeps x in the intersection, so $\gamma \cap K_i = \{x\}$ and the same arc witnesses $x \in \mathrm{Exp}(A_i, K_i; P_i)$. This proves (19). In particular, an *empty* exposure locus stays empty under every same-label enlargement of the packet: port loss, once incurred, is persistent. □

8.4. **The audit theorem.** By the *geometric augmented-packet clauses* of a fixed finite typed assignment we mean clauses (1)–(4) of Definition 7.1, evaluated on the completions determined by that assignment; the typed record itself — component name, proof-use stages, attachment trace, return word, selected sheet, absorb-or-separate decision — is data carried along unchanged, not a geometric assertion to be audited.

Theorem 8.3 (Fixed-Packet Closure and Port Audit; manuscript Theorem 10.14)**.** *For a fixed finite typed assignment, the geometric augmented-packet clauses hold if and only if:*

(1) *the frozen inventory is covered and every raw seed S_ℓ is compact and connected;*
(2) *the unique completions $K_\ell = \mathrm{Fill}(S_\ell)$ are pairwise disjoint and compactly contained in U; and*
(3) $\mathrm{Exp}(A_i, K_i; P_i) \neq \varnothing$ *for every original atom: the unprotected exposure locus is nonempty.*

The completions are independent of processing order. Merger, localization, and empty-exposure defects persist under monotone same-label enlargement. The complete audit is natural under a conformal homeomorphism of Jordan localizations that transports the full packets and protected marking.

Proof. Closure. Lemma 8.1 gives unique, order-independent completions $K_\ell = \mathrm{Fill}(S_\ell)$: every schedule of adjoin-and-fill operations on the pieces of a label computes (16). If clause (1) holds, each raw seed is a compact connected set, so each completion is a full continuum, by the discussion following (12).

Equivalence. Suppose first that clauses (1)–(3) hold. Clause (1) is precisely clause (1) of Definition 7.1: the coverage assertion is common to both, and connectedness of the raw seeds is the connectedness demanded of the atom seeds $A_i \cup E_i$ and obstacle seeds O_ℓ. Writing $\widehat{A}_i := K_i$ for the atom labels and $\widehat{O}_\ell := L_\ell$ for the obstacle labels, clause (2) of the definition holds by (16), and the sets are full continua by the closure paragraph. Clause (2) of the audit makes these completions pairwise disjoint and, being compact subsets of the open set U, compactly contained in U; coverage of Q follows from clause (1), since $Q \subset \bigcup_\ell S_\ell \subset \bigcup_\ell K_\ell$. This is clause (3) of the definition. Finally, clauses (2) and (3) of the audit satisfy the hypotheses of Lemma 8.2, which produces pairwise disjoint Jordan arcs α_i with $\alpha_i(0) = x_i \in A_i \setminus P_i$, $\alpha_i \cap K_i = \{x_i\}$, and positive-parameter parts in the common complement $U \setminus (\bigcup_k K_k \cup \bigcup_\ell L_\ell)$ — exactly clause (4) of the definition, including its final requirement: P_i contains every impression forbidden by the protected marking, so $x_i \notin P_i$ is not the impression of a protected chain.

Conversely, suppose the geometric packet clauses hold. Clause (1) of the definition is clause (1) of the audit, as before. The completed packets of the definition are the unique completions (16) of the assigned seeds, and clause (3) of the definition makes them pairwise disjoint, while $\widehat{A}_i, \widehat{O}_\ell \Subset U$ gives compact containment: clause (2) of the audit. For clause (3) of the audit, fix an atom A_i. The packet's own port arc α_i satisfies $\alpha_i(0) = x_i \in A_i$, $\alpha_i \cap \widehat{A}_i = \{x_i\}$, and maps into U; since $\widehat{A}_i = K_i$, the arc witnesses $x_i \in \mathrm{Exp}(A_i, K_i; P_i)$ provided $x_i \notin P_i$, and this is the final requirement of clause (4): x_i is not the impression of a protected chain, and P_i consists of such impressions. Hence the exposure locus is nonempty, completing the equivalence.

Persistence. Monotonicity of Fill makes the merger witnesses (17) and the localization-escape witnesses persistent under monotone same-label enlargement, as recorded in Section 8.2: the witnessing point lies in the enlarged completions and outside the unchanged localization, respectively. The monotonicity clause (19) of Lemma 8.2 makes empty exposure persistent: the locus of the enlarged packet is a subset of the old locus, so it cannot become nonempty by enlargement. Thus each of the three defect types, once witnessed, is witnessed forever along every monotone same-label enlargement of the seeds.

Naturality. Let $\varphi\colon U \to V$ be an orientation-preserving conformal homeomorphism of Jordan localizations that transports the full packets and the protected marking. By the Carathéodory extension theorem for Jordan domains [9], φ extends to a homeomorphism $\overline{U} \to \overline{V}$ of the closures; restricting to the boundaries gives a homeomorphism $\partial U \to \partial V$ of Jordan curves. The complementary sets $S^2 \setminus U$ and $S^2 \setminus V$ are closed Jordan disks, and by the Schoenflies theorem [8] the boundary homeomorphism extends to a homeomorphism between them; composing with a self-homeomorphism of the target disk that is the identity on its boundary circle and moves the image of ∞ to ∞, we may arrange that the extension fixes ∞. Gluing along ∂U yields a sphere homeomorphism Φ with $\Phi|_U = \varphi$ and $\Phi(\infty) = \infty$. For any compact $K \subset \mathbb{C}$, the homeomorphism Φ carries the components of $S^2 \setminus K$ bijectively onto the components of $S^2 \setminus \Phi(K)$, and it carries the component containing ∞ to the component containing ∞; deleting ∞, it carries the unbounded complementary component of K to the unbounded complementary component of $\Phi(K)$, and the bounded components to the bounded components. Hence Φ commutes with Fill:

$$\Phi\big(\mathrm{Fill}(K)\big) \;=\; \mathrm{Fill}\big(\Phi(K)\big).$$

Consequently Φ carries raw seeds to raw seeds (piece by piece, with the same typed record), completions to completions, merger witnesses (17) to merger witnesses (the point x to $\Phi(x)$, the chambers to the corresponding chambers, seed record to seed record), and localization escapes to localization escapes: $\Phi(\mathbb{C} \setminus U) = \mathbb{C} \setminus V$, so a point of K_ℓ outside U is carried to a point of the transported completion outside V. It also transports the port data: P_i maps to the transported forbidden impression set, and $\gamma \mapsto \Phi \circ \gamma$ is a bijection between the arcs read by (18) in U and those read by the transported locus in V, because $\Phi|_U = \varphi$ is a homeomorphism onto V and $\Phi(\gamma) \cap \Phi(K_i) = \Phi(\gamma \cap K_i)$. Each audit clause therefore holds for the transported assignment if and only if it holds for the original: the complete audit is natural. □

The theorem does not construct the typed assignment. What it proves is that, once the assignment is fixed, a favorable completion order and a global untangling of already existing individual ports are *not* additional degrees of freedom: the completions are canonical by Lemma 8.1, and individual exposure already implies simultaneous localized ports by Lemma 8.2. The

one choice that matters — which inventory piece is assigned to which label — is made upstream and recorded in the typed data. The following example, the merger half of the required failure taxonomy, shows that this choice genuinely matters, and that its defects surface only after fullification.

8.5. **A post-fullification packet merger.**

Example 8.4 (Post-fullification packet merger)**.** Work in the Jordan localization $U := \{z \in \mathbb{C} : |z| < 3\}$, and consider a two-packet system whose typed assignment creates the raw seeds

$$S_1 \;:=\; \{z : |z| = 1\}, \qquad S_2 \;:=\; \left[-\tfrac{1}{4}, \tfrac{1}{4}\right] \subset \mathbb{R},$$

say with both labels typed as separated obstacles recorded for future proof use. Each seed is compact and connected, so clause (1) of Theorem 8.3 holds, and the seeds are pairwise *disjoint*: every point of S_1 has modulus 1, every point of S_2 has modulus at most $\frac{1}{4}$. Consequently every seed-level disjointness audit passes — any inspection of the raw pieces, in any processing order, at any stage of the assignment, sees pairwise disjoint data.

The completions tell a different story. The segment has connected complement, so

$$K_2 \;=\; \mathrm{Fill}(S_2) \;=\; S_2,$$

but the complement of the circle has exactly two components, the bounded open unit disk and the unbounded set $\{|z| > 1\}$, so

$$K_1 \;=\; \mathrm{Fill}(S_1) \;=\; \{z : |z| \le 1\} \;\supset\; S_2.$$

Hence

$$K_1 \cap K_2 \;\supset\; S_2 \;\neq\; \varnothing$$

(indeed $K_1 \cap K_2 = S_2$), and clause (2) of Theorem 8.3 fails: the merger appears only after fullification. The geometric witness (17) at the point $x = 0$ is

$$\big(1,\ 2,\ 0,\ \mathrm{ch}_1(0),\ \mathrm{ch}_2(0)\big), \qquad \mathrm{ch}_1(0) = \{z : |z| < 1\}, \quad \mathrm{ch}_2(0) = \mathsf{seed} :$$

the origin is not seed material of label 1 — it lies in the recorded bounded complementary chamber of S_1, the open unit disk — while for label 2 it lies in the seed itself. See Figure 6.

By monotonicity of Fill, the witness persists under *every* same-label enlargement $S_\ell \subset S'_\ell$: the enlarged completions satisfy $K_1 \subset K'_1$ and $K_2 \subset K'_2$, so $0 \in K'_1 \cap K'_2$ continues to witness a merger. The persistence clause of Theorem 8.3 is therefore sharp: the defect cannot be repaired by growing the seeds. What can remove it is only a *different typed assignment* — for instance, assigning S_2 to packet 1, together with, say, the connecting radius $[\frac{1}{4}, 1]$ so that the label-1 seed $S_1 \cup [\frac{1}{4}, 1] \cup S_2$ remains connected as clause (1) requires; the single completion is then the closed unit disk and no merger remains. The typed record of Definition 7.1 makes this an explicit *re-decision*, not a repair: the record of the segment changes its assignment, and the

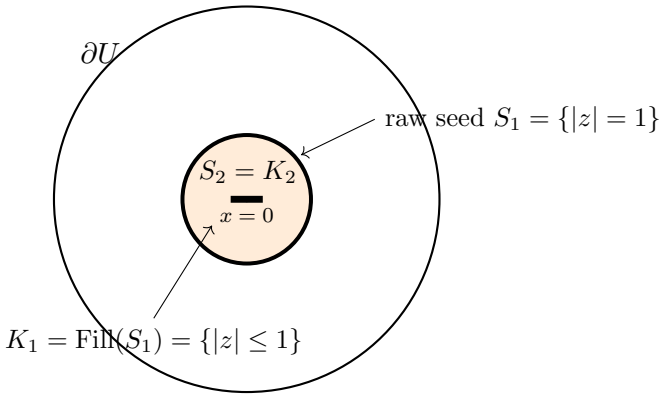


FIGURE 6. The post-fullification merger of Example 8.4 in the localization $U = \{|z| < 3\}$. The raw seeds — the circle S_1 and the segment S_2 — are disjoint, so every seed-level disjointness audit passes. But the completion $K_1 = \mathrm{Fill}(S_1)$ is the closed unit disk (shaded), which contains $S_2 = K_2$: the completed packets merge. The witness point $x = 0$ carries the record $(1, 2, 0, \mathrm{ch}_1(0), \mathrm{ch}_2(0))$ with $\mathrm{ch}_1(0)$ the bounded complementary chamber $\{|z| < 1\}$ of S_1 and $\mathrm{ch}_2(0) = \mathsf{seed}$. By monotonicity of Fill the merger persists under every same-label enlargement of the seeds; only a different typed assignment removes it.

change is visible in the audit trail. Within the fixed assignment, the failure certificate is final.

This is exactly why clause (3) of Definition 7.1 requires disjointness of the *completed* packets, not of the raw seeds. Raw-seed disjointness is not stable under fullification: an audit performed at seed level would certify the present system and hand Theorem 7.4 a family that is not pairwise disjoint, invalidating the duality argument of Lemma 7.3 at its first step. Auditing the completions, by contrast, costs nothing extra — Lemma 8.1 makes the completions canonical — and it converts every failure into a persistent witness (17).

8.6. **The cellwise certificate.** The audit clauses of Theorem 8.3 still require computing each completion and comparing every pair of completions across the localization. The final result of this section decomposes even this: it confines each packet to its own closed Jordan cell, inside which the completion is trapped automatically and the port verification becomes local to the cell. Disjointness of the completions then follows from disjointness of the cells, which is checked on N disks rather than on $\binom{N}{2}$ pairs of filled hulls.

Proposition 8.5 (Cellwise packet-and-port certificate; manuscript Proposition 10.15)**.** *Let U be a Jordan domain and let $D_1, \dots, D_N \Subset U$ be pairwise*

disjoint closed Jordan disks. Suppose every packet has a compact connected raw seed $S_\ell \Subset D_\ell$, the seeds cover the complete frozen inventory, and every original atom $A_i \subset S_i$ has an unprotected point $x_i \in A_i \setminus P_i$ and a Jordan arc $\gamma_i\colon [0,1] \to D_i$ satisfying

$$\gamma_i(0) = x_i, \qquad \gamma_i\big((0,1)\big) \subset D_i \setminus S_i, \qquad \gamma_i(1) \in \partial D_i \tag{20}$$

(condition (9.11a) of [7]). Then $K_\ell = \mathrm{Fill}(S_\ell) \Subset D_\ell$, the completed packets are pairwise disjoint, every $x_i \in \mathrm{Exp}(A_i, K_i; P_i)$, and the packet system has simultaneous ports, one common free component, and a protected regulated connector.

Proof. Step 1: the completion is trapped in its cell. Fix ℓ. By the Schoenflies theorem, $S^2 \setminus D_\ell$ is an open Jordan disk containing ∞, so its planar part $\mathbb{C} \setminus D_\ell$ — an open disk with one point removed — is connected, and it is unbounded. Since $S_\ell \Subset D_\ell$ means $S_\ell \subset \operatorname{int} D_\ell$, the exterior $\mathbb{C} \setminus D_\ell$ is a connected unbounded subset of $\mathbb{C} \setminus S_\ell$ and therefore lies in the unbounded component $\Omega_\infty(S_\ell)$. Consequently every bounded component B of $\mathbb{C} \setminus S_\ell$ is disjoint from $\mathbb{C} \setminus D_\ell$ — distinct components of $\mathbb{C} \setminus S_\ell$ are disjoint — so $B \subset D_\ell$, and $\mathrm{Fill}(S_\ell) \subset D_\ell$. The containment is compact: a point $x \in \partial D_\ell$ does not lie in $S_\ell \subset \operatorname{int} D_\ell$, and every neighborhood of x meets $\mathbb{C} \setminus D_\ell \subset \Omega_\infty(S_\ell)$; since the component of $\mathbb{C} \setminus S_\ell$ containing x is open and contains a small ball around x, that component meets $\Omega_\infty(S_\ell)$ and hence equals it, so $x \in \Omega_\infty(S_\ell)$ and $x \notin \mathrm{Fill}(S_\ell)$. Thus $K_\ell = \mathrm{Fill}(S_\ell) \subset \operatorname{int} D_\ell$, and being compact, $K_\ell \Subset D_\ell$. The cells are pairwise disjoint, so the completions are pairwise disjoint, and $D_\ell \Subset U$ gives $K_\ell \Subset U$.

Step 2: the arc condition certifies exposure. Fix i. The point $\gamma_i(1)$ lies on the Jordan curve ∂D_i, which bounds the closed Jordan disk $S^2 \setminus \operatorname{int} D_i$ containing ∞ in its interior. Straightening this disk by the Schoenflies theorem [8] and taking the preimage of a radius, we obtain a Jordan arc β from $\gamma_i(1)$ to ∞ meeting D_i only at $\gamma_i(1)$. Consider the concatenated set

$$\Gamma \;:=\; \gamma_i\big((0,1]\big) \,\cup\, \big(\beta \setminus \{\infty\}\big),$$

a connected unbounded subset of $\mathbb{C}$. It is disjoint from S_i: the piece $\gamma_i((0,1))$ lies in $D_i \setminus S_i$ by (20); the point $\gamma_i(1)$ lies on ∂D_i, which misses $S_i \subset \operatorname{int} D_i$; and $\beta \setminus \{\gamma_i(1), \infty\}$ lies in $\mathbb{C} \setminus D_i \subset \mathbb{C} \setminus S_i$. A connected unbounded subset of $\mathbb{C} \setminus S_i$ lies in the unbounded component $\Omega_\infty(S_i)$, so in particular

$$\gamma_i\big((0,1]\big) \;\subset\; \Omega_\infty(S_i) \;=\; \mathbb{C} \setminus \mathrm{Fill}(S_i),$$

that is, $\gamma_i((0,1])$ misses K_i. Since $x_i \in A_i \subset S_i \subset K_i$, this gives

$$\gamma_i \cap K_i \;=\; \{x_i\}.$$

The arc γ_i maps into $D_i \subset U$, and $x_i \in A_i \setminus P_i$ by hypothesis, so γ_i is precisely an arc of the form read by (18): every $x_i \in \mathrm{Exp}(A_i, K_i; P_i)$.

Step 3: assembly. The three audit clauses of Theorem 8.3 now hold: clause (1), because the seeds are compact, connected, and cover the complete frozen inventory by hypothesis; clause (2), by Step 1; and clause (3),

by Step 2. Lemma 8.2 turns the individual arcs into pairwise disjoint simultaneous port arcs with positive-parameter parts in the common complement of the completions, and Theorem 8.3 certifies that the geometric augmented-packet clauses of Definition 7.1 hold for this assignment. Theorem 7.4 then determines one common free component of Ω_{free} with simultaneous singleton ports to every original atom and supplies a protected regulated connector. □

This proposition is sufficient only, and we close the section by marking the boundary of what it proves, in the discipline of Section 10. The cellwise certificate solves the planar verification problem once the cells D_ℓ, the complete seeds S_ℓ, and the unprotected arcs γ_i are supplied. The actual virtual theorem contemplated by the source manuscript must still *construct* those cells, complete seeds, and unprotected arcs, and must transport them through the final Wave horizon; that construction is posed in [7] only as an open problem, it is not addressed by anything in this paper, and no clause of Theorem 8.3 or Proposition 8.5 is promoted here to a statement about virtual carriers, the Wave horizon, or Mandelbrot local connectivity.

9. A finite direct-decoration application

The certification framework of the preceding sections was assembled for a purpose, and this final section is its intended consumer. The setting is the *direct-decoration* configuration that the source manuscript [7] extracts from the decoration construction of Kahn and Lyubich [5]: a closed Jordan puzzle piece carrying two disjoint univalent pullbacks of itself, iterated to produce a rooted binary tree of finite pullback fibers accumulating on a Cantor family of end fibers. The resulting rooted-end compactum has *infinitely many* components, so the finite packet technology of Sections 7 and 8 does not apply as it stands; the first task of this section is to extend the nonseparation lemma to nested infinite packets (Lemma 9.2). The second task is geometric: the inverse branches transport a single Jordan cell around the root continuum into a *gap-cell atlas*, a family of pairwise disjoint closed Jordan cells, one around every finite fiber, each isolating its fiber from the entire remaining compactum (Theorem 9.5). The atlas is precisely the dynamical realization of the cellwise criterion of Proposition 8.5: the cells that the cellwise certificate asks the user to supply are here produced by the dynamics itself, at every finite address simultaneously. The final results assemble the package. Packet replacement (Theorem 9.6) shows that fullifying finitely many selected fibers inside their cells preserves the connected complement; the port lemma (Lemma 9.8) shows that a proper compact set of protected impressions always leaves a free port; and the protected-carrier theorem (Theorem 9.10) concludes that every finite selection of decoration addresses, decorated with actual protected prime-end chains at the root, admits a protected regulated connector in the sense of Theorem 5.4 that avoids every unselected finite fiber and every end fiber and preserves the

transported prime ends with their cyclic order. Throughout, *fullification* is the manuscript's term for the passage from a compact set to its filled hull $\mathrm{Fill}(\,\cdot\,)$ of Section 7, and *atom* retains the meaning fixed in Section 5.

Every result of this section is reproduced, with notational streamlining only, from Section 10.11 of the manuscript [7] (Lemma 10.39 through Theorem 10.45); the proofs are the manuscript's proofs, expanded for readability. One background result, the direct-decoration pullback compactification (manuscript Theorem 10.37), is *quoted without proof*, and only three of its conclusions are used. The section closes with the manuscript's own caveat on the strictly finite, direct-scale reach of the application.

9.1. **The direct-decoration setting.** Following Section 10.11 of the manuscript [7], which organizes dynamical data supplied by the decoration source of Kahn–Lyubich [5], fix the following configuration. Let f be a polynomial with

$$K(f) = J(f),$$

an assumption the manuscript makes explicitly, let $q \geq 1$, and let P be a closed Jordan puzzle piece (in the manuscript's normalization, $P = Y^{(n-1)q+1}$). Put $g = f^q$ and let

$$g\colon Q_a \longrightarrow P, \qquad a \in \mathcal{A} := \{0,1\}, \tag{21}$$

be the two *actual univalent pullbacks* of P inside P (equation (HT6) of [7]). Here P, Q_0, Q_1 are closed Jordan puzzle pieces, the pullbacks satisfy $Q_0 \cap Q_1 = \varnothing$, and each proper degree-one restriction extends to a homeomorphism of the *closed* pieces. These closed-piece assertions—not merely disjointness of interiors—are what the arguments below use, and the manuscript records the separation argument for the closed pieces, which we reproduce. Suppose Q_0 and Q_1 met at a point z. Univalence of each restriction makes z a regular point of g, so g is a local homeomorphism at z; the local inverse defined on a small Jordan half-neighborhood of $g(z)$ in P would then belong to both pullback components, contradicting their distinctness. Hence $Q_0 \cap Q_1 = \varnothing$ as closed sets, and proper degree one makes each restriction $g\colon Q_a \to P$ a homeomorphism of closed Jordan pieces. Write

$$\phi_a\colon P \longrightarrow Q_a, \qquad a \in \mathcal{A},$$

for the two inverse branches; each is a homeomorphism of closed Jordan pieces, conformal and orientation-preserving on the interior, with $g \circ \phi_a = \mathrm{id}_P$.

Let $K_0 \Subset \operatorname{int} V_0$ be the *terminal primitive root* of the source construction: a filled Julia set, in particular a nondegenerate full continuum. The source construction makes V_0, Q_0, Q_1 distinct puzzle pieces of the common depth $nq+1$. Their interiors are pairwise disjoint, and K_0 is compactly contained in the interior of the first; hence the *root-gap relation*

$$K_0 \cap (Q_0 \cup Q_1) = \varnothing \tag{22}$$

(equation (HT7) of [7]) holds, and the separation in (22) is separation of *closed* pieces.

Addresses, fibers, and the rooted-end compactum. Let $\mathcal{A}^{<\mathbb{N}}$ denote the set of finite words in the alphabet $\mathcal{A} = \{0, 1\}$, including the empty word ε, and let $\mathcal{A}^{\mathbb{N}}$ denote the set of infinite words. For a finite word $w = a_1 \cdots a_m$ put

$$\phi_w = \phi_{a_1} \circ \cdots \circ \phi_{a_m}, \qquad Q_w = \phi_w(P), \qquad K_w = \phi_w(K_0) \tag{23}$$

(equation (HT8) of [7]), with the conventions $\phi_\varepsilon = \mathrm{id}_P$, $Q_\varepsilon = P$, and $K_\varepsilon = K_0$. For a one-letter word a the cylinder Q_a of (23) agrees with the pullback of (21), so the notation is consistent; the root fiber is written K_0 throughout, and the depth-one fibers are written $\phi_0(K_0)$, $\phi_1(K_0)$ with explicit branches whenever a subscript could be misread as the root label. Each ϕ_w is a homeomorphism of P onto the closed Jordan piece Q_w, and $g^{|w|} \colon Q_w \to P$ is univalent. For $\omega \in \mathcal{A}^{\mathbb{N}}$, write $\omega|m$ for the initial subword of length m and define the *end fiber*

$$E_\omega = \bigcap_{m \geq 1} Q_{\omega|m} \tag{24}$$

(equation (HT9) of [7]).

Three combinatorial consequences of the standing data are used repeatedly, and we record them once with their one-line proofs. *Nesting under extension*: appending a symbol descends one level,

$$Q_{wa} = \phi_w\big(\phi_a(P)\big) = \phi_w(Q_a) \subset \phi_w(P) = Q_w,$$

so each E_ω in (24) is a nested intersection of nonempty compact connected sets, hence a nonempty compact connected set. *Incomparable words have disjoint cylinders*: if neither of u, v is a prefix of the other, write $u = pas$, $v = pbt$ with p the longest common prefix and $a \neq b$; then $Q_u \subset \phi_p(Q_a)$ and $Q_v \subset \phi_p(Q_b)$, and injectivity of ϕ_p together with $Q_0 \cap Q_1 = \varnothing$ gives $Q_u \cap Q_v = \varnothing$. *The root gap pulls back*: for every finite word u, applying the injective map ϕ_u to (22) gives

$$K_u \cap (Q_{u0} \cup Q_{u1}) = \varnothing, \tag{25}$$

so every finite fiber lies in the *gap* of its own cylinder, outside both children.

The object of study is the *rooted-end compactum*

$$H_\partial = \bigcup_{w \in \mathcal{A}^{<\mathbb{N}}} K_w \;\cup\; \bigcup_{\omega \in \mathcal{A}^{\mathbb{N}}} E_\omega \tag{26}$$

(equation (HT11) of [7]). We quote, as background and without proof, the part of the manuscript's compactification theorem that the present section uses.

Theorem 9.1 (Direct-decoration pullback compactification; manuscript Theorem 10.37, quoted background)**.** *The set H_∂ of* (26) *is a compact subset of $J(f)$, and its connected components are exactly the displayed finite*

pullback fibers K_w and end fibers E_ω. Moreover the transport identities

$$g(K_{aw}) = K_w, \qquad g(E_{a\omega}) = E_\omega \qquad (a \in \mathcal{A}) \tag{27}$$

hold, by actual univalent branches of f^q (equation (HT13) of [7]*).*

The proof—together with the compactness of the address space, the upper semicontinuity of the address realization, and the careful distinction between the geometric parent map (deleting the *last* symbol), the dynamical pre-root shift (deleting the *first* symbol), and the separate root arrow τ_{root}—is given in [7, Theorem 10.37] and is not reproduced here; the present section uses only the three properties displayed above. We note for orientation that the two deletion directions are genuinely different maps: nesting under extension is geometric containment, while (27) is dynamical transport with transition time q, and the manuscript keeps them distinct.

9.2. **Fullness and nonseparation.** The packet lemma of Section 7 (Lemma 7.3) handles finitely many pairwise disjoint full continua. The compactum (26) has infinitely many components, so an extension is needed, and the correct one is monotone: an intersection of a decreasing sequence of finite packets does not separate the sphere. This is the single new topological ingredient of the section.

Lemma 9.2 (Nested finite-full-packet nonseparation; manuscript Lemma 10.39)**.** *Let $F_0 \supset F_1 \supset \cdots$ be compact subsets of the sphere $\widehat{\mathbb{C}}$, each a finite union of pairwise disjoint full continua. Then $\bigcap_m F_m$ has connected complement in $\widehat{\mathbb{C}}$. If, moreover, $U \subset \widehat{\mathbb{C}}$ is a Jordan domain with $F_m \Subset U$ for every m, so that each $F_m \cup (\widehat{\mathbb{C}} \setminus U)$ is again a finite union of pairwise disjoint full continua, then*

$$U \setminus \bigcap_m F_m$$

is connected.

Proof. For the first assertion, fix m and let $K_1, \dots, K_N$ be the components of F_m: finitely many pairwise disjoint full continua. Lemma 7.3 (manuscript Lemma 10.10) shows that $\widehat{\mathbb{C}} \setminus F_m$ is connected. It is also nonempty: the finitely many components of F_m are pairwise disjoint proper closed subsets of $\widehat{\mathbb{C}}$, and a connected space is not the union of two or more disjoint nonempty closed sets, so $F_m \neq \widehat{\mathbb{C}}$. Fix a point $p \in \widehat{\mathbb{C}} \setminus F_0$. Because the sets F_m decrease, the open connected complements

$$\widehat{\mathbb{C}} \setminus F_0 \subset \widehat{\mathbb{C}} \setminus F_1 \subset \cdots$$

increase, and every one of them contains the common point p. A union of connected sets sharing a point is connected, and

$$\bigcup_m (\widehat{\mathbb{C}} \setminus F_m) = \widehat{\mathbb{C}} \setminus \bigcap_m F_m,$$

so the complement of the intersection is connected, as claimed.

For the localized assertion, adjoin the exterior continuum. Since U is a Jordan domain, the Schoenflies theorem [8] makes $\widehat{\mathbb{C}} \setminus U$ a closed Jordan disk: a continuum whose complement is the connected set U, hence a full continuum, and it is disjoint from F_m because $F_m \Subset U$. Thus each

$$F'_m = F_m \cup (\widehat{\mathbb{C}} \setminus U)$$

is again a compact set that is a finite union of pairwise disjoint full continua, and the sequence $(F'_m)_m$ is decreasing. The first assertion applied to $(F'_m)_m$ makes

$$\widehat{\mathbb{C}} \setminus \bigcap_m F'_m = \widehat{\mathbb{C}} \setminus \Big(\big(\textstyle\bigcap_m F_m\big) \cup (\widehat{\mathbb{C}} \setminus U) \Big) = U \setminus \textstyle\bigcap_m F_m$$

connected. □

With the monotone packet lemma available, fullness of every fiber and nonseparation of the whole compactum follow from a single decreasing approximation, which will be reused twice more in this section.

Theorem 9.3 (Direct-decoration full fibers and nonseparation; manuscript Theorem 10.40)**.** *Every finite fiber K_w and every end fiber E_ω in Theorem 9.1 is full. Moreover $\widehat{\mathbb{C}} \setminus H_\partial$ is connected.*

Proof. Finite fibers. The root K_0 is full by the standing hypotheses of Section 9.1: it is a filled Julia set, hence a nondegenerate full continuum. Fix a finite word w. The map $\phi_w \colon P \to Q_w$ is a homeomorphism of closed Jordan disks. Its boundary restriction $\partial P \to \partial Q_w$ is a homeomorphism of Jordan curves, which by the Schoenflies theorem [8] extends to a homeomorphism of the complementary closed disks $\widehat{\mathbb{C}} \setminus \operatorname{int} P \to \widehat{\mathbb{C}} \setminus \operatorname{int} Q_w$; gluing the two extensions along the boundary curves produces a sphere homeomorphism $\Phi_w \colon \widehat{\mathbb{C}} \to \widehat{\mathbb{C}}$ with $\Phi_w|_P = \phi_w$, which may be taken orientation-preserving because ϕ_w is orientation-preserving on $\operatorname{int} P$. A sphere homeomorphism carries complements to complements and connected sets to connected sets, so it preserves fullness: $K_w = \Phi_w(K_0)$ is full.

End fibers. Fix $\omega \in \mathcal{A}^{\mathbb{N}}$. By nesting under extension, E_ω is a nested intersection of the nonempty compact connected sets $Q_{\omega|m}$, hence a nonempty compact connected set. Its complement is the increasing union

$$\widehat{\mathbb{C}} \setminus E_\omega = \bigcup_{m \geq 1} \big(\widehat{\mathbb{C}} \setminus Q_{\omega|m}\big)$$

of connected open sets—each $Q_{\omega|m}$ is a closed Jordan disk—all containing ∞. A union of connected sets sharing a point is connected, so E_ω is full.

The decreasing approximants. For $m \geq 0$ define

$$H_m = \bigcup_{|u|<m} K_u \;\cup \bigcup_{|w|=m} Q_w \tag{28}$$

(display (DG1) of [7]); thus $H_0 = P$ and $H_1 = K_0 \cup Q_0 \cup Q_1$. We check that each H_m is a finite union of pairwise disjoint full continua. The terms are full: the fibers K_u by the first step, the cylinders Q_w because they are closed

Jordan disks. Pairwise disjointness splits into three cases, each settled by the combinatorial facts of Section 9.1. Equal-depth cylinders $Q_w, Q_{w'}$ with $w \neq w'$, $|w| = |w'| = m$ belong to incomparable words and are disjoint. Two distinct fibers $K_u, K_{u'}$ with $|u|, |u'| < m$: if u and u' are incomparable their cylinders are disjoint, and $K_u \subset Q_u$, $K_{u'} \subset Q_{u'}$; if u is a strict prefix of u', say $u' = uat$ with $a \in \mathcal{A}$, then $K_{u'} \subset Q_{u'} \subset Q_{ua}$, while the pulled-back root gap (25) keeps K_u outside the child Q_{ua}. A fiber K_u with $|u| < m$ against a cylinder Q_w with $|w| = m$: the words are distinct since $|u| < |w|$; if incomparable, the cylinders containing them are disjoint; if u is a strict prefix of w, say $w = uat'$, then $Q_w \subset Q_{ua}$ and (25) again separates. (The case w a prefix of u cannot occur, as $|w| > |u|$.)

The approximants decrease to H_∂. $H_{m+1} \subset H_m$: a fiber term K_u of H_{m+1} with $|u| < m$ is a term of H_m, and one with $|u| = m$ satisfies $K_u \subset Q_u$, a term of H_m; a cylinder term Q_w of H_{m+1} with $|w| = m + 1$ satisfies $Q_w \subset Q_{w|m}$, a term of H_m. Next, $H_\partial \subset \bigcap_m H_m$: for each m, a fiber K_w with $|w| < m$ is a term of H_m; one with $|w| \geq m$ lies in $Q_w \subset Q_{w|m}$; and every end fiber satisfies $E_\omega \subset Q_{\omega|m}$. Conversely let $x \in \bigcap_m H_m$ and suppose x lies in no finite fiber. Then for every m the membership $x \in H_m$ forces $x \in Q_{w_m}$ for some word w_m of length m. Equal-depth cylinders are pairwise disjoint, and $x \in Q_{w_{m+1}} \subset Q_{w_{m+1}|m}$, so $w_{m+1}|m = w_m$ for every m: the words are coherent truncations of one unique infinite address ω with $\omega|m = w_m$, and $x \in \bigcap_m Q_{\omega|m} = E_\omega$. Outside the explicit finite fibers, a point of every approximant thus determines one unique compatible infinite cylinder address, and

$$\bigcap_m H_m = H_\partial.$$

Conclusion. The sets (28) form a decreasing sequence of compact sets, each a finite union of pairwise disjoint full continua, with intersection H_∂. Lemma 9.2 shows that $\widehat{\mathbb{C}} \setminus H_\partial$ is connected. □

9.3. **The gap-cell atlas.** The next step localizes each finite fiber inside a closed Jordan cell that isolates it from everything else in the compactum. The cell is produced once, at the root, by a classical uniformization argument, and then transported everywhere by the inverse branches.

Lemma 9.4 (Full-continuum Jordan neighborhood; manuscript Lemma 10.41)**.** *If a nondegenerate full continuum K is compactly contained in an open set G, there is a closed Jordan disk D such that*

$$K \subset \operatorname{int} D \subset D \Subset G \tag{29}$$

(display (DG2) of [7]*).*

Proof. The complement $\widehat{\mathbb{C}} \setminus K$ is a simply connected domain containing ∞, and K is nondegenerate, so the domain is conformally a round-disk exterior: there is a conformal homeomorphism

$$\psi \colon \{|z| > 1\} \longrightarrow \widehat{\mathbb{C}} \setminus K$$

with $\psi(\infty)=\infty$ [9]. For $R>1$ the image $\psi(\{|z|=R\})$ of the round circle is an analytic Jordan curve, and we let D_R be its *filled interior*, the closure of the bounded complementary component:

$$D_R=\widehat{\mathbb{C}}\setminus\psi\big(\{|z|>R\}\big).$$

Each D_R is a closed Jordan disk containing K in its interior, and the family decreases as $R\downarrow 1$: for $1<R'<R$ the annulus image $\psi(\{R'<|z|\le R\})$ lies in $D_R\setminus D_{R'}$, giving $D_{R'}\subset\operatorname{int}D_R$. Moreover $\bigcap_{R>1}D_R=K$: a point of $\widehat{\mathbb{C}}\setminus K$ is $\psi(z)$ for a unique $|z|>1$ and lies outside D_R for every $R<|z|$.

Suppose no D_R were compactly contained in G. Then for every $R>1$ the compact set $D_R\cap(\widehat{\mathbb{C}}\setminus G)$ would be nonempty; these compacta decrease with R, so their total intersection

$$\bigcap_{R>1}\big(D_R\cap(\widehat{\mathbb{C}}\setminus G)\big)=K\cap(\widehat{\mathbb{C}}\setminus G)$$

would be nonempty, contradicting $K\Subset G$. Hence some $D=D_R$ satisfies (29). □

Now let G_0 be the component of $\operatorname{int}P\setminus(Q_0\cup Q_1)$ containing K_0. The root-gap relation (22), together with $K_0\Subset\operatorname{int}V_0\subset\operatorname{int}P$, makes K_0 a compact connected subset of the open set $\operatorname{int}P\setminus(Q_0\cup Q_1)$ with positive distance to its complement, so $K_0\Subset G_0$. Choose a closed Jordan disk D for $K=K_0$ and $G=G_0$ by Lemma 9.4, and transport it through every inverse branch:

$$D_w=\phi_w(D),\qquad w\in\mathcal{A}^{<\mathbb{N}},$$

with $D_\varepsilon=D$. These are the *gap cells*, and the family is the *gap-cell atlas* of the decoration data; see Figure 7.

Theorem 9.5 (Direct-decoration gap-cell atlas; manuscript Theorem 10.42)**.** *The closed Jordan cells D_w, $w\in\mathcal{A}^{<\mathbb{N}}$, are pairwise disjoint, including for ancestor–descendant pairs of words, and satisfy*

$$K_w\subset\operatorname{int}D_w,\qquad D_w\Subset\operatorname{int}Q_w\setminus(Q_{w0}\cup Q_{w1}),\tag{30}$$

$$D_w\cap H_\partial=K_w,\qquad g(D_{aw})=D_w\tag{31}$$

(displays (DG3) and (DG4) of [7]*). If $S_w\Subset\operatorname{int}D_w$ is compact and connected, then*

$$\operatorname{Fill}(S_w)\Subset\operatorname{int}D_w\tag{32}$$

(display (DG5) of [7]*).*

Proof. Position (30). At the root, $K_0\subset\operatorname{int}D$ and $D\Subset G_0\subset\operatorname{int}P\setminus(Q_0\cup Q_1)$ by construction. The map ϕ_w is a homeomorphism of the closed Jordan piece P onto the closed Jordan piece Q_w carrying $\operatorname{int}P$ onto $\operatorname{int}Q_w$ and carrying Q_a onto Q_{wa} for $a\in\mathcal{A}$. Applying ϕ_w to the root inclusions gives $K_w=\phi_w(K_0)\subset\phi_w(\operatorname{int}D)=\operatorname{int}D_w$ (the sphere extension Φ_w of the proof of Theorem 9.3 carries the open disk $\operatorname{int}D$ onto the open disk $\operatorname{int}D_w$) and

$$D_w=\phi_w(D)\Subset\phi_w\big(\operatorname{int}P\setminus(Q_0\cup Q_1)\big)=\operatorname{int}Q_w\setminus(Q_{w0}\cup Q_{w1}).$$

Pairwise disjointness. Let $u \neq v$ be finite words. If u and v are incomparable, their cylinders are disjoint and $D_u \subset Q_u$, $D_v \subset Q_v$. If v is a strict descendant of u, write $v = uat$ with $a \in \mathcal{A}$ and t a (possibly empty) word. Then

$$D_v \subset Q_v \subset Q_{ua},$$

the child of Q_u with label a, while by (30) the cell D_u lies outside both children $Q_{u0} \cup Q_{u1}$ of Q_u. Hence $D_u \cap D_v = \varnothing$. This dichotomy—incomparable words are separated by their cylinders, comparable words by the gap position of the ancestor cell—proves disjointness for every pair.

Isolation: the first identity of (31). The inclusion $K_w \subset D_w \cap H_\partial$ holds by (30). For the converse we show that every component of H_∂ other than K_w misses D_w; by Theorem 9.1 those components are the finite fibers K_u, $u \neq w$, and the end fibers E_ω. The same dichotomy applies. If u and w are incomparable, then $K_u \subset Q_u$ and $D_w \subset Q_w$ lie in disjoint cylinders. If u is a strict prefix of w, say $w = uat'$, then $D_w \subset Q_w \subset Q_{ua}$ while (25) keeps K_u outside Q_{ua}. If w is a strict prefix of u, say $u = wat$, then $K_u \subset Q_u \subset Q_{wa}$, while D_w misses both children of Q_w by (30). For an end fiber E_ω: if w is a prefix of ω, then $E_\omega \subset Q_{\omega|(|w|+1)} = Q_{wa}$ for the symbol $a = \omega_{|w|+1}$, a child of Q_w, which D_w misses; if w is not a prefix of ω, let $m \leq |w|$ be the first index at which ω differs from w; then $\omega|m$ and w are incomparable, so $E_\omega \subset Q_{\omega|m}$ is disjoint from $Q_w \supset D_w$. Hence $D_w \cap H_\partial = K_w$.

Transport: the second identity of (31). For $a \in \mathcal{A}$ and any finite word w,

$$D_{aw} = \phi_{aw}(D) = \phi_a\big(\phi_w(D)\big) = \phi_a(D_w),$$

so $g(D_{aw}) = g\big(\phi_a(D_w)\big) = D_w$ by $g \circ \phi_a = \mathrm{id}_P$.

Fill containment (32). Let $S_w \Subset \operatorname{int} D_w$ be compact and connected. Choose a Jordan curve $C' \subset \operatorname{int} D_w$ close enough to ∂D_w that the closed annulus A between C' and ∂D_w (including both boundary curves) misses S_w; this is possible because $S_w \Subset \operatorname{int} D_w$ has positive distance to ∂D_w. Write D'_w for the closed Jordan disk bounded by C', so that $D'_w \subset \operatorname{int} D_w$ and $\operatorname{int} D_w \setminus A = \operatorname{int} D'_w$. The set

$$A \cup \big(\mathbb{C} \setminus \operatorname{int} D_w\big)$$

is connected (the collar is attached to ∂D_w), unbounded, and disjoint from S_w, so it lies in the unbounded component of $\mathbb{C} \setminus S_w$. Every bounded component B of $\mathbb{C} \setminus S_w$ therefore misses the collar and the exterior, whence $B \subset \operatorname{int} D_w \setminus A = \operatorname{int} D'_w$; and $S_w \subset \operatorname{int} D'_w$ for the same reason. Thus $\mathrm{Fill}(S_w)$, which is S_w together with the bounded complementary components, is contained in the compact set $D'_w \subset \operatorname{int} D_w$; in particular $\mathrm{Fill}(S_w) \Subset \operatorname{int} D_w$. □

The atlas should be compared clause for clause with Proposition 8.5: the cells D_w are the pairwise disjoint closed Jordan disks demanded there, the fibers K_w are the raw seeds, the fill containment (32) is the localization clause, and the exit arcs will be supplied by Lemma 9.8 below. What the dynamics adds to the static cellwise picture is the isolation identity $D_w \cap$

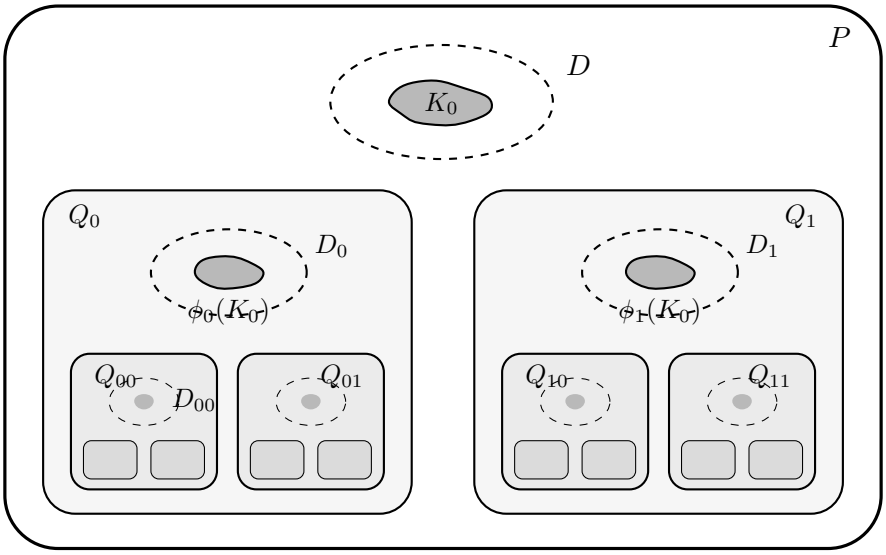


FIGURE 7. The direct-decoration gap-cell atlas through depth two. The closed Jordan piece P contains the two disjoint closed univalent pullbacks Q_0 and Q_1, each of which contains its own next-level cylinders Q_{a0}, Q_{a1}, and so on. The root continuum K_0 (blob) lies in the root gap $P \setminus (Q_0 \cup Q_1)$, and the dashed closed Jordan cell D supplied by Lemma 9.4 isolates it: $K_0 \subset \operatorname{int} D \subset D \Subset \operatorname{int} P \setminus (Q_0 \cup Q_1)$. The inverse branches transport the picture into every cylinder: the fiber $K_w = \phi_w(K_0)$ lies in the gap of its own cylinder Q_w, surrounded by the dashed cell $D_w = \phi_w(D) \Subset \operatorname{int} Q_w \setminus (Q_{w0} \cup Q_{w1})$; the depth-one and depth-two cells D_0, D_1, D_{00}, ... are shown. The cells are pairwise disjoint, including for ancestor–descendant words, and each one isolates its fiber from the entire rooted-end compactum: $D_w \cap H_\partial = K_w$ (Theorem 9.5), so every unselected finite fiber and every end fiber misses D_w. Dynamics transports the atlas by $g(D_{aw}) = D_w$.

$H_\partial = K_w$, which controls not only the finitely many packets that a certificate names but *all* fibers of the infinite compactum at once, and the transport identity $g(D_{aw}) = D_w$, which makes the atlas equivariant.

9.4. **Packet replacement.** An augmented packet replaces an atom by the filled hull of the atom together with its protected collars. In the decoration setting the atoms are finite fibers, and the replacement must not disturb the connected complement guaranteed by Theorem 9.3—otherwise the free-component criterion of Theorem 6.3 could fail after fullification for reasons

invisible in the certificate. The next theorem rules this out: fullifying finitely many selected fibers inside their cells leaves the complement connected.

Theorem 9.6 (Finite direct-decoration packet replacement; manuscript Theorem 10.43)**.** *Let $F \subset \mathcal{A}^{<\mathbb{N}}$ be finite and let*

$$K_w \subset L_w \Subset \operatorname{int} D_w, \qquad w \in F, \tag{33}$$

with every L_w full (display (DG6) of [7]). Then replacing the selected fibers K_w by L_w in H_∂, that is, forming

$$H' = \Big(H_\partial \setminus \bigcup_{w\in F} K_w\Big) \cup \bigcup_{w\in F} L_w,$$

gives a compact set with connected sphere complement.

Proof. The replacement is compact. The isolation identity (31) makes each selected K_w clopen in H_∂: it is relatively closed because $K_w = D_w \cap H_\partial$ with D_w closed, and relatively open because $K_w = \operatorname{int} D_w \cap H_\partial$ by (30) and the same identity. Hence $H_\partial \setminus \bigcup_{w\in F} K_w$ is a relatively closed subset of the compact set H_∂, so it is compact, and its union H' with the finitely many compacta L_w is compact.

Cells meet deep approximants only in their fibers. Fix $w \in F$ and let $m > |w|$. We claim

$$D_w \cap H_m = K_w, \tag{34}$$

where H_m is the approximant (28). The inclusion $\supset$ holds because K_w is a term of H_m (as $|w| < m$) and $K_w \subset D_w$. For $\subset$, run the dichotomy of the proof of Theorem 9.5 over the terms of H_m. A fiber term K_u, $|u| < m$, $u \neq w$: incomparable words have disjoint cylinders; if u is a strict prefix of w then $D_w \subset Q_{ua}$ for the appropriate child while (25) keeps K_u out of both children; if w is a strict prefix of u then $K_u \subset Q_{wa}$ for some $a \in \mathcal{A}$, while D_w misses both children of Q_w by (30). A cylinder term Q_v, $|v| = m > |w|$: if v and w are incomparable their cylinders are disjoint; otherwise w is a strict prefix of v, so $Q_v \subset Q_{wa}$ for some a, which D_w misses. This proves (34).

Modified approximants. Put $M = \max_{w\in F} |w|$ and, for $m > M$, make the same replacements in the approximants:

$$H'_m = \Big(H_m \setminus \bigcup_{w\in F} K_w\Big) \cup \bigcup_{w\in F} L_w.$$

Each H'_m is again a finite union of pairwise disjoint full continua. Indeed, the terms of H_m other than the selected fibers are unchanged, and each selected term K_w is replaced by the full continuum L_w; disjointness survives because, by (34),

$$L_w \subset \operatorname{int} D_w \quad \text{and} \quad D_w \cap H_m = K_w,$$

so L_w meets no unselected term of H_m, and distinct selected replacements $L_w, L_{w'}$ lie in the disjoint cells $D_w, D_{w'}$ of Theorem 9.5. The modified approximants decrease: the unreplaced parts decrease as in the proof of Theorem 9.3, and the replaced terms L_w are constant in m. Their intersection is the modified rooted-end compactum:

$$\bigcap_{m>M} H'_m = \Big(\bigcap_{m>M} H_m \setminus \bigcup_{w\in F} K_w \Big) \cup \bigcup_{w\in F} L_w = H',$$

where the first equality uses (34) twice: a point of D_w lies in H'_m exactly when it lies in L_w, and a point outside every selected cell lies in H'_m exactly when it lies in H_m.

Conclusion. The sets H'_m, $m > M$, form a decreasing sequence of compact sets, each a finite union of pairwise disjoint full continua, with intersection H'. Lemma 9.2 shows that $\widehat{\mathbb{C}} \setminus H'$ is connected. □

Remark 9.7. The proof yields a localized supplement that Theorem 9.10 uses. Fix any Jordan domain U with $P \Subset U$—for instance a round disk containing P. Every approximant, modified or not, lies in P, hence is compactly contained in U, so the localized clause of Lemma 9.2 applies to the sequence $(H'_m)_{m>M}$ and shows that $U \setminus H'$ is connected: the replaced compactum has one *common* complementary component inside the localization, not merely a connected complement on the sphere.

9.5. **Ports at proper protected impressions.** The free-component criterion requires each atom to retain a free singleton port not equal to a protected impression. In the decoration setting the atom is a fiber K_w, the protected data are finitely many prime-end crosscut chains, and the port must avoid not only the chains' collars but the entire fullification of the fiber together with those collars. The next lemma produces such a port from a purely metric selection, under the sole hypothesis that the protected impressions do not exhaust the boundary.

Lemma 9.8 (Proper protected impressions leave a port; manuscript Lemma 10.44)**.** *Let $K \Subset \operatorname{int} D$ be a nondegenerate full continuum in a closed Jordan disk D, and let $P_{\mathrm{prot}} \subsetneq \partial K$ be compact. Then there are a point $x \in \partial K \setminus P_{\mathrm{prot}}$ and a Jordan arc α from x to ∂D, otherwise in $\operatorname{int} D \setminus K$. Moreover, finitely many protected crosscut-chain collars whose impressions have union contained in P_{prot} can be truncated so that, for the connected union S of K with the truncated closed collars,*

$$\alpha \cap \operatorname{Fill}(S) = \{x\} \tag{35}$$

(display (DG7) of [7]*).*

Proof. Metric selection of the port point. Since P_{prot} is a proper compact subset of ∂K, choose $y \in \partial K \setminus P_{\mathrm{prot}}$ and put $\delta = \operatorname{dist}(y, P_{\mathrm{prot}}) > 0$ (with $\delta = +\infty$ if $P_{\mathrm{prot}} = \varnothing$). Because y is a boundary point of K and $K \Subset \operatorname{int} D$, every small ball around y contains points of $\operatorname{int} D \setminus K$; choose

$$z \in \operatorname{int} D \setminus K, \qquad |z - y| < \delta/4.$$

Let $x \in K$ minimize the distance to z; the minimum is positive since $z \notin K$ and K is compact. Every point p of the half-open segment $(x, z]$ satisfies $|p - z| < |x - z| = \operatorname{dist}(z, K)$, so $(x, z]$ misses K; in particular x is a limit of points of the complement, so $x \in \partial K$. The triangle inequality and the minimizing property (applied against the competitor $y \in K$) give

$$|x - y| \le |x - z| + |z - y| \le 2|z - y| < \delta/2,$$

hence $\operatorname{dist}(x, P_{\mathrm{prot}}) \ge \delta - |x - y| > \delta/2$ and $x \notin P_{\mathrm{prot}}$.

Construction of the arc. The complement $\mathbb{C}\setminus K$ of the full continuum is an open connected subset of the plane, hence arcwise connected [8]. Join z by an arc in $\mathbb{C} \setminus K$ to some point outside D, and concatenate with the segment from x to z. The concatenation is a path from x leaving D, contained in $(\mathbb{C}\setminus K)\cup\{x\}$ and meeting K only at its initial point; extract from it a Jordan arc from x to the far endpoint, which again meets K only at x. Since the arc starts in $\operatorname{int} D$ and ends outside D, the parameter set of points on ∂D is nonempty and closed; truncating at the *first hit* on ∂D yields a Jordan arc α from x to a point of ∂D with $\alpha \setminus \{x, \alpha(1)\} \subset \operatorname{int} D$ and $\alpha \cap K = \{x\}$. This proves the first assertion.

Truncation of the collars. Let the finitely many protected chains be $(\gamma_{j,n}, D_{j,n})_{n\ge1}$, prime-end crosscut chains of $\widehat{\mathbb{C}} \setminus K$ whose impressions $I_j = \bigcap_n \overline{D_{j,n}}$ satisfy $\bigcup_j I_j \subset P_{\mathrm{prot}}$. Since $\alpha\cap K = \{x\}$ and $x \notin P_{\mathrm{prot}}$, the compact sets α and P_{prot} are disjoint and have positive distance $2\rho > 0$. Shrinking ρ if necessary, choose for each j an open neighborhood W_j of I_j contained in the ρ-neighborhood of P_{prot}, in $\operatorname{int} D$ (possible since $I_j \subset \partial K \subset K \Subset \operatorname{int} D$), and in the prescribed neighborhood of I_j if one was given. The closures $\overline{D_{j,n}}$ are nested compact sets with intersection I_j, so for every j there is n_j with $\overline{D_{j,n_j}} \subset W_j$: otherwise compactness would place a point of $\bigcap_n \overline{D_{j,n}} = I_j$ outside W_j. Truncate the j-th chain at depth n_j and set

$$S = K \cup \bigcup_j \overline{D_{j,n_j}}.$$

Each closed collar meets K in its nonempty impression $I_j \subset \overline{D_{j,n_j}} \cap K$, so S is a compact connected set, and $S \Subset \operatorname{int} D$ by the choice of the W_j. By construction the truncated collars lie in the ρ-neighborhood of P_{prot} and therefore miss α; hence

$$\alpha \cap S = \alpha \cap K = \{x\}.$$

Extension to infinity and (35). Extend α beyond its endpoint on ∂D by an arc to ∞ in $\mathbb{C} \setminus \operatorname{int} D$ (for instance, along the image of a radial ray of the exterior of D). The set

$$\beta = \alpha\big((0, 1]\big) \cup (\text{the extension})$$

is connected and unbounded and misses S: the extension misses $S \Subset \operatorname{int} D$, and $\alpha((0, 1])$ misses S by the previous step. Hence β lies in the unbounded component of $\mathbb{C}\setminus S$, and no point of β belongs to $\operatorname{Fill}(S)$, which is S together

with the *bounded* complementary components. Every point of α except x lies in β, and $x \in K \subset S \subset \mathrm{Fill}(S)$; therefore $\alpha \cap \mathrm{Fill}(S) = \{x\}$, which is (35). □

Remark 9.9. The proof gives slightly more than the statement: the truncation can confine the truncated closed collars to any prescribed open neighborhoods of their impressions, since nested compact crosscut-chain closures have intersection equal to their impressions and sufficiently deep tails therefore lie in any such neighborhoods. This supplement is established by the proof above, following the manuscript's own proof of Lemma 10.44; the manuscript's printed statement ends at (35), and we use the supplement below with this explicit attribution.

9.6. **The protected-carrier theorem.** The pieces are now assembled. The theorem below is the finite, checkable statement toward which the whole paper has been building: at any fixed decoration depth, any finite selection of fibers, decorated by actual protected prime-end chains at the root whose impressions do not exhaust the root boundary, is joined by a protected regulated connector that respects every piece of the frozen inventory—including the infinitely many unselected fibers and end fibers.

Theorem 9.10 (Finite direct-decoration protected carrier; manuscript Theorem 10.45)**.** *Fix a finite set $F \subset \mathcal{A}^{<\mathbb{N}}$ of direct-decoration addresses, and fix finitely many actual protected prime-end crosscut chains of $\widehat{\mathbb{C}} \setminus K_0$ at the root whose total impression is a proper compact subset of ∂K_0. Then sufficiently deep collar tails of the protected chains, and the free port supplied by Lemma 9.8, can be transported through the inverse branches ϕ_w, $w \in F$, so that the resulting packet fullifications remain in the pairwise disjoint cells D_w, avoid every unselected finite fiber and every end fiber, and admit a protected regulated connector preserving the transported prime ends and their cyclic order.*

Proof. Step 1: root data. Discarding duplicates, assume the given chains define pairwise distinct prime ends: two equivalent chains protect the same access, so nothing is lost. Apply Lemma 9.8 with $K = K_0$, with D the root cell of the atlas construction (so $K_0 \Subset \operatorname{int} D$ by (30) at the empty word), and with P_{prot} the union of the impressions of the given protected chains—a finite union of compacta, compact, and a proper subset of ∂K_0 by hypothesis. The lemma yields a port point $x \in \partial K_0 \setminus P_{\mathrm{prot}}$, a Jordan arc α from x to ∂D meeting K_0 only at x, and truncations of the protected chains, with closed collar tails in prescribed neighborhoods of their impressions (Remark 9.9), such that the connected union S of K_0 with the truncated closed collars satisfies $S \Subset \operatorname{int} D$ and $\alpha \cap \mathrm{Fill}(S) = \{x\}$. Because the chains define pairwise distinct prime ends, their deep selected domains are eventually pairwise disjoint, so after a further truncation—which keeps every conclusion, the deeper collars being nested in the previous ones—the closed collar tails may in addition be taken pairwise disjoint, exactly as

in the truncation normalization of Section 5. By the fill containment (32) applied at the empty word,

$$\mathrm{Fill}(S) \Subset \operatorname{int} D.$$

Step 2: transport through the inverse branches. For $w \in F$ let $\Phi_w \colon \widehat{\mathbb{C}} \to \widehat{\mathbb{C}}$ be the orientation-preserving sphere extension of ϕ_w constructed in the proof of Theorem 9.3. Define the transported data at address w:

$$S_w = \Phi_w(S), \qquad \alpha_w = \Phi_w(\alpha), \qquad x_w = \phi_w(x),$$

together with the transported protected chains $\Phi_w(\gamma_{j,n}, D_{j,n})_{n \geq n_j}$. As in the proof of Proposition 5.6, an orientation-preserving sphere homeomorphism carries crosscuts of $\widehat{\mathbb{C}} \setminus K_0$ to crosscuts of $\widehat{\mathbb{C}} \setminus K_w$, null chains to null chains, selected sides to selected sides, and preserves the cyclic order of prime ends; so the transported chains are prime-end crosscut chains of $\widehat{\mathbb{C}} \setminus K_w$, pairwise distinct, with impressions $\Phi_w(I_j) \subset \partial K_w$, and S_w is the connected union of K_w with their truncated closed collars. Fullification commutes with the transport: Φ_w maps $\widehat{\mathbb{C}} \setminus P$ onto $\widehat{\mathbb{C}} \setminus Q_w$, hence maps the complementary component of S containing $\widehat{\mathbb{C}} \setminus P$—the unbounded one—onto the complementary component of S_w containing $\widehat{\mathbb{C}} \setminus Q_w$—again the unbounded one—and maps the remaining (bounded) complementary components bijectively onto the remaining ones; therefore

$$L_w := \mathrm{Fill}(S_w) = \Phi_w\big(\mathrm{Fill}(S)\big).$$

Applying Φ_w to the Step-1 inclusions and to (35) gives

$$K_w \subset L_w \Subset \operatorname{int} D_w, \qquad \alpha_w \cap L_w = \{x_w\},$$

with α_w a Jordan arc from x_w to ∂D_w, otherwise in $\operatorname{int} D_w \setminus K_w$. Each L_w is full: the filled hull of a continuum is a full continuum.

Step 3: localization and isolation. Theorem 9.5 now localizes and isolates the packets. The cells D_w, $w \in F$, are pairwise disjoint, so the packet fullifications $L_w \subset D_w$ are pairwise disjoint and each port arc $\alpha_w \subset D_w$ meets every other packet's cell not at all. By the isolation identity (31), every component of H_∂ other than K_w misses D_w; hence every L_w, and every α_w, avoids every unselected finite fiber and every end fiber.

Step 4: one common free component after fullification. The data (33) of Theorem 9.6 hold for the family $(L_w)_{w \in F}$, so the replaced compactum

$$H' = \Big(H_\partial \setminus \bigcup_{w \in F} K_w\Big) \cup \bigcup_{w \in F} L_w$$

is compact with connected sphere complement; and by Remark 9.7, for a fixed Jordan localization U with $P \Subset U$ the set $U \setminus H'$ is connected. Freeze the inventory of Section 6 as follows: the original atoms are the selected fibers K_w; the closed terminal protected collars are the truncated collar tails, absorbed into the completed packets L_w; the obstacles are the unselected finite fibers and the end fibers, that is, $H_\partial \setminus \bigcup_{w \in F} K_w$; the critical-value exclusion is empty, the branch system (21) being univalent; and the

localization-forbidden set is $\widehat{\mathbb{C}} \setminus U$. The raw frozen set Q of Section 6 consists of the listed pieces themselves; the completed packets L_w enlarge the selected atoms with their truncated collars and the bounded complementary components added by Fill. The set $U \setminus H'$ is connected by Remark 9.7, it lies in $\Omega_{\mathrm{free}} = U \setminus Q$ because $H' \supset Q \cap U$, and it is in fact an entire component Ω_0 of Ω_{free}: its relative boundary in U lies in H', while the finitely many fill components of the packets $L_w \setminus S_w$, the only points of H' not in Q, are open subsets of $U \setminus Q$ separate from $U \setminus H'$. So the component $\Omega_0 = U \setminus H'$ of the frozen free domain contains every port interior, as the next paragraph verifies.

The port arcs land in it. Fix $w \in F$. The cell identity $D_w \cap H_\partial = K_w$ and the disjointness of the cells give $D_w \cap H' = L_w$, and $\alpha_w \cap L_w = \{x_w\} = \{\alpha_w(0)\}$; since $\alpha_w \subset D_w \subset U$, the half-open arc $\alpha_w((0,1])$ lies in $U \setminus H' = \Omega_0$. The arcs α_w, $w \in F$, are pairwise disjoint, lying in pairwise disjoint cells. Thus Ω_0 has simultaneous singleton ports to every required atom K_w, with each port point x_w a free point not equal to a protected impression ($x \notin P_{\mathrm{prot}}$ transports to $x_w \notin \Phi_w(P_{\mathrm{prot}})$).

Step 5: conclusion by the criterion and the realization theorem. By Theorem 6.3, the frozen data admit a protected ribbon-routing certificate: the single free component Ω_0 has simultaneous singleton ports to every required atom. Theorem 5.4 realizes the certificate by a compact connected full carrier that contains every selected fiber K_w, has singleton attachment trace $\{x_w\}$ at each, misses the frozen obstacle set—in particular every unselected finite fiber and every end fiber—and preserves every transported protected chain as a prime-end chain of the enlarged complement, with injective survival and unchanged cyclic order. The cyclic order in question is the transported one: by Step 2 the transport Φ_w is orientation-preserving and carries the root cyclic order of the protected chains to the cyclic order at K_w, and Theorem 5.4(4) preserves the latter. This is the asserted protected regulated connector. □

We reproduce the manuscript's caveat, recorded immediately after Theorem 10.45 of [7], in its own warning environment; the Scope section returns to it.

Warning 9.11. The protected-carrier theorem, Theorem 9.10, is finite and direct-scale only. It does not construct a second satellite portrait, an end-fiber return map, a uniform incidence bound, or future fjord, dam, and attachment data.

This section is the framework's intended mode of use, and it is worth stating plainly what has and has not been done. What has been done is finite and checkable: at a fixed decoration depth, for an explicitly listed selection of addresses and an explicitly listed family of protected root accesses, Theorem 9.10 produces a protected regulated connector whose every property—localization in the cells, isolation from the unselected and

end fibers, survival and cyclic order of the protected prime ends—is certified by the finite data of the gap-cell atlas and the packet audit, and each such certificate persists under the monotone operations of Section 8. What has not been done is everything infinite or virtual. The manuscript's open problems 10.50 (the virtual cylinder target-address selector), 10.51 (disconnected marking transfer to actual bases and iterates on the cylinders and end fibers of Theorem 9.1), and 10.52 (virtual Cantor-package realization at every intermediate scale) are stated as open in [7], remain open, and are not advanced by anything in this section: no return map on end fibers is constructed, no end fiber is identified with a small Julia set, and no selection of target addresses for virtual cylinders is made. The finite theorem is offered as what it is—a template showing that, whenever the missing virtual data are someday supplied, the planar routing they require is already exact, certified, and persistent.

10. Scope and limitations

The results of this paper form a closed package, and the manuscript [7] disciplines its use with an explicit warning (its Warning 10.59), whose discipline we retain here in full. The package has exactly five load-bearing statements, and their division of labor should be read precisely. Theorem 5.4 proves realization *from* a certificate: given a protected ribbon-routing certificate, it produces a compact connected full carrier with singleton attachment traces, surviving protected chains, injective survival, and preserved cyclic order. Theorem 6.3 characterizes *when* such a certificate exists: exactly when one component of the frozen free domain has simultaneous singleton ports to every required atom, with the persistence clause that a separation visible in the frozen data can never be repaired by monotone connector enlargement. Theorem 7.4 is *one sufficient planar route* to that criterion: an augmented-atom packet system forces a single common free component. Theorem 8.3 removes completion order and global port untangling from that route: the packet completions are order-independent, the port checks localize, and merger, localization, and port-loss defects are persistent certificates of failure. Finally, Theorem 9.10 instantiates the entire package at finite direct-decoration scale, where the cells of the atlas play the role of the cellwise data of Proposition 8.5 and the nested packet lemma absorbs the infinitely many unselected fibers.

The mathematical ingredients of all five statements are classical: Janiszewski's separation theorem and the plane topology of continua [8, 10], prime ends [9], filled hulls, and Čech cohomology with the compacta form of Alexander duality [3], in the tradition of planar arguments long standard in complex dynamics [1]. The contribution is not a new theorem about any of these ingredients; it is the assembled certification framework—explicitly protected accesses with declared sides, a frozen future inventory fixed before any connector is chosen, typed packets with order-independent completions, and

monotone, persistent failure certificates. The paper is proof technology, and its value is exactly the conversion it performs: informal planar-routing arguments of the form "connect these continua while keeping these accesses and avoiding that set," which are traditionally re-derived ad hoc inside larger proofs, become exact, persistent, checkable certificates that can be audited clause by clause, transported by univalent maps, and reused without reproof.

Warning 10.1. Theorem 5.4 proves realization from a certificate. Theorem 6.3 characterizes when that certificate exists, Theorem 7.4 proves one sufficient planar route, and Theorem 8.3 removes completion order and port untangling from that route. Theorem 9.10 instantiates the package at finite direct-decoration scale, and it is finite and direct-scale only (Warning 9.11). None of these results constructs the virtual Cantor package, the complete Wave horizon, the analytic covering or immersion maps, a period-independent incidence bound or future-faithful compression, or any width, compactness, rigidity, or MLC conclusion. The open problems of the source manuscript—10.6, 10.50–10.52, 10.55, 10.57, and 10.58 in the numbering of [7]—remain open and are not advanced here: the virtual canonical interface carrier, the manuscript's first protected-tower clause, and MLC itself remain open.

For the reader consulting the manuscript, the open problems named in the warning are, briefly: 10.6, the virtual protected routing certificate for the lamination-lift candidate carrier; 10.50, the virtual cylinder target-address selector; 10.51, the transfer of the disconnected marking to actual bases and actual iterates on the cylinders and end fibers of the pullback compactification; 10.52, the virtual Cantor-package realization at every intermediate scale; and 10.58, the virtual common free component through the complete Wave horizon, together with the related problems 10.55 and 10.57. On the last of these the manuscript is explicit that neither the connected-hull construction of [2] nor the primitive-category Hubbard-tree results of [4] supply the completed free domain that the problem demands; nothing in the present paper does so either, and nothing here is conditional on those problems: every theorem above is proved outright in its finite planar setting.

Two further boundaries deserve restatement. First, the only intentional divergence from the printed manuscript is the repair of Lemma 10.7 and the correspondingly rewritten proof of Theorem 10.8, recorded in Remark 6.2; every other statement is a faithful transcription. Second, no result of this paper asserts local connectivity of any atom, of any fiber, or of any Julia set, and no result should be quoted as progress toward MLC. What may be quoted is the technology: certificates that realize, criteria that are exact, audits that are order-free, and one finite dynamical instantiation in which all of it is checkable.

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Appendix A. Crosswalk to the source manuscript

All results below are reproduced, with notational streamlining and the single recorded repair, from the source research manuscript [7] (build of 2026-07-27). No result labelled open or conditional there has been promoted here.

Present result	Source manuscript	Note
Theorem 3.1	Theorem 6.12	verbatim
Proposition 3.2	Proposition 6.15	verbatim
Definition 5.1	Definition 10.1	verbatim
Lemma 5.2	Lemma 10.2	verbatim
Lemma 5.3	Lemma 10.3	verbatim
Theorem 5.4	Theorem 10.4	verbatim
Proposition 5.6	Proposition 10.5	verbatim
Lemma 6.1	Lemma 10.7	*repaired; see Remark 6.2*
Theorem 6.3	Theorem 10.8	proof rewritten to the repaired lemma
Definition 7.1	Definition 10.9	verbatim
Lemma 7.3	Lemma 10.10	verbatim
Theorem 7.4	Theorem 10.11	verbatim
Lemma 8.1	Lemma 10.12	verbatim
Lemma 8.2	Lemma 10.13	verbatim
Theorem 8.3	Theorem 10.14	verbatim
Proposition 8.5	Proposition 10.15	verbatim
Lemma 9.2	Lemma 10.39	verbatim
Theorem 9.3	Theorem 10.40	verbatim
Lemma 9.4	Lemma 10.41	verbatim
Theorem 9.5	Theorem 10.42	verbatim
Theorem 9.6	Theorem 10.43	verbatim
Lemma 9.8	Lemma 10.44	verbatim
Theorem 9.10	Theorem 10.45	verbatim

The examples of Sections 6 and 8 (Examples 6.4 and 8.4) and the comparison Section 4 are new expository material added for this paper; they promote no manuscript claim.

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